
EHS 2018/19 – 2019/20

Dictionary of derived variables

Dictionary of derived variables

This document contains a series of tables that provide the reader with information on the derived variables contained in the five main EHS derived datasets; generalfs19.sav, interviewfs19.sav, general 18+19.sav, interview 18+19.sav and physical 18+19.sav.

- Interviewfs19.sav and generalfs19.sav relate to the full household sample for survey year 2019/20. These datasets comprise of cases where the home was occupied and where an interview was secured.
- Interview 18+19.sav relates to the household sub-sample for survey years 2018/19 and 2019/20 combined. This dataset contains cases where, as well as an interview with the household, a physical survey inspection was also secured.
- General 18+19.sav and physical 18+19.sav relate to the dwelling sample for survey years 2018/19 and 2019/20 combined. These datasets comprise of cases where the home was occupied or vacant and where a physical inspection was carried out.

This document is split into three chapters. The first chapter covers all variables in generalfs19.sav and general 18+19.sav. The second chapter comprises all variables in interviewfs19.sav and interview 18+19.sav. All variables in physical 18+19.sav are included in chapter three.

The derived variables are in alphabetical order within each chapter and each table corresponds to a derived variable, clearly labelling which data file(s) the variable belongs to.

Files in the End User Licence dataset have the suffix '_EUL' e.g. Interviewfs19_EUL.sav. Corresponding files in the Special Licence dataset have the suffix '_SL' e.g. Interviewfs19_SL.sav.

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Chapter 1

Variables in Generalfs19.sav and General 18+19.sav

Aagfh19**Generalfs19.sav**

Variable name	Aagfh19
Variable label	Household weight (2019-20)
Units	Households
Description	Combined weighting and grossing factor designed to gross up the full sample cases to the total number of households in England.
Type	Numerical
Measure	Scale
Value labels	N/A
Missing data	None
File location	Generalfs19.sav
Validation	Checks against results from previous years for key variables See English Housing Survey 2019-20 Technical Report, chapter 6.
Key inputs	See English Housing Survey 2019-20 Technical Report, chapter 6
Modelling	See English Housing Survey 2019-20 Technical Report, chapter 6
Imputation	None
Key dependencies	None
Introduction date	2008
Date of key modelling changes	The weighting methodology was revised in 2013-14. For more information see English Housing Survey 2019-20 Technical Report, chapter 6 and the weighting methodology paper (see link below): https://www.gov.uk/government/uploads/system/uploads/attachment_data/file/406771/EHS_2013-14_weighting_methodology_paper_FINAL.pdf
Other relevant information	This variable is available in the datasets provided under the End User Licence and the Special Licence.
Previous variable	Aagfh16

Aagpd1819**General 18+19.sav**

Variable name	Aagpd1819
Variable label	Rounded dwelling weight for 2 year Physical Survey sample
Units	Dwellings
Description	Combined weighting and grossing factor designed to gross up the paired sample cases to the total number of dwellings in England.
Type	Numerical
Measure	Scale
Value labels	N/A
Missing data	None
File location	General 18+19.sav
Validation	Checks against results from previous years for key variables. See English Housing Survey 2019-20 Technical Report, chapter 6.
Key inputs	See English Housing Survey 2019-20 Technical Report, chapter 6.
Modelling	See English Housing Survey 2019-20 Technical Report, chapter 6
Imputation	None
Key dependencies	None
Introduction date	2008
Date of key modelling changes	The weighting methodology was revised in 2013-14. For more information see English Housing Survey 2019-20 Technical Report, chapter 6 and the weighting methodology paper (see link below): https://www.gov.uk/government/uploads/system/uploads/attachment_data/file/406771/EHS_2013-14_weighting_methodology_paper_FINAL.pdf The grossing methodology was previously revised in 2008 resulting in a higher proportion of flatted dwellings than under the earlier EHCS methodology. See Appendix A to the EHS Household Report 2008/09 and the EHS 2008 Housing Stock Report respectively for further information.
Other relevant information	This variable is available in the datasets provided under the End User Licence and the Special Licence.
Previous variable	Aagpd1516

Aagph1819**General 18+19.sav**

Variable name	Aagph1819
Variable label	Rounded Household weight for 2-year paired sample
Units	Households
Description	Combined weighting and grossing factor designed to gross up the paired sample cases to the total number of households in England.
Type	Numerical
Measure	Scale
Value labels	N/A
Missing data	None
File location	General 18+19.sav
Validation	Checks against results from previous years for key variables See English Housing Survey 2019-20 Technical Report, chapter 6.
Key inputs	See English Housing Survey 2019-20 Technical Report, chapter 6
Modelling	See English Housing Survey 2019-20 Technical Report, chapter 6
Imputation	None
Key dependencies	None
Introduction date	2008
Date of key modelling changes	The weighting methodology was revised in 2013-14. For more information see English Housing Survey 2019-20 Technical Report, chapter 6 and the weighting methodology paper (see link below): https://www.gov.uk/government/uploads/system/uploads/attachment_data/file/406771/EHS_2013-14_weighting_methodology_paper_FINAL.pdf
Other relevant information	This variable is available in the datasets provided under the End User Licence and the Special Licence.
Previous variable	Aagph1516

Fimonth**Generalfs19.sav, General 18+19.sav**

Variable name	Fimonth
Variable label	Interview survey month (interviewed)
Units	N/A
Description	Month in which the interview survey fieldwork was completed.
Type	Numeric
Measure	Ordinal
Value labels	1 "January" 2 "February" 3 "March" 4 "April" 5 "May" 6 "June" 7 "July" 8 "August" 9 "September" 10 "October" 11 "November" 12 "December"
Missing data	None
File location	generalfs19.sav and general 18+19.sav
Validation	None
Key inputs	None
Modelling	None
Imputation	None
Key dependencies	None
Introduction date	2016
Date of key modelling changes	None
Other relevant information	This variable is only available in the datasets provided under the Special Licence.
Previous variable	N/A

Fiqtr**Generalfs19.sav, General 18+19.sav**

Variable name	Fiqtr
Variable label	Interview survey quarter (interviewed)
Units	N/A
Description	Yearly quarter in which the interview survey fieldwork was completed.
Type	Numeric
Measure	Ordinal
Value labels	1 "quarter 1 (Apr-Jun) 2 "quarter 2 (Jul-Sep)" 3 "quarter 3 (Oct-Dec)" 4 "quarter 4 (Jan-Mar)"
Missing data	None
File location	generalfs19.sav and general 18+19.sav
Validation	None
Key inputs	None
Modelling	None
Imputation	None
Key dependencies	None
Introduction date	2016
Date of key modelling changes	None
Other relevant information	This variable is only available in the datasets provided under the Special Licence.
Previous variable	N/A

Fiyear**Generalfs19.sav, General 18+19.sav**

Variable name	Fiyear
Variable label	Interview survey year (interviewed)
Units	N/A
Description	Year the interview survey fieldwork was completed.
Type	Numeric
Measure	Ordinal
Value labels	None
Missing data	None
File location	generalfs19.sav and general 18+19.sav
Validation	None
Key inputs	None
Modelling	None
Imputation	None
Key dependencies	None
Introduction date	2016
Date of key modelling changes	None
Other relevant information	This variable is only available in the datasets provided under the Special Licence.
Previous variable	N/A

Fmonth**Generalfs19.sav, General 18+19.sav**

Variable name	Fmonth
Variable label	Interview survey month (issued)
Units	N/A
Description	Month in which the interview survey fieldwork was allocated.
Type	Numeric
Measure	Ordinal
Value labels	1 "January" 2 "February" 3 "March" 4 "April" 5 "May" 6 "June" 7 "July" 8 "August" 9 "September" 10 "October" 11 "November" 12 "December"
Missing data	None
File location	generalfs19.sav and general 18+19.sav
Validation	None
Key inputs	None
Modelling	None
Imputation	None
Key dependencies	'lenown', 'vaclngth'
Introduction date	2008
Date of key modelling changes	None
Other relevant information	This variable is only available in the datasets provided under the Special Licence.
Previous variable	N/A

Fqtr**Generalfs19.sav, General 18+19.sav**

Variable name	Fqtr
Variable label	Interview survey quarter (issued)
Units	N/A
Description	Yearly quarter in which the interview survey fieldwork was allocated.
Type	Numeric
Measure	Ordinal
Value labels	1 "quarter 1 (Apr-Jun) 2 "quarter 2 (Jul-Sep)" 3 "quarter 3 (Oct-Dec)" 4 "quarter 4 (Jan-Mar)"
Missing data	None
File location	generalfs19.sav and general 18+19.sav
Validation	None
Key inputs	None
Modelling	None
Imputation	None
Key dependencies	None
Introduction date	2008
Date of key modelling changes	None
Other relevant information	This variable is only available in the datasets provided under the Special Licence.
Previous variable	N/A

Fyear**Generalfs19.sav, General 18+19.sav**

Variable name	Fyear
Variable label	Interview survey year (issued)
Units	N/A
Description	Year the interview survey fieldwork was allocated.
Type	Numeric
Measure	Ordinal
Value labels	None
Missing data	None
File location	generalfs19.sav and general 18+19.sav
Validation	None
Key inputs	None
Modelling	None
Imputation	None
Key dependencies	None
Introduction date	2008
Date of key modelling changes	None
Other relevant information	This variable is only available in the datasets provided under the Special Licence.
Previous variable	N/A

GorEHS**Generalfs19.sav, General 18+19.sav**

Variable name	GorEHS
Variable label	Government office region EHS version
Units	N/A
Description	Government Office Region. Categories are in the same order as in EHS reports.
Type	Numeric
Measure	Nominal
Value labels	1 "North East" 2 "North West" 4 "Yorkshire and the Humber" 5 "East Midlands" 6 "West Midlands" 7 "East" 8 "London" 9 "South East" 10 "South West"
Missing data	None
File location	general 18+19.sav and generalfs19.sav
Validation	Cases are checked for any GOR outside of England as sampling can pick up cases on Welsh or Scottish border which have been assigned a GOR outside England (when in fact they are inside England). These cases have their GOR recoded to the correct English region.
Key inputs	'Gor' (taken from EHS sample file)
Modelling	None
Imputation	None
Key dependencies	'govreg1' region3x'
Introduction date	2008
Date of key modelling changes	None
Other relevant information	This variable is available in the datasets provided under the End User Licence and the Special Licence.
Previous variable	EHCS - 'goregx' (1991-2007) SEH - 'Govreg2'

Govreg1**Generalfs19.sav, General 18+19.sav**

Variable name	Govreg1
Variable label	Government office region, grouped
Units	N/A
Description	Government Office Region – Four category grouping of regions.
Type	Numeric
Measure	Nominal
Value labels	1 “North” 2 “Midlands” 3 “London” 4 “Rest of South”
Missing data	None
File location	generalfs19.sav and general 18+19.sav
Validation	Crosstabulation against GorEHS to make sure that these two variables are consistent. Cases are checked for any GOR outside of England as sampling can pick up cases on Welsh or Scottish border which have been assigned a GOR outside England (when in fact they are inside England). These cases have their GOR recoded to the correct English region.
Key inputs	‘gorEHS’
Modelling	None
Imputation	None
Key dependencies	None
Introduction date	2008
Date of key modelling changes	None
Other relevant information	This variable is available in the datasets provided under the End User Licence and the Special Licence.
Previous variable	N/A

Imd1510**Generalfs19.sav, General 18+19.sav**

Variable name	IMD1510	
Variable label	IMD 2015 decile ranking of areas (lower layer SOA)	
Units	N/A	
Description	Deprivation - Lower Layer Super Output Areas (LSOAs) scored and ranked by the 2015 Index of Multiple Deprivation (IMD).	
Type	Numeric	
Measure	Ordinal	
Value labels	1 "most deprived 10% of areas" 2 "2nd" through to 9 "9th" 10 "least deprived 10% of areas"	
Missing data	None	
File location	generalfs19.sav and general 18+19.sav	
Validation	Frequency checks.	
Key inputs	MHCLG index of multiple deprivation available on the MHCLG web site.	
Modelling	Matched "LSOAcodes" from "IMD15 lookup table to MHCLG file.	
Imputation	None	
Key dependencies	None	
Introduction date	2015	
Date of key modelling changes	None	
Other relevant information	This variable is only available in the dataset provided under the Special Licence.	
Previous variable	IMD1010 (EHCS - 'imd1007' and 'imd1004')	

Imd1910**Generalfs19.sav**

Variable name	IMD1910
Variable label	IMD 2019 decile ranking of areas (lower layer SOA)
Units	N/A
Description	Deprivation - Lower Layer Super Output Areas (LSOAs) scored and ranked by the 2019 Index of Multiple Deprivation (IMD).
Type	Numeric
Measure	Ordinal
Value labels	1 "most deprived 10% of areas" 2 "2nd" through to 9 "9th" 10 "least deprived 10% of areas"
Missing data	None
File location	generalfs19.sav
Validation	Frequency checks.
Key inputs	MHCLG index of multiple deprivation available on the MHCLG web site.
Modelling	Matched "LSOAcodes" from "IMD19 lookup table to MHCLG file.
Imputation	None
Key dependencies	None
Introduction date	2019
Date of key modelling changes	None
Other relevant information	This variable is only available in the dataset provided under the Special Licence.
Previous variable	IMD1510

Paired

Generalfs19.sav

Variable name	Paired
Variable label	Whether paired sample case
Units	N/A
Description	Whether case forms part of the paired dwelling and household data set.
Type	Numeric
Measure	Nominal
Value labels	0 "not paired" 1 "paired"
Missing data	None
File location	generalfs19.sav
Validation	Extensive checks of dwelling type, tenure and vacancy to ensure that interview and physical survey data relates to the same dwelling and that an interview has been carried out where the property was occupied.
Key inputs	casecat (not provided in dataset)
Modelling	None
Imputation	None
Key dependencies	'Aagpd1819', 'aagph1819' (available in the combined dataset only)
Introduction date	2008
Date of key modelling changes	None
Other relevant information	This variable is available in the datasets provided under the End User Licence and the Special Licence.
Previous variable	N/A

Pmonth**Generalfs19.sav, General 18+19.sav**

Variable name	Pmonth
Variable label	Physical survey fieldwork month (completed)
Units	N/A
Description	Month in which the physical survey fieldwork was completed
Type	Numeric
Measure	Ordinal
Value labels	1 "January" 2 "February" 3 "March" 4 "April" 5 "May" 6 "June" 7 "July" 8 "August" 9 "September" 10 "October" 11 "November" 12 "December"
Missing data	None
File location	generalfs19.sav and general 18+19.sav
Validation	None
Key inputs	None
Modelling	None
Imputation	None
Key dependencies	None
Introduction date	2016
Date of key modelling changes	None
Other relevant information	This variable is only available in the datasets provided under the Special Licence.
Previous variable	N/A

Pqtr**Generalfs19.sav, General 18+19.sav**

Variable name	Pqtr
Variable label	Physical survey fieldwork quarter (completed)
Units	N/A
Description	Yearly quarter in which the physical survey fieldwork was completed
Type	Numeric
Measure	Ordinal
Value labels	1 "quarter 1 (Apr-Jun) 2 "quarter 2 (Jul-Sep)" 3 "quarter 3 (Oct-Dec)" 4 "quarter 4 (Jan-Mar)"
Missing data	None
File location	generalfs19.sav and general 18+19.sav
Validation	None
Key inputs	None
Modelling	None
Imputation	None
Key dependencies	None
Introduction date	2016
Date of key modelling changes	None
Other relevant information	This variable is only available in the datasets provided under the Special Licence.
Previous variable	N/A

Pyear**Generalfs19.sav, General 18+19.sav**

Variable name	Pyear
Variable label	Physical survey fieldwork year (completed)
Units	N/A
Description	Year the physical survey fieldwork was completed
Type	Numeric
Measure	Ordinal
Value labels	None
Missing data	None
File location	generalfs19.sav and general 18+19.sav
Validation	None
Key inputs	None
Modelling	None
Imputation	None
Key dependencies	None
Introduction date	2016
Date of key modelling changes	None
Other relevant information	This variable is only available in the datasets provided under the Special Licence.
Previous variable	N/A

Region3x**Generalfs19.sav, General 18+19.sav**

Variable name	Region3x
Variable label	Overall region of England
Units	N/A
Description	Region of England – 3 categories.
Type	Numeric
Measure	Nominal
Value labels	1 “northern regions” 2 “London and South East Regions” 3 “rest of England”
Missing data	None
File location	generalfs19.sav and general 18+19.sav
Validation	Frequency checks and consistency checks against time series.
Key inputs	‘gorEHS’
Modelling	None
Imputation	None
Key dependencies	None
Introduction date	1991
Date of key modelling changes	None
Other relevant information	This variable is available in the datasets provided under the End User Licence and Special Licence.
Previous variable	N/A

Ru11combin**Generalfs19.sav, General 18+19.sav**

Variable name	Ru11combin
Variable label	Rurality classification – combined (2011 COA)
Units	N/A
Description	Rurality classification - combined (2011 COA).
Type	Numeric
Measure	Nominal
Value labels	1 “Urban major conurbation: 10,000 + less sparsely populated” 2 “Urban minor conurbation: 10,000 + less sparsely populated” 3 “Urban city and town: 10,000 + less sparsely populated” 4 “Urban city and town in a sparse setting: 10,000+” 5 “Rural town and fringe: less sparsely populated” 6 “Rural town and fringe in a sparse setting” 7 “Rural village: less sparsely populated” 8 “Rural village in a sparse setting” 9 “Rural hamlet and isolated dwellings: less sparsely populated” 10 “Rural hamlet and isolated dwellings in a sparse setting”
Missing data	None
File location	generalfs19.sav and general 18+19.sav
Validation	Frequency checks and consistency checks against time series.
Key inputs	‘ru11morph’ and ‘ru11contxt’
Modelling	‘ru11combin’ is created by matching in a rurality file from DEFRA using the census11 geographic indicator.
Imputation	None
Key dependencies	None
Introduction date	2015
Date of key modelling changes	None
Other relevant information	This variable is only available in the datasets provided under the Special Licence.
Previous variable	2001 ‘rucombin’

Ru11contxt**Generalfs19.sav, General 18+19.sav**

Variable name	Ru11contxt
Variable label	Rurality classification – context (2011 COA)
Units	N/A
Description	Rurality classification - context (2011 COA).
Type	Numeric
Measure	Nominal
Value labels	0 “less sparse” 1 “sparse”
Missing data	None
File location	generalfs19.sav and general 18+19.sav
Validation	Frequency checks and consistency checks against time series.
Key inputs	rurality (COA) table
Modelling	‘ru11contxt’ is created by matching in a rurality file from DEFRA using the census11 geographic indicator.
Imputation	None
Key dependencies	None
Introduction date	2015
Date of key modelling changes	None
Other relevant information	This variable is only available in the dataset provided under the Special Licence.
Previous variable	2001 ‘rucontxt’

Ru11morph**Generalfs19.sav, General 18+19.sav**

Variable name	Ru11morph
Variable label	Rurality classification – morphology (2011 COA)
Units	N/A
Description	Rurality classification - morphology (2011 COA).
Type	Numeric
Measure	Nominal
Value labels	1 "urban > 10k" 2 "town and fringe" 3 "village" 4 "hamlets and isolated dwellings"
Missing data	None
File location	generalfs19.sav and general 18+19.sav
Validation	Frequency checks and consistency checks against time series.
Key inputs	rurality (COA) table
Modelling	'ru11morph' is created by matching in a rurality file from DEFRA using the census11 geographic indicator.
Imputation	None
Key dependencies	None
Introduction date	2015
Date of key modelling changes	None
Other relevant information	This variable is only available in the dataset provided under the Special Licence.
Previous variable	2001 'rumorph'

Tenure2x**Generalfs19.sav, General 18+19.sav**

Variable name	Tenure2x
Variable label	Tenure
Units	N/A
Description	Tenure – 2 categories
Type	Numeric
Measure	Nominal
Value labels	1 “private” 2 “social”
Missing data	None
File location	generalfs19.sav and general 18+19.sav
Validation	Crosschecks with other tenure variables and consistency checks against time series.
Key inputs	tenure4x
Modelling	None
Imputation	None
Key dependencies	None
Introduction date	1991
Date of key modelling changes	2008
Other relevant information	This variable is available in the datasets provided under the End User Licence and the Special Licence.
Previous variable	N/A

Tenure4x**Generalfs19.sav, General 18+19.sav**

Variable name	Tenure4x
Variable label	Tenure
Units	N/A
Description	Tenure – 4 categories.
Type	Numeric
Measure	Nominal
Value labels	1 “owner occupied” 2 “private rented” 3 “local authority” 4 “housing association”
Missing data	None
File location	generalfs19.sav and general 18+19.sav
Validation	Crosschecks with other tenure variables and consistency checks against time series.
Key inputs	‘tenfin4’ ‘finten4’ and ‘tenure1’
Modelling	None
Imputation	None
Key dependencies	‘tenure8x’, ‘tenure2x’
Introduction date	1996 (in 1991 only tenure7x and tenure2x existed)
Date of key modelling changes	2008
Other relevant information	This variable is available in the datasets provided under the End User Licence and the Special Licence.
Previous variable	N/A

Tenure8x**Generalfs19.sav, General 18+19.sav**

Variable name	Tenure8x
Variable label	Tenure
Units	N/A
Description	Tenure – 8 categories.
Type	Numeric
Measure	Nominal
Value labels	1 “owner occupied - occupied” 2 “private rented - occupied” 3 “local authority - occupied” 4 “housing association – occupied” 5 “owner occupied – vacant” 6 “private rented – vacant” 7 “local authority – vacant” 8 “housing association – vacant”
Missing data	None
File location	generalfs19.sav and general 18+19.sav
Validation	Crosschecks with other vacancy and tenure variables and consistency checks against time series.
Key inputs	‘tenure4x’, ‘vacantx’
Modelling	None
Imputation	None
Key dependencies	‘lenres’, ‘lenown’
Introduction date	1996
Date of key modelling changes	2008
Other relevant information	This variable is available in the datasets provided under the End User Licence and the Special Licence.
Previous variable	In 1991, ‘tenure7x’ did not distinguish between vacant dwellings that were owner occupied and those that were private rented.

Vacantx**General 18+19.sav**

Variable name	Vacantx
Variable label	Whether occupied or vacant
Units	N/A
Description	Whether the dwelling was occupied or vacant at the time of the interviewer's first visit.
Type	Numeric
Measure	Nominal
Value labels	1 "occupied" 2 "vacant"
Missing data	None
File location	general 18+19.sav
Validation	Checks with information from the physical survey on occupancy and when any occupants moved in.
Key inputs	'casecat' (not provided in dataset)
Modelling	None
Imputation	None
Key dependencies	'tenure8x', 'vaclngth'
Introduction date	1991
Date of key modelling changes	2008
Other relevant information	This variable is available in the datasets provided under the End User Licence and the Special Licence.
Previous variable	N/A

VacIngh

General 18+19.sav

Variable name	VacIngh
Variable label	Length of vacancy
Units	N/A
Description	Length of time the dwelling has been vacant at the time of the interviewer's visit.
Type	Numeric
Measure	Nominal
Value labels	-9 "not applicable – occupied" -8 "vacant but unknown duration" 1 "short term (less than 6 months)" 2 "long term (6 months or more)"
Missing data	Yes
File location	general 18+19.sav
Validation	Checks with information from the physical survey on occupancy and when any occupants moved in.
Key inputs	'vacantx', 'vacmonth', 'vacyear' 'vnlstyr', 'vnseas', 'fodvacny', 'fodvacnm', 'fyear', 'fmonth' 'frecL1mt to frecL5mt', 'fodoccup'
Modelling	None – based on a combination of the above variables.
Imputation	If length of vacancy is unknown but both interview and physical surveys recorded as vacant and there was at least 6 months between them, case is coded as 'long term (6 months or more)'.
Key dependencies	None
Introduction date	2007
Date of key modelling changes	2007
Other relevant information	A significant proportion of vacant dwellings are coded as 'vacant but unknown duration'. This variable is only available in the dataset provided under the Special Licence.
Previous variable	EHCS variable 'vacprob' – this has same categories and was created in the same way.

Chapter 2

Variables in Interviewfs19.sav and Interview 18+19.sav

Accomhh Interviewfs19.sav, Interview 18+19.sav

Variable name	Accomhh
Variable label	Type of accommodation for household
Units	N/A
Description	Type of dwelling occupied by the household.
Type	Numeric
Measure	Nominal
Value labels	-9 "does not apply" -8 "no answer" 1 "detached house or bungalow" 2 "semi-detached" 3 "terrace/end of terrace" 4 "purpose built flat/maisonette" 5 "flat conversion/rooms" 6 "caravan or boat" 7 "other"
Missing data	None
File location	interviewfs19.sav and interview 18+19.sav
Validation	Frequency checks and consistency checks against time series
Key inputs	'accom', 'hsetype', 'FltTyp', 'accoth'
Modelling	Combines the data from the 4 raw variables above to create the accommodation categories.
Imputation	Unknowns distributed based on year built and predominant type for that year
Key dependencies	'accomhh1'
Introduction date	2008
Date of key modelling changes	Modelled the same way as SEH method.
Other relevant information	SEH variable. This information is based on household perception, and hence there may be some discrepancies between this variable and the dwelling type variables on physical 18+19.sav – 'dwtype8x', 'dwtypenx' etc which are based on surveyor judgement. This variable is only available in the dataset provided under the Special Licence.
Previous variable	N/A

Accomhh1**Interviewfs19.sav, Interview 18+19.sav**

Variable name	Accomhh1
Variable label	Type of accommodation for household & if not self-contained
Unit	N/A
Description	Type of accommodation for household including category for those where accommodation is not self-contained.
Type	Numeric
Measure	Nominal
Value labels	-9 "does not apply" -8 "no answer" 1 "detached house or bungalow" 2 "semi-detached" 3 "terrace/end of terrace" 4 "purpose built flat/maisonette" 5 "flat conversion/rooms" 6 "caravan or boat" 7 "other" 8 "non self contained"
Missing data	None
File location	interviewfs19.sav and interview 18+19.sav
Validation	Frequency checks and consistency checks against time series
Key inputs	'sharer', 'accomhh'
Modelling	Uses the derived variable 'accomhh' to establish type and separately identifies all those who share any part of their accommodation using 'sharer'.
Imputation	None
Key dependencies	None
Introduction date	2008
Date of key modelling changes	Modelled the same way as SEH method.
Other relevant information	SEH variable. This information is based on household perception, and hence there may be some discrepancies between this variable and the dwelling type variables on physical 18+19.sav – 'dwtype8x', 'dwtypenx' etc which are based on surveyor judgement. This variable is available in the datasets provided under the End User Licence and the Special Licence.
Previous variable	N/A

Agehrp2x**Interviewfs19.sav, Interview 18+19.sav**

Variable name	Agehrp2x
Variable label	Age of HRP – 2 band
Units	N/A
Description	Age of household reference person – 2 categories.
Type	Numeric
Measure	Ordinal
Value labels	-9 "does not apply" -8 "no answer" 1 "under 60 years" 2 "60 or over"
Missing data	None
File location	interviewfs19.sav and interview 18+19.sav
Validation	Frequency checks and consistency checks against time series.
Key inputs	'agehrpx'
Modelling	None – recodes agehrpx into two bands.
Imputation	None.
Key dependencies	None
Introduction date	2008
Date of key modelling changes	None
Other relevant information	This variable is available in the datasets provided under the End User Licence and the Special Licence.
Previous variable	Former SEH variable named 'agehrp1' Former EHCS variable name 'hrpage2x' In 1996, the EHCS variable was based on head of household (HOH) rather than household reference person (HRP). Before 2001, the SEH variable was also based on HOH.

Agehrp4x**Interviewfs19.sav, Interview 18+19.sav**

Variable name	Agehrp4x
Variable label	Age of HRP – 4 band
Units	N/A
Description	Age of household reference person – 4 categories.
Type	Numeric
Measure	Ordinal
Value labels	-9 "does not apply" -8 "no answer" 1 "16 - 29" 2 "30 - 44" 3 "45 - 64" 4 "65 or over"
Missing data	None
File location	interviewfs19.sav and interview 18+19.sav
Validation	Frequency checks and consistency checks against time series.
Key inputs	'agehrpx'
Modelling	None – recodes agehrpx into four bands.
Imputation	None
Key dependencies	None
Introduction date	2008
Date of key modelling changes	None
Other relevant information	This variable is available in the datasets provided under the End User Licence and the Special Licence.
Previous variable	Former EHCS variable name 'hrpage4x' In 1996, the EHCS variable was based on head of household (HOH) rather than household reference person (HRP). Before 2001, the SEH variable was also based on HOH.

Agehrp6x**Interviewfs19.sav, Interview 18+19.sav**

Variable name	Agehrp6x
Variable label	Age of HRP – 6 band
Units	N/A
Description	Age of household reference person – 6 categories.
Type	Numeric
Measure	Ordinal
Value labels	-9 "does not apply" -8 "no answer" 1 "16 - 24" 2 "25 - 34" 3 "35 - 44" 4 "45 - 54" 5 "55 - 64" 6 "65 or over"
Missing data	None
File location	interviewfs19.sav and interview 18+19.sav
Validation	Frequency checks and consistency checks against time series.
Key inputs	'agehrpx'
Modelling	None – recodes agehrpx into six bands.
Imputation	None
Key dependencies	None
Introduction date	2008
Date of key modelling changes	None
Other relevant information	Former SEH variable named 'agehrp5' Former EHCS variable name 'hrpage6x' This variable is available in the datasets provided under the End User Licence and the Special Licence.
Previous variable	Former SEH variable named 'agehrp5' Former EHCS variable name 'hrpage6x' In 1996, the EHCS variable was based on head of household (HOH) rather than household reference person (HRP). Before 2001, the SEH variable was also based on HOH.

Agehrpx**Interviewfs19.sav, Interview 18+19.sav**

Variable name	Agehrpx
Variable label	Age of HRP – continuous
Units	Years
Description	Age of the Household Reference Person.
Type	Numeric
Measure	Scale
Value labels	-9 "does not apply" -8 "no answer" 85 "85 or over"
Missing data	None
File location	interviewfs19.sav and interview 18+19.sav
Validation	Frequency checks and consistency checks against time series.
Key inputs	'age', 'reltohrp2'
Modelling	None – based on the age of the HRP as provided in the interview survey.
Imputation	Any missing cases for 'reltohrp2' were imputed using other information in the dataset.
Key dependencies	'agehrp2x', 'agehrp4x', 'agehrp6x', 'agehrp8x', 'ager'
Introduction date	2008
Date of key modelling changes	Uses same methodology as former EHCS – previous variable name - 'hrpagex'.
Other relevant information	For anonymisation purposes, where the age of the HRP is 85 or more, their age is capped at '85 or more'. This variable is only available in the dataset provided under the Special Licence.
Previous variable	Former SEH variable named 'agehrp' Former EHCS variable name 'hrpagex' In 1996, the EHCS variable was based on head of household (HOH) rather than household reference person (HRP). Before 2001, the SEH variable was also based on HOH.

Agen16**Interviewfs19.sav, Interview 18+19.sav**

Variable name	Agen16
Variable label	Number of persons under 16 in household
Units	N/A
Description	Number of persons under 16 in household.
Type	Numeric
Measure	Scale
Value labels	-9 "does not apply" -8 "no answer"
Missing data	None
File location	interviewfs19.sav and interview 18+19.sav
Validation	Frequency checks and consistency checks against time series.
Key inputs	'age'
Modelling	Sums the number of household members aged under 16.
Imputation	None
Key dependencies	None
Introduction date	2008
Date of key modelling changes	2008 – some differences from old SEH variable.
Other relevant information	Includes all those living in halls of residence. This variable is only available in the dataset provided under the Special Licence.
Previous variable	SEH – 'Nage16'

Ageoldbx**Interviewfs19.sav, Interview 18+19.sav**

Variable name	Ageoldbx
Variable label	Age of oldest person in household – banded
Units	N/A
Description	Age of oldest person in household – 7 categories.
Type	Numeric
Measure	Ordinal
Value labels	-9 "does not apply" -8 "no answer" 1 "16 - 24" 2 "25 - 34" 3 "35 - 49" 4 "50 - 59" 5 "60 - 74" 6 "75 - 84" 7 "85 or over"
Missing data	None
File location	interviewfs19.sav and interview 18+19.sav
Validation	Frequency checks and consistency checks against time series.
Key inputs	'ageoldx'
Modelling	None – recodes ageoldx into 7 bands.
Imputation	None
Key dependencies	None
Introduction date	2008
Date of key modelling changes	None
Other relevant information	Excludes those living in halls of residence unless they are the HRP or partner of the HRP. This variable is available in the datasets provided under the End User Licence and the Special Licence.
Previous variable	Former SEH variable – 'Maxageg' Former EHCS variable – 'Poldbx'

Ageoldx**Interviewfs19.sav, Interview 18+19.sav**

Variable name	Ageoldx
Variable label	Age of oldest person in household
Units	Years
Description	Age of oldest person in household.
Type	Numeric
Measure	Scale
Value labels	-9 "does not apply" -8 "no answer" 85 "85 or over"
Missing data	None
File location	interviewfs19.sav and interview 18+19.sav
Validation	Frequency checks and consistency checks against time series
Key inputs	'age'
Modelling	None – identifies the maximum age of the household members.
Imputation	None
Key dependencies	'ageoldbx'
Introduction date	2008
Date of key modelling changes	Based on former EHCS methodology.
Other relevant information	Excludes those living away from home in halls of residence if not HRP or HRP's partner. For anonymisation purposes, where the age of the oldest person in the household is over 85, their age is capped at 85 or over. This variable is only available in the dataset provided under the Special Licence.
Previous variable	Former SEH variable –'Ageold' Former EHCS variable –'Poldx'

Agepartx**Interviewfs19.sav, Interview 18+19.sav**

Variable name	Agepartx
Variable label	Age of partner – continuous
Units	Years
Description	Age of household reference person's partner.
Type	Numeric
Measure	Scale
Value labels	-9 "does not apply" -8 "no answer" 0 "no partner of HRP" 85 "85 or over"
Missing data	None
File location	interviewfs19.sav and interview 18+19.sav
Validation	Frequency checks and consistency checks against time series.
Key inputs	'age', 'reltohrp2'
Modelling	None – based on the age of the partner as provided in the interview survey.
Imputation	Any missing cases for 'reltohrp2' were imputed using other information in the dataset.
Key dependencies	'olderx'
Introduction date	2008
Date of key modelling changes	None
Other relevant information	For anonymisation purposes, where the age of the partner is 85 or more, their age is capped at '85 or over'. This variable is only available in the dataset provided under the Special Licence.
Previous variable	Former SEH variable – 'ageprt' Former EHCS variable – 'Psagex'

Agepart6x**Interviewfs19.sav, Interview 18+19.sav**

Variable name	Agepart6x
Variable label	Age of partner – 6 band
Units	Years
Description	Age of household reference person's partner -6 categories
Type	Numeric
Measure	Ordinal
Value labels	-9 "no partner of HRP" -8 "no answer" 1 "16 - 24" 2 "25 - 34" 3 "35 - 44" 4 "45 - 54" 5 "55 - 64" 6 "65 or over"
Missing data	None
File location	interviewfs19.sav and interview 18+19.sav
Validation	Frequency checks and consistency checks against time series.
Key inputs	Agepartx
Modelling	None –recodes age of the partner (agepartx) into 6 categories
Imputation	None
Key dependencies	None
Introduction date	2014
Date of key modelling changes	None
Other relevant information	This variable is available in the datasets provided under the End User Licence and the Special Licence.
Previous variable	None

Ager

Interviewfs19.sav, Interview 18+19.sav

Variable name	Ager
Variable label	Report age categories
Units	N/A
Description	Variable indicating age of HRP used in EHS household reporting.
Type	Numeric
Measure	Ordinal
Value labels	-9 "does not apply" -8 "no answer" 1 "16 to 24" 2 "25 to 34" 3 "35 to 44" 4 "45 to 64" 5 "65 to 74" 6 "75 or over"
Missing data	None
File location	Interviewfs19.sav and interview 18+19.sav
Validation	Frequency checks and consistency checks against time series.
Key inputs	'agehrpx'
Modelling	None – recodes agehrpx into the above bands.
Imputation	None
Key dependencies	None
Introduction date	2008
Date of key modelling changes	Although now created using EHS variable 'agehrpx' to identify age group of HRP, this should be the same as SEH method.
Other relevant information	This variable is available in the datasets provided under the End User Licence and the Special Licence.
Previous variable	Former SEH variable – 'age4rep' – before 2001 this was based on head of household (HOH) rather than HRP.

AHCinceq Interviewfs19.sav, Interview 18+19.sav

Variable name	AHCinceq
Variable label	AHC equivalised weekly income (modified OECD scale)
Units	£'s
Description	Weekly net household income after equivalisation factors have been applied and certain housing costs have been deducted. For more details see the section on Household derived indicators, Chapter 5 of the English Housing Survey 2019-20 Technical Report, Annex 5.4.
Type	Numeric
Measure	Scale (continuous variable)
Value labels	1920.00 "£1,920 per week or more"
Missing data	None
File location	interviewfs19.sav and interview 18+19.sav
Validation	The SPSS code contains frequency checks on the input variables. The results from the EHS on the final AHCinceq variable are compared to the equivalent equivalisation results in the DWP Household Below Average Income reports.
Key inputs	The derived variables 'BHCinceq', 'mortwx' and 'rentExS'. OECD equivalisation factors.
Modelling	The AHC income measure takes into account all of the components of the BHC income measure (see 'BHCinceq') but then deducts certain costs - weekly rent (see 'rentExS') and weekly mortgage payments (see 'mortwx'). To derive the AHC equivalised weekly income, the AHC weekly income is divided by the AHC equivalisation factor that is specific to the household (which is derived based on the OECD equivalisation factors). For more details see the section on Household derived indicators, Chapter 5 of the English Housing Survey 2019-20 Technical Report, Annex 5.4.
Imputation	None (as the variable is based upon derived variables with no missing data).
Key dependencies	'AHCinceqv5' and 'AHCinceqv60h'
Introduction date	1996
Date of key modelling changes	From 2011/12 data the EHS AHC equivalised income variable deducts the derived rentExS variable rather than the previously used rentwx variable. Using rentExS in the AHC equivalised income calculations puts all rented properties on a level playing field, i.e. the total weekly rent payable for the property, excluding in all cases the cost of services. 2013 – From 2013/14, Council Tax Support replaced Council Tax benefit in the derivation of BHCinceq.

Other relevant information	The source code for the modelling also calculates the variables 'BHCinceq', 'BHCinceqv5', 'BHCinceqv60h', 'AHCinceqv5' and 'AHCinceqv60h'. Excludes those living in halls of residence unless they are the HRP or partner of HRP. Note, no cases have a measured AHC income greater than the measured BHC income. However, different equivalisation factors are applied to the BHC income measure than the AHC income measure. When the equivalisation factors are applied to the measured BHC income and AHC income, this can result in an equivalised AHC income, greater than the equivalised BHC income. For anonymisation purposes, a weekly income of £1,920 or more is capped at '£1,920 per week or more'. This variable is available in the datasets provided under the End User Licence and the Special Licence.
Previous variable	N/A

AHCinceqv5 Interviewfs19.sav, Interview 18+19.sav

Variable name	AHCinceqv5
Variable label	AHC equivalised income quintiles (weighted by peoplegross)
Units	None
Description	This is a banded version of 'AHCinceq'. The first quintile relates to the households with the lowest 20% of AHC equivalised weekly incomes and the fifth quintile relates to the households with the highest 20% of AHC equivalised weekly incomes.
Type	Numeric
Measure	Ordinal
Value labels	1 "1st quintile (lowest)" 2 "2nd quintile" 3 "3rd quintile" 4 "4th quintile" 5 "5th quintile (highest)"
Missing data	None
File location	interviewfs19.sav and interview 18+19.sav
Validation	Simple frequency checks against results from previous years. See also 'AHCinceq'.
Key inputs	'AHCinceq'
Modelling	Using the full sample equivalised income dataset weighted by peoplegross (which is the person level grossing factor derived by multiplying the household grossing factor by the number of people in the household) the 'AHCinceqv5' ranks and groups the AHC equivalised weekly income into five equal sized categories (quintiles). The quintiles are not re-calculated for the paired dataset.
Imputation	None
Key dependencies	None
Introduction date	1996
Date of key modelling changes	None
Other relevant information	The source code for the modelling also calculates the variables 'BHCinceq', 'BHCinceqv5', 'BHCinceqv60h', 'AHCinceqv5' and 'AHCinceqv60h'. Excludes those living in halls of residence unless they are the HRP or partner of HRP. This variable is available in the datasets provided under the End User Licence and the Special Licence.
Previous variable	N/A

AHCinceqv60h Interviewfs19.sav, Interview 18+19.sav

Variable name	AHCinceqv60h
Variable label	AHC: below 60% of median income (weighted by peoplegross)
Units	None
Description	Whether the household's After Housing Cost equivalised income ('AHCinceq') is less than 60% of the median value of 'AHCinceq'.
Type	Numeric
Measure	Nominal
Value labels	0 "not below threshold" 1 "below threshold"
Missing data	None
File location	interviewfs19.sav and interview 18+19.sav
Validation	The results from the EHS on the final 'AHCinceqv60h' variable are compared to the equivalent equivalisation results in the DWP Household Below Average Income reports.
Key inputs	AHCinceq
Modelling	The full sample equivalised income dataset is weighted by 'peoplegross' (which is the person level grossing factor derived by multiplying the household grossing factor by the number of people in the household) and the median 'AHCinceq' is derived from the EHS data. If the household has an income less than or equal to 60% of the median amount then the value for 'AHCinceqv60h' is set to 1 "below threshold". If the household has an income greater than 60% of the median amount then the value for 'AHCinceqv60h' is set to 0 "not below threshold".
Imputation	None
Key dependencies	None
Introduction date	1996
Date of key modelling changes	None
Other relevant information	The source code for the modelling also calculates the variables 'BHCinceq', 'BHCinceqv5', 'BHCinceqv60h', 'AHCinceqv5' and 'AHCinceqv60h'. Excludes those living in halls of residence unless they are the HRP or partner of HRP. This variable is available in the datasets provided under the End User Licence and the Special Licence.
Previous variable	N/A

ALLincx**Interviewfs19.sav, Interview 18+19.sav**

Variable name	ALLincx
Variable label	Annual gross income of the HRP and partner inc. income from housing benefit and LHA
Units	£'s
Description	Gross annual income of the Household Reference Person (HRP) and any partner from wages, pensions, other private sources, savings and state benefits. This income variable (unlike JOINTINCx) does include housing related benefits that help pay towards rent as income.
Type	Numeric
Measure	Scale
Value labels	100000.00 "£100,000 or more"
Missing data	None
File location	interviewfs19.sav and interview 18+19.sav
Validation	Frequency checks. If the household is not in receipt of housing related benefits that help pay towards rent (housbenx) then a check is made to ensure that ALLincx equals JOINTINCx. If the household is in receipt of housing related benefits that help pay towards rent (housbenx) then a check is made to ensure that ALLincx is greater than JOINTINCx.
Key inputs	JOINTINCx and amthbenx
Modelling	This variable simply adds together the two derived variables annual gross income of the HRP and partner (JOINTINCx) to an annualised housing benefit amount (amthbenx * 52).
Imputation	None
Key dependencies	None
Introduction date	2010/11 data (interviewfs10.sav) 2010+2011 data (interview 10+11.sav)
Date of key modelling changes	None
Other relevant information	The source code for the modelling also calculates 'hhvulx' and 'hhbensx'. For anonymisation purposes, an annual income of £100,000 or more is capped at '£100,000 or more'. This variable is available in the datasets provided under the End User Licence and the Special Licence.
Previous variable	N/A

Amthbenx Interviewfs19.sav, Interview 18+19.sav

Variable name	Amthbenx
Variable label	Weekly housing benefit
Units	£'s
Description	Total weekly amount of housing benefit or Local Housing Allowance (LHA) or Universal Credit received by the householder to help pay for all or part of their rent. This variable only applies to households that rent their home, including households in a shared ownership scheme.
Type	Numeric
Measure	Scale
Value labels	-9.00 "not applicable - owner occupier and not shared ownership"
Missing data	None
File location	interviewfs19.sav and interview 18+19.sav
Validation	The SPSS code contains internal checks of input and interim variables and checks to ensure missing data has been imputed correctly. Housing benefit amounts are checked in conjunction with the validation of the extreme rent outliers (see rentwtx). The final variable is compared with results from previous years overall and by key dwelling descriptor variables.
Key inputs	Most of the Rent and Housing Benefit (QLandB) section in the EHS interview survey from the variable 'Ldjoint' to 'SerInc' (see the EHS questionnaire documentation for the full list) and also variables 'RentHOLA' and 'RentHolWks'. Also, the derived variable 'hhincx' and the interim amounts for individual state benefits and savings and the derived LHA bedroom standard variables LHAqx and LHAstdx, which are key to deriving theoretical housing benefit amounts for missing values.
Modelling	'Amthbenx' and the total weekly rent payable ('rentwtx') are calculated in conjunction with one another in the modelling process. The variables are derived predominantly from the householder's response to the questions in the rent and housing benefit module. Households that rent their home but do not provide any rent/housing benefit information are assigned a theoretical total weekly rent payable (see rentwtx for more information). Renting households that live rent-free are not asked the module on Rent and Housing benefit and for these cases their rent and housing benefit amount are set to zero. For more details see the section on Household derived indicators, Chapter 5 of the English Housing Survey 2019-20 Technical Report, Annex 5.4.
Imputation	Households that receive housing benefit/LHA/Universal Credit towards their rent but do not provide an amount or households that do not know if they receive housing benefit/LHA/Universal Credit are assigned an estimated housing benefit/LHA/Universal Credit amount in the following ways:

	• If the household states that they are in receipt of full housing related benefits towards their rent then the weekly housing related benefit is set to their total weekly rent payable excluding the cost of services (as housing related benefits do not cover the cost of services such as heating and regular meals). • If the household states that they are in receipt of partial housing related benefit payments towards their rent, or if they do not know if their housing benefit/LHA covers all or some of their rent, then a housing related benefit amount is imputed based on their total rent amount payable (excluding the cost of services) and their theoretical entitlement to housing benefit receipt. There are a selected number of Universal Credit cases where the missing weekly housing related benefit amount received by the household is calculated earlier on in the income modelling process, rather than following the above rules. Where this applies, the housing related benefit value is brought into the rent and housing benefit modelling to calculate the final amounts and for these cases, the rentflg is set to a value of 2 'rent/housing related benefits overwritten'.
Key dependencies	'ALLincx', 'housbenx', 'hhbensx', 'hhvulx', 'BHCinceq' and 'AHCinceq'.
Introduction date	1996
Date of key modelling changes	2007 – The rent and housing benefit module was revised in the EHCS 2007 interview survey and the modelling was adjusted accordingly. 2008 – From the EHS 2008 the variable 'amthbenx' incorporates housing benefit and Local Housing Allowance. From the EHS 2013-14, the interview survey was modified so that Universal Credit was included in the question on whether the household receives any help towards their rent in the rent and housing benefit module (Hbena2). 2013 – From the 2013 modelling of EHS data, the theoretical entitlement to partial housing related benefits includes the modelling of the under-occupation charge for working-age social tenants as introduced by government in April 2013. 2016 – From the 2016 modelling of the EHS data, Universal Credit was split into the housing component and the income component, to avoid double counting. 2019 – From 2019 modelling, for Universal Credit cases that report a rent holiday and provide an amount of Universal Credit housing element that helps towards rent during the interview survey, only the rent is adjusted to incorporate the rent free weeks, and not the Universal Credit housing element amount. This aligns the modelling to how Universal Credit housing element payments are issued to claimants.
Other relevant information	The source code used for modelling also calculates the variables 'rentwxx', 'rentflg', 'rentExS' and 'housbenx'. In 2010/11 data a new income variable 'ALLincx' was created that simply adds together the two derived variables annual gross income

	of the HRP and partner (JOINTINCx) to an annualised housing benefit/LHA amount (amthbenx*52). 2012 – From EHS 2012/13 data there were no cases where housing benefit covered more than the rent. As of April 2012 this was no longer an option and Local Housing Allowance was capped at the rent amount. This variable is available in the datasets provided under the End User Licence and the Special Licence.
Previous variable	N/A

Atspaany**Interviewfs19.sav, Interview 18+19.sav**

Variable name	Atspaany
Variable label	Is any household member at state pension age?
Units	N/A
Description	Identifies whether any household member is at state pensionable age (SPA)
Type	Numeric
Measure	Nominal
Value labels	-8 "unknown" 0 "no" 1 "yes"
Missing data	None
File location	interviewfs19.sav and interview 18+19.sav
Validation	Frequency checks and crosschecks
Key inputs	'dteofbth', 'sex', 'age' plus fieldwork dates
Modelling	From 6th April, 2010, DWP introduced a change to the SPA timetable which meant that, for women, the SPA increased gradually from 60 to 65 for those born between April 1950 and December 1953. From December 2018, the SPA for both males and females began increasing incrementally to 66 for those born in or after December 1953, increasing with date of birth (more information on this can be found on DWP web site). The modelling looks at the date of interview, and date of birth of a respondent to calculate their exact age in years and months.
Imputation	For males and females aged of 65 affected by DWP changes, where data on date of birth was missing, likelihood of being at SPA was imputed based on proportion of months likely to be retired within key survey dates. Prior to 2016, such cases were imputed based on results from those with date of birth data available.
Key dependencies	None
Introduction date	2013
Date of key modelling changes	None
Other relevant information	By October 2020, SPA for all males and females will be 66. The State Pension Age will then continue to increase to 67 by 2026. This variable is available in the datasets provided under the End User Licence and the Special Licence.
Previous variable	None

Atspahrp**Interviewfs19.sav, Interview 18+19.sav**

Variable name	Atspahrp
Variable label	Is HRP at state pension age?
Units	N/A
Description	Identifies whether HRP is at state pensionable age (SPA)
Type	Numeric
Measure	Nominal
Value labels	-8 "unknown" 0 "no" 1 "yes"
Missing data	None
File location	interviewfs19.sav and interview 18+19.sav
Validation	Frequency checks and crosschecks
Key inputs	'reltohrp2', 'dteofbth', 'sex', 'age' plus fieldwork dates
Modelling	From 6th April, 2010, DWP introduced a change to the SPA timetable which meant that, for women, the SPA increased gradually from 60 to 65 for those born between April 1950 and December 1953. From December 2018, the SPA for both males and females began increasing incrementally to 66 for those born in or after December 1953, increasing with date of birth (more information on this can be found on DWP web site). The modelling looks at the date of interview, and date of birth of a respondent to calculate their exact age in years and months.
Imputation	For males and females aged of 65 affected by DWP changes, where data on date of birth was missing, likelihood of being at SPA was imputed based on proportion of months likely to be retired within key survey dates. Prior to 2016, such cases were imputed based on results from those with date of birth data available.
Key dependencies	None
Introduction date	2013
Date of key modelling changes	None
Other relevant information	By October 2020, SPA for all males and females will be 66. The State Pension Age will then continue to increase to 67 by 2026. This variable is available in the datasets provided under the End User Licence and the Special Licence.
Previous variable	None

Atspaprt**Interviewfs19.sav, Interview 18+19.sav**

Variable name	Atspaprt
Variable label	Is partner at state pension age?
Units	N/A
Description	Identifies whether partner of HRP is at state pensionable age (SPA)
Type	Numeric
Measure	Nominal
Value labels	-9 "no partner" -8 "unknown" 0 "no" 1 "yes"
Missing data	None
File location	interviewfs19.sav and interview 18+19.sav
Validation	Frequency checks and crosschecks
Key inputs	Partner
Modelling	From 6th April, 2010, DWP introduced a change to the SPA timetable which meant that, for women, the SPA increased gradually from 60 to 65 for those born between April 1950 and December 1953. From December 2018, the SPA for both males and females began increasing incrementally to 66 for those born in or after December 1953, increasing with date of birth (more information on this can be found on DWP web site). The modelling looks at the date of interview, and date of birth of a respondent to calculate their exact age in years and months.
Imputation	For males and females aged of 65 affected by DWP changes, where data on date of birth was missing, likelihood of being at SPA was imputed based on proportion of months likely to be retired within key survey dates. Prior to 2016, such cases were imputed based on results from those with date of birth data available.
Key dependencies	None
Introduction date	2013
Date of key modelling changes	None
Other relevant information	By October 2020, SPA for all males and females will be 66. The State Pension Age will then continue to increase to 67 by 2026. This variable is available in the datasets provided under the End User Licence and the Special Licence.
Previous variable	None

Bedrqx**Interviewfs19.sav, Interview 18+19.sav**

Variable name	Bedrqx
Variable label	Number of bedrooms required by the household (2006 definition)
Units	Rooms
Description	Number of bedrooms that the household theoretically requires based on the guidelines below under 'modelling'.
Type	Numeric
Measure	Scale
Value labels	-9 "does not apply" -8 "no answer" 5 "5 or more"
Missing data	None
File location	interviewfs19.sav and interview 18+19.sav
Validation	Frequency checks.
Key inputs	'sex', 'age', and all relationship variables
Modelling	Uses 2006 definition of the Bedroom Standard to calculate number of bedrooms household requires a) 1 bedroom for couple (no age base). b) 1 bedroom for single person aged 21 or more. c) 1 bedroom for paired adolescents (10-20) same sex. d) 1 bedroom for paired children under 10. e) 1 bedroom for unpaired adolescents and unpaired child of the same sex. f) 1 bedroom for any remaining unpaired adolescent or child.
Imputation	Missing values for relationship variables imputed using other information in the dataset.
Key dependencies	'bedstdx'
Introduction date	2008
Date of key modelling changes	None
Other relevant information	Excludes those living in halls of residence unless they are the HRP or partner of HRP. For anonymisation purposes, a household that requires 5 or more bedrooms is capped at '5 or more'. This variable is available in the datasets provided under the End User Licence and Special Licence.
Previous variable	SEH – 'newbed' EHCS – 'bedrqx' – but this was modelled slightly differently to the SEH and EHS variables.

Bedstdx Interviewfs19.sav, Interview 18+19.sav

Variable name	Bedstdx
Variable label	Bedroom standard (2006 definition)
Units	N/A
Description	Whether household is overcrowded as measured by the bedroom standard.
Type	Numeric
Measure	Ordinal
Value labels	-9 "does not apply" -8 "no answer" 1 "two or more below standard" 2 "one below standard" 3 "at standard" 4 "one above standard" 5 "two or more above standard"
Missing data	None
File location	interviewfs19.sav and interview 18+19.sav
Validation	Frequency checks.
Key inputs	'bedrqx', 'nbedsx'
Modelling	Deducts number of bedrooms each household has from number of bedrooms required based on 2006 Bedroom Standard definition (see 'bedrqx').
Imputation	None
Key dependencies	None
Introduction date	2008
Date of key modelling changes	None
Other relevant information	This variable is available in the datasets provided under the End User Licence and the Special Licence.
Previous variable	SEH – 'bedstnd' EHCS – 'bedstdx' – but this was modelled slightly differently to EHS and SEH.

BHCinceq Interviewfs19.sav, Interview 18+19.sav

Variable name	BHCinceq
Variable label	BHC equivalised weekly income (modified OECD scale)
Units	£'s
Description	Weekly net household income after equivalisation factors have been applied before certain housing costs have been deducted. For more details see the section on Household derived indicators, Chapter 5 of the English Housing Survey 2019-20 Technical Report, Annex 5.4.
Type	Numeric
Measure	Scale
Value labels	1920.00 "£1,920 per week or more"
Missing data	None
File location	interviewfs19.sav and interview 18+19.sav
Validation	The SPSS code contains frequency checks on the input variables. The results from the EHS on the final BHCinceq variable are compared to the equivalent equivalisation results in the DWP Household Below Average Income reports.
Key inputs	The derived variables 'hhincx' and 'amthbenx'. OECD equivalisation factors.
Modelling	The income components that make up the EHS equivalised BHC income variable include; net income of the HRP and any partner, net income from additional adults in the household, winter fuel payment and the addition of council tax support and housing related benefits that help pay towards rent. The income from these specified sources are added together and the amount of council tax payable is deducted. This measured household income is then divided by the BHC equivalisation factor that is specific to the household (which is derived based on the OECD equivalisation factors). For more details see the section on Household derived indicators, Chapter 5 of the English Housing Survey 2019-20 Technical Report, Annex 5.4.
Imputation	None (as the variable is based upon derived and interim variables with no missing data).
Key dependencies	'BHCinceqv5', 'BHCinceqv60h' and 'AHCinceq'
Introduction date	1996
Date of key modelling changes	2013 – From 2013/14, Council Tax Support (also known as Council Tax Reduction) replaced Council Tax benefit.
Other relevant information	The source code used for modelling also calculates the variables 'BHCinceq', 'BHCinceqv5', 'BHCinceqv60h', 'AHCinceqv5' and 'AHCinceqv60h'. Excludes those living in halls of residence unless they are the HRP or partner of HRP. Note, no cases have a measured AHC income greater than the measured BHC income. However, different equivalisation factors are applied to the BHC income measure than the AHC income measure. When the equivalisation factors are applied

	to the measured BHC income and AHC income, this can result in an equivalised AHC income, greater than the equivalised BHC income. For anonymisation purposes, a weekly income of £1,920 or more is capped at '£1,920 per week or more'. This variable is available in the datasets provided under the End User Licence and the Special Licence.
Previous variable	N/A

BHCinceqv5 Interviewfs19.sav, Interview 18+19.sav

Variable name	BHCinceqv5
Variable label	BHC equivalised income quintiles (weighted by peoplegross)
Units	None
Description	This is a banded version of 'BHCinceq'. The first quintile relates to the households with the lowest 20% of BHC equivalised weekly incomes and the fifth quintile relates to the households with the highest 20% of BHC equivalised weekly incomes.
Type	Numeric
Measure	Ordinal
Value labels	1 "1st quintile (lowest)" 2 "2nd quintile" 3 "3rd quintile" 4 "4th quintile" 5 "5th quintile (highest)"
Missing data	None
File location	interviewfs19.sav and interview 18+19.sav
Validation	Simple frequency checks with results from previous years. See also 'BHCinceq'.
Key inputs	'BHCinceq'
Modelling	Using the full sample equivalised income dataset weighted by 'peoplegross' (which is the person level grossing factor derived by multiplying the household grossing factor by the number of people in the household) the 'BHCinceqv5' ranks and groups the BHC equivalised weekly income into five equal sized categories (quintiles). The quintiles are not re-calculated for the paired dataset.
Imputation	None
Key dependencies	None
Introduction date	1996
Date of key modelling changes	None
Other relevant information	The source code used for modelling also calculates the variables 'BHCinceq', 'BHCinceqv5', 'BHCinceqv60h', 'AHCinceqv5' and 'AHCinceqv60h'. Excludes those living in halls of residence unless they are the HRP or partner of HRP. This variable is available in the datasets provided under the End User Licence and the Special Licence.
Previous variable	N/A

BHCinceqv60h Interviewfs19.sav, Interview 18+19.sav

Variable name	BHCinceqv60h
Variable label	BHC: below 60% of median income (weighted by peoplegross)
Units	None
Description	This variable identifies whether the household's Before Housing Cost equivalised income ('BHCinceq') is less than 60% of the median value of 'BHCinceq'.
Type	Numeric
Measure	Nominal
Value labels	0 "not below threshold" 1 "below threshold"
Missing data	None
File location	interviewfs19.sav and interview 18+19.sav
Validation	The results from the EHS on the final BHCinceqv60h variable are compared to the equivalent equivalisation results in the DWP Household Below Average Income reports. Frequency checks with results from the previous years.
Key inputs	'BHCinceq'
Modelling	The full sample equivalised income dataset is weighted by 'peoplegross' (which is the person level grossing factor derived by multiplying the household grossing factor by the number of people in the household) and the median 'BHCinceq' is derived from the EHS data. If the household has an income less than or equal to 60% of the median amount then the value for 'BHCinceqv60h' is set to 1 "below threshold". If the household has an income greater than 60% of the median amount then the value for 'BHCinceqv60h' is set to 0 "not below threshold".
Imputation	None
Key dependencies	None
Introduction date	1996
Date of key modelling changes	None
Other relevant information	The source code used for modelling also calculates the variables 'BHCinceq', 'BHCinceqv5', 'BHCinceqv60h', 'AHCinceqv5' and 'AHCinceqv60h'. Excludes those living in halls of residence unless they are the HRP or partner of HRP. This variable is available in the datasets provided under the End User Licence and the Special Licence.
Previous variable	N/A

Buyfresh**Interviewfs19.sav, Interview 18+19.sav**

Variable name	Buyfresh
Variable label	Year HRP bought present accommodation
Units	N/A
Description	Year of purchasing current dwelling (owner occupiers only).
Type	Numeric
Measure	Ordinal
Value labels	-9 "does not apply" -8 "no answer" 3 "up to 1964" 4 "1965-1974" 5 "1975-1979" 6 "1980-1984" 7 "1985" 8 "1986" 9 "1987" 10 "1988" 11 "1989" 12 "1990" 13 "1991" 14 "1992" 15 "1993" 16 "1994" 17 "1995" 18 "1996" 19 "1997" 20 "1998" 21 "1999" 22 "2000" 23 "2001" 24 "2002" 25 "2003" 26 "2004" 27 "2005" 28 "2006" 29 "2007" 30 "2008" 31 "2009" 32 "2010" 33 "2011" 34 "2012" 35 "2013" 36 "2014" 37 "2015" 38 "2016" 39 "2017" 40 "2018" 41 "2019"

	42 "2020"
Missing data	None
File location	interviewfs19.sav and interview 18+19.sav
Validation	Frequency checks and consistency checks.
Key inputs	'onbyyear', 'tenex'
Modelling	Derived from the above two variables.
Imputation	None
Key dependencies	None
Introduction date	2008
Date of key modelling changes	None
Other relevant information	Originates from SEH. For anonymisation purposes, the three categories 'up to 1944', '1945-1954' and '1955-1964' are combined as 'up to 1964' This variable is only available in the dataset provided under the Special Licence.
Previous variable	Previous SEH variable was "ybought3"

Cohabhrp Interviewfs19.sav, Interview 18+19.sav

Variable name	Cohabhrp
Variable label	If HRP is cohabiting
Units	N/A
Description	Whether HRP is living with partner.
Type	Numeric
Measure	Nominal
Value labels	-9 "does not apply" -8 "no answer" 1 "cohabiting" 2 "not cohabiting" 3 "same sex couple"
Missing data	None
File location	interviewfs19.sav and interview 18+19.sav
Validation	Frequency checks and consistency checks against time series.
Key inputs	'reltohrp2', 'livwth' 'sex'
Modelling	Based on the above 3 variables modelling same sex couple using 'sex' of HRP and partner.
Imputation	Any missing data for 'reltohrp2' imputed using other information from the data set.
Key dependencies	None
Introduction date	2008
Date of key modelling changes	From 2013/14 'livwth' category (3) for same sex couples was removed, now modelled using sex of HRP and partner.
Other relevant information	Originates from SEH. May conflict with other indicators see the section on Household derived indicators, Chapter 5 of the English Housing Survey 2019-20 Technical Report Annex 5.4. This variable is only available in the dataset provided under the Special Licence.
Previous variable	'cohabhrp' from SEH

Cohabprt**Interviewfs19.sav, Interview 18+19.sav**

Variable name	Cohabprt
Variable label	If HRP's partner is cohabiting
Units	N/A
Description	Whether HRP partner is cohabiting with HRP.
Type	Numeric
Measure	Nominal
Value labels	-9 "does not apply" -8 "no answer" 1 "cohabiting" 2 "not cohabiting" 3 "same sex couple "
Missing data	None
File location	interviewfs19.sav and interview 18+19.sav
Validation	Frequency check and crosstab against cohabhrp.
Key inputs	'reltop2', 'livwth', 'sex'
Modelling	Based on the above 3 variables modelling same sex couple using 'sex' of HRP and partner.
Imputation	Any missing data for 'reltop2' imputed using other information from the dataset.
Key dependencies	None
Introduction date	2008
Date of key modelling changes	From 2013/14 'livwth' category (3) for same sex couples removed so now modelled using sex of HRP and partner.
Other relevant information	Originates from SEH. May conflict with other indicators see the section on Household derived indicators, Chapter 5 of the English Housing Survey 2019-20 Technical Report Annex 5.4. This variable is only available in the dataset provided under the Special Licence.
Previous variable	'cohabprt' from SEH

Emphrp3x**Interviewfs19.sav, Interview 18+19.sav**

Variable name	Emphrp3x
Variable label	Working status of HRP (primary) - 3 categories
Units	N/A
Description	Working status of HRP (primary) - 3 categories.
Type	Numeric
Measure	Nominal
Value labels	-9 "does not apply" -8 "no answer" 1 "working" 2 "unemployed" 3 "inactive"
Missing data	None
File location	interviewfs19.sav and interview 18+19.sav
Validation	Frequency checks and consistency checks.
Key inputs	'emphrpx'
Modelling	Simple recoding of 'emphrpx'.
Imputation	See 'emphrpx' for more details
Key dependencies	None
Introduction date	2008
Date of key modelling changes	None
Other relevant information	See 'emphrpx' for more details. This variable is available in the datasets provided under the End User Licence and the Special Licence.
Previous variable	Former EHCS variable – 'hrpemp3x' In 1991 and 1996, the EHCS variable was based on head of household (HOH) rather than household reference person (HRP). Before 2001, the SEH variable was also based on HOH.

Emphrpx**Interviewfs19.sav, Interview 18+19.sav**

Variable name	Emphrpx
Variable label	Employment status (primary) of HRP
Units	N/A
Description	Employment status (primary) of HRP
Type	Numeric
Measure	Nominal
Value labels	-9 "does not apply" -8 "no answer" ' 1 "full-time work" 2 "part-time work" 3 "retired" 4 "unemployed" 5 "full-time education" 6 "other inactive"
Missing data	None
File location	interviewfs19.sav and interview 18+19.sav
Validation	Frequency checks, consistency checks and time series checks.
Key inputs	Raw data on working status, EconAct (allows multiple responses: EconFull to EconOthr), 'age', 'sex', 'reltohrp2'
Modelling	EHS allows respondents to tick more than one category to describe working status of all persons aged 16 or over e.g. retired and working part-time. The modelling assumes which of these is the 'primary' status giving priority to 1) full-time student then 2) retired then 3) registered unemployed. For example, full-time students who are working (part-time or full-time) or who are described as unemployed etc. are all classed as full-time students. Those described as 'retired' and nothing else are classed as retired irrespective of their age. However, for those described as 'retired' and something else, e.g. working or long-term sick, classification depends on their age. If they are at or above state retirement age, then they are classed as retired, if not then they are put into the other category (working part-time etc.). From 2017, those who are of retirement age and classed as 'working' but have not provided employment/self-employment income nor worked in previous 7 days, are reclassified as 'retired'.
Imputation	Missing cases are imputed based on age and, as from 2015, whether received income from work, i.e. if under 18, set to 'full-time student'; if 18 and received income from work, set according to education/employer details; if 19 or more and are at or above state retirement age, set to 'retired'; if 19 or more but below state retirement age, and in receipt of money from work, set to part-time work, if 19 or more but below state

	retirement age, set to 'at home, not seeking work' if no income from work.
Key dependencies	'emphrp3x', 'hhempx'
Introduction date	2008
Modelling changes	Amendment to how missing cases are assigned if aged 19 or more but below state retirement age
Other relevant information	From 2012, further analysis was done on cases where 'EconOthr=1', by examining open code responses to variables EconActOth1 to EconActOth16 (in other.sav) and recoding into correct responses for EconFull to EconStdtd. If no open code response was recorded where 'EconOthr=1', then set to 'retired' if at/over state retirement age, or set to 'not seeking work' if below state retirement age. This variable is available in the datasets provided under the End User Licence and the Special Licence.
Previous variable	Former SEH variable – 'Econprt' but modelling differences ref retirement age. Former EHCS variable – 'hrpemp6x'. Prior to 2001, the EHCS variable was based on head of household (hoh) rather than household reference person (HRP). Before 2001, the SEH variable was also based on HOH.

Empprt3x**Interviewfs19.sav, Interview 18+19.sav**

Variable name	Empprt3x
Variable label	Working status of partner (primary) - 3 categories
Units	N/A
Description	Working status of partner (primary) - 3 categories
Type	Numeric
Measure	Nominal
Value labels	-9 "does not apply" -8 "no answer" 1 "working" 2 "unemployed" 3 "inactive"
Missing data	None
File location	interviewfs19.sav and interview 18+19.sav
Validation	Frequency checks and consistency checks.
Key inputs	'empprt3x'
Modelling	Simple recoding of 'empprt3x'
Imputation	See 'empprt3x' for more details
Key dependencies	None
Introduction date	2008
Date of key modelling changes	None
Other relevant information	See 'empprt3x' for more details. This variable is available in the datasets provided under the End User Licence and the Special Licence.
Previous variable	N/A

Empprtx**Interviewfs19.sav, Interview 18+19.sav**

Variable name	Empprtx
Variable label	Employment status (primary) of partner
Units	N/A
Description	Employment status (primary) of partner.
Type	Numeric
Measure	Nominal
Value labels	-9 "does not apply" -8 "no answer" 1 "full-time work" 2 "part-time work" 3 "retired" 4 "unemployed" 5 "full-time education" 6 "other inactive"
Missing data	None
File location	interviewfs19.sav and interview 18+19.sav
Validation	Frequency checks, consistency checks and time series checks.
Key inputs	Raw data on working status, EconAct (allows multiple responses: EconFull to EconOthr), 'age', 'sex', 'reltohrp2'
Modelling	EHS allows respondents to tick more than one category to describe working status of all persons aged 16 or over e.g. retired and working part-time. The modelling assumes which of these is the 'primary' status giving priority to 1) full-time student then 2) retired then 3) registered unemployed. For example, full-time students who are working (part-time or full-time) or who describe themselves as unemployed etc. are all classed as full-time students. Those described as 'retired' and nothing else are classed as retired irrespective of their age. However, for those described as 'retired' and something else, e.g. working or long-term sick, classification depends on their age. If they are at or above state retirement age, then they are classed as retired, if not then they are put into the other category (working part-time etc.). From 2017, those who are of retirement age and classed as 'working' but have not provided employment/self-employment income nor worked in previous 7 days, are reclassified as 'retired'.
Imputation	Missing cases are imputed based on age and, as from 2015, whether received income from work, i.e. if under 18, set to 'full-time student'; if 18 and received income from work, set according to education/employer details; if 19 or more and are at or above state retirement age, set to 'retired'; if 19 or more but below state retirement age, and in receipt of money from work, set to part-time work, if 19 or more but below state

	retirement age, set to 'at home, not seeking work' if no income from work.
Key dependencies	'hhempx', 'empprt3x'
Introduction date	2008
Date of key modelling changes	None
Other relevant information	From 2012, further analysis was done on cases where 'EconOthr=1', by examining open code responses to variables EconActOth1 to EconActOth16 (in other.sav) and recoding into correct responses for EconFull to EconStd. If no open code response was recorded where 'EconOthr=1', then set to 'retired' if at/over state retirement age, or recode to 'not seeking work' if below state retirement age. This variable is available in the datasets provided under the End User Licence and the Special Licence.
Previous variable	N/A

Equityr

Interviewfs19.sav, Interview 18+19.sav

Variable name	Equityr
Variable label	Equity in home (based on respondent valuation only)
Units	£'s
Description	Amount of equity in the home for owner occupiers. This is based on the respondent's estimate of the value of the dwelling minus the amount of outstanding mortgage (and other loans on the property).
Type	Numeric
Measure	Scale
Value labels	-99999999 "not applicable – tenant" -88888888 "unknown" 1000000 "£1,000,000 or more"
Missing data	Yes
File location	interviewfs19.sav and interview 18+19.sav
Validation	There is no external source that could be used to help validate this data. All validation is therefore internal. It includes checks and cross checks of raw input and interim variables. They also include comparing the final variable with results from previous years overall and by key household descriptor variables.
Key inputs	'currbal', 'outstand1', 'outstand2', 'prptval1', 'prptval2'
Modelling	Equity is calculated for owner occupiers by deducting outstanding mortgage from current valuation (as estimated by respondent). For more details see the section on Household derived indicators, Chapter 5 of the English Housing Survey 2019-20 Technical Report Annex 4
Imputation	If 'prptval1'/'prptval2' missing, use latest MHCLG average house prices by region and dwelling type (Table 511)
Key dependencies	None
Introduction date	2009
Date of key modelling changes	Predecessor to this variable, "equity" was calculated using valuation from market value survey. As this survey discontinued in 2010, household estimates of market value were used instead, and imputed using annual MHCLG average house prices by region and dwelling type (Table 511) where unknown.
Other relevant information	Source code also calculates 'mortwx'. For anonymisation purposes, an equity value of £1,000,000 or more is capped at '£1,000,000 or more'. This variable is available in the datasets provided under the End User Licence and the Special Licence.
Previous variable	'Equity'

Equityr5**Interviewfs19.sav, Interview 18+19.sav**

Variable name	Equityr5
Variable label	Equity in home (based on respondent valuation only)
Units	N/A
Description	Amount of equity in the home for owner occupiers - banded. This is based on the respondent's estimate of the value of the dwelling minus the amount of mortgage outstanding (and other loans on the property).
Type	Numeric
Measure	Ordinal
Value labels	-9 "not applicable - tenant" -8 "unknown" 1 "less than £50,000" 2 "£50,000 to £80,000" 3 "£80,000 to £120,000" 4 "£120,000 to £180,000" 5 "over £180,000"
Missing data	Yes
File location	interviewfs19.sav and interview 18+19.sav
Validation	Check results with previous years
Key inputs	'equityr'
Modelling	None – ranks equityr into five categories.
Imputation	None
Key dependencies	None
Introduction date	2009
Date of key modelling changes	Predecessor to this variable, "equity5" was calculated using valuation from market value survey. As this survey discontinued in 2010, household estimates of market value were used instead, and imputed using annual MHCLG average house prices by region and dwelling type (Table 511) where unknown.
Other relevant information	Source code also calculates 'mortwvx'. This variable is available in the datasets provided under the End User Licence and the Special Licence.
Previous variable	'equity5'

Ethhrp2x**Interviewfs19.sav, Interview 18+19.sav**

Variable name	Ethhrp2x
Variable label	Ethnic origin of HRP - 2 categories
Units	N/A
Description	Ethnic origin of HRP - 2 categories.
Type	Numeric
Measure	Nominal
Value labels	-9 "does not apply" -8 "no answer" 1 "white" 2 "ethnic minority"
Missing data	None
File location	interviewfs19.sav and interview 18+19.sav
Validation	Frequency checks and consistency checks against time series
Key inputs	'ethE', 'ethhrp4x', 'ethhrp8x'
Modelling	None – recodes the derived variable ethhrp4x.
Imputation	See 'ethhrp8x'
Key dependencies	None
Introduction date	2008
Date of key modelling changes	None
Other relevant information	The "white" category includes "white British" and "all other white backgrounds". This variable is available in the datasets provided under the End User Licence and the Special Licence.
Previous variable	Former EHCS variable 'hrpeth2x' Prior to 2001, the EHCS variable was based on head of household (HOH) rather than household reference person (HRP). Before 2001, the SEH variable was also based on HOH.

Ethhrp4x**Interviewfs19.sav, Interview 18+19.sav**

Variable name	Ethhrp4x
Variable label	Ethnic origin of HRP - 4 categories
Units	N/A
Description	Ethnic origin of HRP - 4 categories.
Type	Numeric
Measure	Nominal
Value labels	-9 "does not apply" -8 "no answer" 1 "white" 2 "black" 3 "Asian" 4 "other"
Missing data	None
File location	interviewfs19.sav and interview 18+19.sav
Validation	Frequency checks and consistency checks.
Key inputs	'ethE', 'ethhrp8x'
Modelling	None – recodes the derived variable ethhrp8x.
Imputation	See 'ethhrp8x'
Key dependencies	None
Introduction date	2008
Date of key modelling changes	None
Other relevant information	The "white" category includes "white British" and "all other white backgrounds". This variable is only available in the dataset provided under the Special Licence.
Previous variable	Former EHCS variable 'hrpeth4x' Prior to 2001, the EHCS variable was based on head of household (HOH) rather than household reference person (HRP). Before 2001, the SEH variable was also based on HOH.

Ethhrp8x**Interviewfs19.sav, Interview 18+19.sav**

Variable name	Ethhrp8x
Variable label	Ethnic origin of HRP - 8 categories
Units	N/A
Description	Ethnic origin of HRP - 8 categories.
Type	Numeric
Measure	Nominal
Value labels	-9 "does not apply" -8 "no answer" 1 "white" 2 "black" 3 "Indian" 4 "Pakistani or Bangladeshi" 5 "other Asian" 6 "Chinese" 7 "mixed" 8 "other"
Missing data	None
File location	interviewfs19.sav and interview 18+19.sav
Validation	Frequency checks and consistency checks.
Key inputs	'EthE, 'reltohrp2'
Modelling	None
Imputation	Where there is no response, ethnicity is assumed to be the same as that given for partner. If there is no partner or partner's ethnicity is also missing, assumed to be white British.
Key dependencies	'ethhrp4x' and 'ethhrp2x'
Introduction date	2008
Date of key modelling changes	Methodology similar to how EHCS variable 'ethhrpx' was derived. This variable is only available in the dataset provided under the Special Licence.
Other relevant information	N/A
Previous variable	Former SEH variable – 'ethhrp' – note this had missing values. Former EHCS variable 'hrpeth8x'. Prior to 2001, the EHCS and SEH variables were based on head of household (HOH) rather than household reference person (HRP). Before 2001, the SEH variable was also based on HOH.

Ethprt2x**Interviewfs19.sav, Interview 18+19.sav**

Variable name	Ethprt2x
Variable label	Ethnic group of HRP's partner – 2 categories
Units	N/A
Description	Ethnic origin of HRP's partner
Type	Numeric
Measure	Nominal
Value labels	-9 "does not apply" -8 "no answer" 1 "white" 2 "ethnic minority"
Missing data	None
File location	interviewfs19.sav and interview 18+19.sav
Validation	Frequency checks and consistency checks.
Key inputs	'ethprt8x'
Modelling	None
Imputation	Where there is no response, ethnicity is assumed to be the same as that given for the HRP. If HRP's ethnicity is also missing, assumed to be white British.
Key dependencies	None
Introduction date	2014
Date of key modelling changes	None
Other relevant information	New variable from 2014. The "white" category includes "white British" and "all other white backgrounds". This variable is available in the datasets provided under the End User Licence and the Special Licence.
Previous variable	Ethprt

Ethprt8x**Interviewfs19.sav, Interview 18+19.sav**

Variable name	Ethprt8x
Variable label	Ethnic group of HRP's partner – 8 categories
Units	N/A
Description	Ethnic origin of HRP's partner.
Type	Numeric
Measure	Nominal
Value labels	-9 “does not apply” -8 “no answer” 1 “white” 2 “black” 3 “Indian” 4 “Pakistani or Bangladeshi” 5 “other Asian” 6 “Chinese” 7 “mixed” 8 “other”
Missing data	None
File location	interviewfs19.sav and interview 18+19.sav
Validation	Frequency checks and consistency checks.
Key inputs	'EthE', 'rektop2'
Modelling	None
Imputation	Where there is no response, ethnicity is assumed to be the same as that given for the HRP. If HRP's ethnicity is also missing, assumed to be white British.
Key dependencies	None
Introduction date	2014
Date of key modelling changes	'ethprt' was created in 2008 using EHS method for 'ethhrp8x'. EHS method is used to be consistent with method for Ethhrp8x. Created prior to 2008 using SEH method.
Other relevant information	From 2008-2013, the variable for ethnicity of the hrp's partner was 'ethprt'. Categories were condensed in 2014 and 'ethprt8x' was created. Categories consistent with ethhrp8x. This variable is only available in the dataset provided under the Special Licence.
Previous variable	Former SEH variable – 'ethprt'

Famnumx Interviewfs19.sav, Interview 18+19.sav

Variable name	Famnumx
Variable label	Number of family units in household
Units	Number
Description	Number of family units in the household. See EHS Glossary for definition of 'family unit'.
Type	Numeric
Measure	Scale
Value labels	-9 "does not apply" -8 "no answer" 5 "5 or more"
Missing data	None
File location	interviewfs19.sav and interview 18+19.sav
Validation	Frequency checks and consistency checks against time series.
Key inputs	'afam', 'reltohrp2'
Modelling	None
Imputation	Any missing cases for 'reltohrp2' were imputed using other information in the dataset.
Key dependencies	'otherfam', 'nounits1'
Introduction date	1991
Date of key modelling changes	None
Other relevant information	This excludes those living in halls of residence who are not the HRP or partner (unlike old SEH definition). Also, same sex couples are counted as separate family units, despite being treated as a couple living together in other variables (e.g. 'loncoupx', 'bedstdx', 'hhcompix'). Therefore, there may be some discontinuity when comparing these variables with SEH. Originates from EHCS. May conflict with other indicators see the section on Household derived indicators, Chapter 5 of the English Housing Survey 2019-20 Technical Report Annex 5.4. For anonymisation purposes, a household with 5 or more family units is capped at '5 or more'. This variable is only available in the dataset provided under the Special Licence
Previous variable	N/A

FreeLeas**Interviewfs19.sav, Interview 18+19.sav**

Variable name	FreeLeas
Variable label	Freehold or leasehold
Units	N/A
Description	Whether property is owned on a freehold or leasehold basis. This is a simplified version of variable 'owntype'.
Type	Numeric
Measure	Nominal
Value labels	-9 "does not apply" -8 "no answer" 1 "freehold" 2 "leasehold"
Missing data	Yes
File location	interviewfs19.sav and interview 18+19.sav
Validation	Frequency checks and consistency checks.
Key inputs	'accom', 'Lease2', 'CHlease2', 'leaseothr', 'FreeHld1New', 'tenure4', 'tenure1'
Modelling	Combines the above variables to create the derived variable.
Imputation	Park homes and boats are recorded as no answer. Unknowns recoded to 'does not apply' if tenure1 says rented from employer. From 2015, if freeLeas unknown for a flat and lease says leasehold, freeLeas recoded to "leasehold".
Key dependencies	None
Introduction date	2008
Date of key modelling changes	2008
Other relevant information	This variable is available in the datasets provided under the End User Licence and the Special Licence.
Previous variable	EHCS only - 'freehold'. But modelled differently as questions asked were changed.

Ftbuyer**Interviewfs19.sav, Interview 18+19.sav**

Variable name	Ftbuyer
Variable label	If first-time buyer
Units	N/A
Description	Whether this is the first time that the HRP and/or partner have ever owned a home.
Type	Numeric
Measure	Nominal
Value labels	-9 "does not apply" -8 "no answer" 1 "first-time buyer" 2 "owner but has owned previously" 3 "non-owner"
Missing data	Yes
File location	interviewfs19.sav and interview 18+19.sav
Validation	Frequency checks and consistency checks.
Key inputs	'ownpr', 'ten1', 'prevac', 'prevacn'
Modelling	Derived from a combination of the above variables
Imputation	Unknown cases set to 'first time buyer' if bought with 100% mortgage and no savings (sournone), or set to 'owned previously' if sold previous house (soursale), otherwise left as unknown.
Key dependencies	None
Introduction date	2008
Date of key modelling changes	N/A
Other relevant information	This variable is available in the datasets provided under the End User Licence and the Special Licence.
Previous variable	N/A

GrossA Interviewfs19.sav, Interview 18+19.sav

Variable name	GrossA
Variable label	Age group & sex of youngest person in household
Units	N/A
Description	Age group & sex of youngest person in household.
Type	Numeric
Measure	Nominal
Value labels	-9 "does not apply" -8 "no answer" 1 "0-4" 2 "5-15" 3 "16-19 male" 4 "16-19 female" 5 "20-24 male" 6 "20-24 female" 7 "25-29 male" 8 "25-29 female" 9 "30-44 male" 10 "30-44 female" 11 "45-64 male" 12 "45-64 female" 13 "65-74 male" 14 "65-74 female" 15 "75 or over male" 16 "75 or over female"
Missing data	None
File location	interviewfs19.sav and interview 18+19.sav
Validation	Frequency checks and consistency checks against time series.
Key inputs	'age', 'sex'
Modelling	None – derived from the above variables.
Imputation	None
Key dependencies	None
Introduction date	2008
Date of key modelling changes	Same as former SEH methodology
Other relevant information	Excludes those living in halls of residence unless they are HRP or partner. This variable is only available in the dataset provided under the Special Licence.
Previous variable	Former SEH variable – GrossA

Hatentp3**Interviewfs19.sav, Interview 18+19.sav**

Variable name	Hatentp3
Variable label	Type of housing association tenancy
Units	N/A
Description	Type of housing association tenancy.
Type	Numeric
Measure	Nominal
Value labels	-9 "does not apply" -8 "no answer" 1 "assured tenancy" 2 "secure tenancy (offered up to and including 1988)" 3 "assured shorthold tenancy" 4 "fixed term tenancy (from April 2011)" 5 "starter tenancy" 6 "other tenancy"
Missing data	Yes
File location	interviewfs19.sav and interview 18+19.sav
Validation	Frequency checks and consistency checks.
Key inputs	'tentypSR2, srtenlen2, srtenlen2b, tenex', 'tenure1'
Modelling	Information is taken from a combination of srtenlen2, srtenlen2b and tentypSR2 where tenure ('tenex' and 'tenure1') is classed as housing association
Imputation	None
Key dependencies	None
Introduction date	2014
Date of key modelling changes	None
Other relevant information	This is created using different input variables to the previous variable 'hatentp2', and there is an extra category 'assured shorthold tenancy' so comparing these variables is not advisable. In 2017, a continuous length of social rented tenancy variable was introduced (srtenlen2) in addition to the existing banded variable (srtenlen2b). This variable is available in the datasets provided under the End User Licence and the Special Licence.
Previous variable	SEH variable - 'Hatentp' and EHS variable 'Hatentp2'

Hhbensx**Interviewfs19.sav, Interview 18+19.sav**

Variable name	Hhbensx
Variable label	Household on means tested bens or tax credits with a relevant income below the threshold
Units	None
Description	Whether the HRP or any partner are in receipt of certain means tested benefits or tax credits where their relevant income is below the threshold.
Type	Numeric
Measure	Nominal
Value labels	1 "yes" 2 "no"
Missing data	None
File location	interviewfs19.sav and interview 18+19.sav
Validation	The final variable is compared with results from previous years overall and by key household descriptor variables.
Key inputs	Interim means tested benefit/tax credit variables calculated within the income modelling and the derived variables 'housbenx' and 'JOINTINCx'.
Modelling	A household is modelled to be in receipt of means tested benefits or tax credits with a relevant income below the threshold if any of the following conditions apply to the HRP and any partner: • In receipt of income support or income based Job Seekers Allowance/incapacity benefit that included an income support component • In receipt of income based Employment and Support Allowance • In receipt of Universal Credit • In receipt of housing related benefits that help pay towards rent • Any household on a low income that has had their income imputed up to their basic income support entitlement • In receipt of pension credit • In receipt of child tax credit * • In receipt of working tax credit * * those in receipt of either working tax credit (modelled to be accompanied by a disability premium), or child tax credit, are only included in the flag if the HRP and any partner has an assessed gross income below the Child Tax Credit first income threshold (entitled to child tax credit only) which was £16,105 in 2018/19 and 2019/20. If their income is below this threshold then they are considered to be vulnerable as referred to in the PSA 7 document (associated with Decent Homes).

Imputation	None
Key dependencies	'Hhvulx'
Introduction date	1991
Date of key modelling changes	For 2003/04 EHS data child tax credit and working tax credit were included in the flag. For 2006/07 EHS data, pension credit was included in the flag. For 2010/11 EHS data, income based Employment and Support was included in the flag. For 2011/12 EHS data, income based JSA was used as a qualifying benefit as opposed to all JSA claimants (a refinement incorporated due to newly collected information on the EHS interview survey). For 2013/14 EHS data, Universal Credit was included in the flag.
Other relevant information	The source code used for the modelling also calculates 'hhvulx'. This variable is available in the datasets provided under the End User Licence and the Special Licence.
Previous variable	N/A

Hhcomp1**Interviewfs19.sav, Interview 18+19.sav**

Variable name	Hhcomp1
Variable label	Household composition
Units	N/A
Description	Household composition focussing on HRP (7 categories).
Type	Numeric
Measure	Nominal
Value labels	-9 "does not apply" -8 "no answer" 1 "married/cohabiting couple" 2 "lone parent, male HRP" 3 "lone parent, female HRP" 4 "multi-family household, male HRP" 5 "multi-family household, female HRP" 6 "1 male" 7 "1 female"
Missing data	None
File location	interviewfs19.sav and interview 18+19.sav
Validation	Frequency check and crosstab with 'livwith' to check same sex couples are treated as couples, crosschecks with hhtype11 and nsing.
Key inputs	'hhsizex', 'sexhrp', 'agehrpx', 'ncouple', 'nlpar', 'child', 'nxdepch', 'ndepchild', "
Modelling	Based on SEH method of deriving household composition. For more details see the section on Household derived indicators, Chapter 5 of the English Housing Survey 2019-20 Technical Report Annex 5.4.
Imputation	None
Key dependencies	None
Introduction date	2008
Date of key modelling changes	None
Other relevant information	Same sex couples are classed as couples where one of the partners is the HRP. This variable is available in the datasets provided under the End User Licence and Special Licence.
Previous variable	N/A

Hhcompx Interviewfs19.sav, Interview 18+19.sav

Variable name	Hhcompx
Variable label	Household composition
Units	N/A
Description	Household composition (7 categories).
Type	Numeric
Measure	Nominal
Value labels	-9 "does not apply" -8 "no answer" 1 "couple, no dependent child(ren) under 60" 2 "couple, no dependent child(ren) aged 60 or over" 3 "couple with dependent child(ren)" 4 "lone parent with dependent child(ren)" 5 "other multi-person households" 6 "one person under 60" 7 "one person aged 60 or over"
Missing data	None
File location	interviewfs19.sav and interview 18+19.sav
Validation	Crosschecks with hhtype11, hhtype7 and hhtype6.
Key inputs	hhtype6, agehrp
Modelling	Based on EHCS method of deriving household composition which treats same sex couples as couples. Dependent children are classed as those that are aged under 16 or aged 16-18 and in full-time education. Couple based households with independent children only are classed as 'couples' whereas lone parents with independent children only are classed as 'other multi-person households'. For more details see the section on Household derived indicators, Chapter 5 of the English Housing Survey 2019-20 Technical Report Annex 5.4.
Imputation	Recoded in some cases where HRP is child of a couple and the couple is not identified.
Key dependencies	None
Introduction date	1991
Date of key modelling changes	None
Other relevant information	This variable is available in the datasets provided under the End User Licence and Special Licence.
Previous variable	EHCS variable – 'hhcomp'

Hhempx**Interviewfs19.sav, Interview 18+19.sav**

Variable name	Hhempx
Variable label	Employment status of HRP and partner combined
Units	N/A
Description	Employment status of HRP and partner combined.
Type	Numeric
Measure	Nominal
Value labels	-9 "does not apply" -8 "no answer" 1 "1 or more work full-time" 2 "1 or more work part-time" 3 "none working, one or more retired" 4 "none working and none retired"
Missing data	None
File location	interviewfs19.sav and interview 18+19.sav
Validation	Frequency checks and consistency checks.
Key inputs	'emphrpx' 'empprt看'
Modelling	None – based on the above variables
Imputation	See 'emphrpx' and 'empprt看' for more details
Key dependencies	None
Introduction date	2008
Date of key modelling changes	None
Other relevant information	See 'emphrpx' and 'empprt看' for more details. This variable is available in the datasets provided under the End User Licence and the Special Licence.
Previous variable	Former EHCS variable – 'hhempx'

Hhinc5x**Interviewfs19.sav, Interview 18+19.sav**

Variable name	Hhinc5x
Variable label	All households - income in 5 bands
Units	None
Description	This is a banded version of 'hhincx' which is ranked and grouped into five equal sized categories. The first quintile relates to the households with the lowest 20% of weekly incomes and the fifth quintile relates to the households with the highest 20% of weekly incomes.
Type	Numeric
Measure	Ordinal
Value labels	-9 "not applicable" 1 "lowest 20%" 2 "quintile 2" 3 "quintile 3" 4 "quintile 4" 5 "highest 20%"
Missing data	None
File location	interviewfs19.sav and interview 18+19.sav
Validation	Checks with results from previous years.
Key inputs	'Hhincx'
Modelling	None – ranks hhincx into five equal sized categories.
Imputation	None
Key dependencies	None
Introduction date	1991
Date of key modelling changes	None
Other relevant information	This variable is available in the datasets provided under the End User Licence and the Special Licence.
Previous variable	N/A

Hhincflg**Interviewfs19.sav, Interview 18+19.sav**

Variable name	Hhincflg
Variable label	Imputations used to create net total household income
Units	None
Description	Whether any imputation of data occurs in the derivation of the annual net income of the HRP and partner – 'hhincx' (and 'hhinc5x' that is derived from it).
Type	Numeric
Measure	Nominal
Value labels	0 "none, all data OK" 1 "some private sources imputed" 2 "some benefits imputed or changed" 3 "some private sources and some benefits imputed" 4 "household total imputed using group median" 5 "was below basic IS/PC - imputed using group median" 6 "was below basic IS/PC - imputed using basic IS/PC" 7 "was below basic IS/PC - imputed using basic IS/PC plus disability premiums"
Missing data	None
File location	interviewfs19.sav and interview 18+19.sav
Validation	Checks with results from previous years.
Key inputs	Interim variables created within the income modelling associated with imputing missing data e.g. 'estbens' (any benefit amounts imputed?), 'empesthg' (main job employee HRP - gross estimated?), 'impmedian' (Income imputed using group median?) etc.
Modelling	None - combines interim variables associated with missing income data.
Imputation	The imputation flag also incorporated imputed savings into the 'hhincflg', i.e. if the respondent's savings amount was imputed it has been classed as a private income source imputation (prior to 2013/14 modelling, imputed savings did not feed into 'hhincflg').
Key dependencies	None
Introduction date	1991
Date of key modelling changes	None
Other relevant information	The source code used for modelling also calculates the variables 'hhincx' and 'JOINTINCx'. From the EHS 2013/14 data, the imputation flag also incorporates imputed savings, i.e. if the respondent's savings amount is imputed, it is now classed as a private income

	source imputation (previously imputed savings did not feed into 'hhincflg'). This variable is only available in the dataset provided under the Special Licence.
Previous variable	N/A

Hhincx**Interviewfs19.sav, Interview 18+19.sav**

Variable name	Hhincx
Variable label	EHS Basic Income (annual net household income (HRP + Partner) including savings)
Units	£'s
Description	The EHS Basic Income refers to the annual net income of the Household Reference Person (HRP) and any partner from wages, pensions, other private sources, savings and state benefits. Amounts are net i.e. after the deduction of Income Tax and National Insurance where applicable. This income variable does not include any housing related benefits/allowances as collected in the 'rent and housing benefit' module and 'council tax and utilities' module of the EHS interview questionnaire.
Type	Numeric
Measure	Scale
Value labels	100000.00 "£100,000 or more"
Missing data	None
File location	interviewfs19.sav and interview 18+19.sav
Validation	The SPSS code contains internal checks of input and interim variables. Checks are made throughout the income modelling to ensure missing data has been imputed correctly. Interim private and benefit income variables and the final variable are compared with results from previous years overall and by key household descriptor variables.
Key inputs	The raw EHS interview survey data collected from the module Subjective economic status ('QInter.QTecStat') on the EHS Interview Survey Questionnaire documentation to the module Savings and investments ('QSavings'), excluding the section on Income support and mortgage interest (QISMI).
Modelling	A total combined gross income of the HRP/partner is calculated for each household that provided income data (private income + benefit income + savings income). Income Tax and National Insurance payable are calculated, where appropriate, and these amounts are deducted to give total net annual household income. Low incomes on the dataset are identified and brought up to at least basic income support/pension credit (the justification for this being that it is likely that they are under-reporting their income for some reason; either deliberately or by mistake). Missing incomes of the HRP and partner are assigned based on the median income of key groups from the weighted EHS dataset. For more details see the section on Household derived indicators, Chapter 5 of the English Housing Survey 2019-20 Technical Report Annex 5.4.
Imputation	There are two forms of imputation; 1. Imputation of missing income amounts for specified private income sources, benefit income sources and savings (for more details see the section on Household derived

	indicators, Chapter 5 of the English Housing Survey 2019-20 Technical Report Annex 5.4), and 2. Imputation of the overall household income. Households where the total HRP and partner income is missing are estimated using the median net income for households as defined by working status, social economic group and whether a partner of the HRP is present in the household.
Key dependencies	'hhinc5x', 'amthbenx', 'BHCinceq' and 'AHCinceq'. Although not dependent on 'hhincx' itself, the gross income equivalent variable ('JOINTINCx') is dependent on the interim variables used to calculate 'hhincx'.
Introduction date	1991
Date of key modelling changes	The definition of income has not changed for this variable, but the income variable is updated periodically to incorporate new interview survey questionnaire changes, policy updates and modelling improvements. E.g. the EHS 2013-14 survey collected income information surrounding the newly introduced Universal Credit and Personal Independence Payment. These benefits have been incorporated into the calculation of state benefit income, an income component of hhincx. Although key modelling changes would be noted in this section, details on relatively minor modelling changes are incorporated into the text of Chapter 5 of the English Housing Survey 2019-20 Technical Report Annex 5.4.
Other relevant information	This variable is the net equivalent of the derived variable 'JOINTINCx'. The source code used for modelling also calculates the variable 'JOINTINCx'. For anonymisation purposes, an annual income of £100,000 or more is capped at £'100,000 or more'. This variable is available in the datasets provided under the End User Licence and the Special Licence.
Previous variable	N/A

Hhltsick**Interviewfs19.sav, Interview 18+19.sav**

Variable name	Hhltsick
Variable label	Anyone in household have long term illness or disability?
Units	N/A
Description	Whether anyone in household has long term illness or disability that limits their activities.
Type	Numeric
Measure	Nominal
Value labels	-9 "does not apply" -8 "no answer" 1 "yes" 2 "no"
Missing data	Yes
File location	interviewfs19.sav and interview 18+19.sav
Validation	Frequency checks and consistency checks and comparisons with previous years.).
Key inputs	'LsIII2', 'IIILim2', 'Hallres'
Modelling	Derived from a combination of the above.
Imputation	If LsIII2 unknown or IIILim2 unknown, then 'hhltsick' is set to unknown. Where hhltsick is unknown, 'hhltsick' is imputed based on receipt of disability benefit. If benefit receipt unknown, then 'hhltsick' left as 'unknown'
Key dependencies	None
Introduction date	1996
Date of key modelling changes	None
Other relevant information	Variable relates to all people in household who are not living away in halls of residence/boarding school. IIILim2 and LsIII2 reworded for the 2012/13 survey, which may have resulted in outcomes looking different to previous years. Imputation assumptions for missing data slightly different to 2011 method and earlier. Originates from EHCS. This variable is available in the datasets provided under the End User Licence and the Special Licence.
Previous variable	N/A

Hhsizex**Interviewfs19.sav, Interview 18+19.sav**

Variable name	Hhsizex
Variable label	Number of persons in the household
Units	Number
Description	Number of people in the household (continuous).
Type	Numeric
Measure	Scale
Value labels – Special Licence	-9 "does not apply" -8 "no answer"
Value labels – End User Licence	-9 "does not apply" -8 "no answer" 9 "9 or more"
Missing data	None
File location	interviewfs19.sav and interview 18+19.sav
Validation	Various crosschecks with household composition variables, frequency checks and consistency checks against time series.
Key inputs	'dvhsizex', 'persno', 'hallres' and 'age'
Modelling	None
Imputation	None
Key dependencies	'bedrqx', 'bedstdx', overcrowding component of HHSRS
Introduction date	1991
Date of key modelling changes	None
Other relevant information	Household members aged 16+ living away in halls of residence/boarding school, who are not the HRP or partner are not included. Originates from EHCS. For anonymisation purposes, a household with nine or more household members is capped at '9 or more' for the End User Licence dataset. This variable is available in the datasets provided under the End User Licence and Special Licence.
Previous variable	N/A

Hhtype6**Interviewfs19.sav, Interview 18+19.sav**

Variable name	Hhtype6
Variable label	Household type - 6 categories
Units	N/A
Description	Household type (6 categories).
Type	Numeric
Measure	Nominal
Value labels	-9 "does not apply" -8 "no answer" 1 "couple, no dependent child(ren)" 2 "couple with dependent child(ren)" 3 "lone parent with dependent child(ren)" 4 "other multi-person households" 5 "one person under 60" 6 "one person aged 60 or over"
Missing data	None
File location	interviewfs19.sav and interview 18+19.sav
Validation	Various crosschecks with other household composition variables, frequency checks and consistency checks against time series.
Key inputs	'hhtype7', 'hhtype11', 'agehrp'
Modelling	Based on EHCS method of deriving household composition which treats same sex couples as couples. Dependent children are classed as those that are aged under 16 or aged 16-18 and in full-time education. Couple based households with independent children only are classed as 'couples' whereas lone parents with independent children only are classed as 'other multi-person households'. For more details see the section on Household derived indicators, Chapter 5 of the English Housing Survey 2019-20 Technical Report Annex 5.4.
Imputation	Recoded in some cases where HRP is child of a couple and the couple is not identified.
Key dependencies	'hhcomp'
Introduction date	1991
Date of key modelling changes	None
Other relevant information	Household members aged 16+ living away in halls of residence/boarding school, who are not the HRP or partner are not included. This variable is available in the datasets provided under the End User Licence and Special Licence.
Previous variable	N/A

Hhtype7**Interviewfs19.sav, Interview 18+19.sav**

Variable name	Hhtype7
Variable label	Household type - 7 categories
Units	N/A
Description	Household type (7 categories).
Type	Numeric
Measure	Nominal
Value labels	-9 "does not apply" -8 "no answer" 1 "couple, no dependent child(ren)" 2 "couple with dependent child(ren)" 3 "lone parent with dependent child(ren)" 4 "other multi-person households" 5 "one male" 6 "one female" 7 "one person (sex unknown)"
Missing data	None
File location	interviewfs19.sav and interview 18+19.sav
Validation	Various crosschecks with other household composition variables, frequency checks and consistency checks against time series.
Key inputs	'hhtype11'
Modelling	Based on EHCS method of deriving household composition which treats same sex couples as couples. Dependent children are classed as those that are aged under 16 or aged 16-18 and in full-time education. Couple based households with independent children only are classed as 'couples' whereas lone parents with independent children only are classed as 'other multi-person households'. For more details see the section on Household derived indicators, Chapter 5 of the English Housing Survey 2019-20 Technical Report Annex 5.4.
Imputation	Recoded in some cases where HRP is child of a couple and the couple is not identified.
Key dependencies	hhtype6, hhcompx
Introduction date	1991
Date of key modelling changes	None
Other relevant information	Household members aged 16+ living away in halls of residence/boarding school, who are not the HRP or partner are not included. This variable is available in the datasets provided under the End User Licence and Special Licence.

Previous variable	N/A
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Hhtype11**Interviewfs19.sav, Interview 18+19.sav**

Variable name	Hhtype11
Variable label	Household type - full 11 categories
Units	N/A
Description	Household type (11 categories).
Type	Numeric
Measure	Nominal
Value labels	-9 "does not apply" -8 "no answer" 1 "couple with no child(ren)" 2 "couple with dependent child(ren) only" 3 "couple with dependent and independent children" 4 "couple with independent child(ren) only" 5 "lone parent with dependent child(ren) only" 6 "lone parent with dependent and independent children" 7 "lone parent with independent child(ren) only" 8 "two or more families" 9 "lone person sharing with other lone persons" 10 "one male" 11 "one female"
Missing data	None
File location	interviewfs19.sav and interview 18+19.sav
Validation	Various crosschecks with other household type variables, frequency checks and consistency checks against time series.
Key inputs	'persno', 'sex', 'reltohrp2' 'reltop2', 'age', employment status variables are used to establish if the person is a full-time student or not, (see 'emphrpx' for further details on modelling employment status for each person), 'livwth', 'agehrp', 'otherfam', 'othfam1p', 'loncoupx'.
Modelling	Based on EHCS method of deriving household composition which treats same sex couples as couples. Dependent children are classed as those that are aged under 16 or aged 16-18 and in full-time education. Couple based households with independent children only are classed as 'couples' whereas lone parents with independent children only are classed as 'other multi-person households'. For more details see the section on Household derived indicators, Chapter 5 of the English Housing Survey 2019-20 Technical Report Annex 5.4.
Imputation	Any missing values for were imputed from other information in the dataset. Recoded in some cases where HRP is child of a couple and the couple is not identified.
Key dependencies	'hhtype7', 'hhtype6', 'hhcompX'
Introduction date	1991

Date of key modelling changes	None
Other relevant information	Household members aged 16+ living away in halls of residence/boarding school, who are not the HRP or partner are not included. This variable is available in the datasets provided under the End User Licence and Special Licence.
Previous variable	N/A

Hhvulx**Interviewfs19.sav, Interview 18+19.sav**

Variable name	Hhvulx
Variable label	Household vulnerable - on means tested or certain disability related benefits?
Units	None
Description	Whether the HRP and any partner are in receipt of at least one of the principal means tested or disability related benefits.
Type	Numeric
Measure	Nominal
Value labels	1 "yes" 2 "no"
Missing data	None
File location	interviewfs19.sav and interview 18+19.sav
Validation	The final variable is compared with results from previous years overall and by key household descriptor variables.
Key inputs	The derived variable 'hhbensx' and interim disability benefit variables as calculated within the income modelling.
Modelling	A household is modelled to be vulnerable if any of the following conditions apply to the HRP and any partner: • In receipt of means tested benefits or tax credits with a relevant income below the threshold (see 'hhbensx') • Attendance allowance • Disability Living Allowance • Personal Independence Payment • Industrial injuries disablement benefit or • War disablement pension.
Imputation	None
Key dependencies	None
Introduction date	1991
Date of key modelling changes	For 2003/4 EHS data, child tax credit and working tax credit were included in the flag. For 2006/7 EHS data, pension credit was included in the flag. For 2010/11 EHS data, income based Employment and Support was included in the flag. For 2011/12 EHS data, income based JSA was used as a qualifying benefit as opposed to all JSA claimants (a refinement incorporated due to newly collected information on the EHS interview survey). For 2013/14 EHS data, Personal Independence Payment (PIP) was included in the flag (PIP started to replace Disability Living Allowance for people aged 16 to 64 from 8 April 2013).
Other information	The source code used for modelling also calculates 'hhbensx'. This variable is available in the datasets provided under the End User Licence and the Special Licence.

Previous variable	N/A
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Hhwhch**Interviewfs19.sav, Interview 18+19.sav**

Variable name	Hhwhch
Variable label	Anyone in household uses a wheelchair
Units	None
Description	Whether anyone in household is a wheelchair user and whether this use is inside or outside the home
Type	Numeric
Measure	Nominal
Value labels	-8 "no answer" 0 "no household members use a wheelchair" 1 "uses wheelchair all the time" 2 "uses wheelchair indoors only" 3 "uses wheelchair outdoors only"
Missing data	Yes
File location	interviewfs19.sav and interview 18+19.sav
Validation	Crosschecks with hhltsick.
Key inputs	'LsIII2', 'WhChair', 'WhFreq', 'Hallres'
Modelling	Derived from a combination of the above.
Imputation	None
Key dependencies	None
Introduction date	2013
Date of key modelling changes	N/A
Other information	Where there is more than one household member that uses a wheelchair: • use of a wheelchair all the time (option 1) takes precedence over options 2 and 3 (indoors/outdoors). • use of a wheelchair indoors (option 2) takes precedence over option 3 (outdoors only). This variable is available in the datasets provided under the End User Licence and the Special Licence.
Previous variable	N/A

Housbenx Interviewfs19.sav, Interview 18+19.sav

Variable name	Housbenx
Variable label	Household (HRP + partner) receives any housing benefit?
Units	N/A
Description	A yes/no flag that outlines whether the HRP and any partner are in receipt of housing benefit or Local Housing Allowance (LHA) or Universal Credit that helps pay towards their rent.
Type	Numeric
Measure	Nominal
Value labels	-9 "not applicable - owner occupier and not shared ownership" 1 "yes" 2 "no"
Missing data	None
File location	interviewfs19.sav and interview 18+19.sav
Validation	The SPSS code contains internal checks to ensure that the correct households have been assigned to receipt of housing benefit/LHA/housing element of Universal Credit. The overall results from the final variable are compared with results from previous years.
Key inputs	'amthbenx'
Modelling	Recoding of 'amthbenx'.
Imputation	None
Key dependencies	None
Introduction date	1991
Date of key modelling changes	2008 – From the EHS 2008 the 'housbenx' flag relates to both housing benefit and Local Housing Allowance. 2013 – From the EHS 2013 the 'housbenx' flag also relates to households in receipt of Universal Credit that helps pay towards their rent.
Other relevant information	Source code used for the modelling also calculates the variables 'rentwxx', 'rentflg', 'rentExS' and 'amthbenx'. This variable is available in the datasets provided under the End User Licence and the Special Licence.
Previous variable	N/A

HYEARGRx**Interviewfs19.sav, Interview 18+19.sav**

Variable name	HYEARGRx
Variable label	Household gross annual income (inc. income from all adult household members)
Units	£'s
Description	An extension of the gross income of the HRP and any partner. This variable represents the household gross income of ALL adults living within the household.
Type	Numeric
Measure	Scale
Value labels	100000.00 "£100,000 or more"
Missing data	None
File location	interviewfs19.sav and interview 18+19.sav
Validation	The SPSS code used for modelling contains internal checks of input and interim variables. Checks in the code are made to ensure that the missing income values for additional adults in the household are imputed correctly. The final variable is compared with results from previous years.
Key inputs	The derived variable 'JOINTINCx' and the raw interview survey variables 'GrossTel', 'GrssTime' and TelBand.
Modelling	This variable uses 'JOINTINCx' for the gross income of the HRP and partner and adds on the income from additional adult household members to create 'HYEARGRx'. The EHS collects information on the income of all household members aged 16 or over. An additional adult (for the purpose of income contribution to the household) is a household member over the age of the 18 (or 16 if they are not in full-time education). However, for income modelling purposes, dependent children aged 19 that are studying for A-Levels or equivalent are not considered as an additional adult. Additional adults are identified in the dataset and where provided in the EHS their income information used. Where the income data for additional adults is missing, an income amount is imputed (see the Imputation category below). Where there are additional adults in the household, the additional adult income is added to the gross income of the HRP and partner ('JOINTINCx') to create the annual gross household income, 'HYEARGRx'. Where there are no additional adults in the household, 'HYEARGRx' is set to 'JOINTINCx'.
Imputation	Where the income data for additional adults is missing, an income amount is imputed using a hot-decking approach. Cases in the dataset that provided an income amount are matched to cases in the dataset with missing values based on similarities in age, gender, working status and in some cases social economic group and geographical location. Any cases that are not matched during the hot-decking process are

	imputed to a sample median based on working status, and for some working status categories with larger samples, age (banded) and gender. Improvement for 2015 modelling: If the additional adult selected receipt of an income source, with a missing income value, then it is only hotdecked against cases with an income greater than zero.
Key dependencies	None
Introduction date	2008
Date of key modelling changes	2008 – this variable is modelled in a very different way from the old SEH variable with the same name so results from the two surveys are not directly comparable.
Other relevant information	Excludes those living in halls of residence unless they are the HRP or partner. For anonymisation purposes, an annual income of £100,000 or more is capped at '£100,000 or more'. This variable is available in the datasets provided under the End User Licence and the Special Licence.
Previous variable	SEH - HYEARGRx

JOINTINCx**Interviewfs19.sav, Interview 18+19.sav**

Variable name	JOINTINCx
Variable label	Annual gross income of the HRP and partner
Units	£'s
Description	Gross annual income of the Household Reference Person (HRP) and any partner from wages, pensions, other private sources, savings and state benefits. This income variable does not include any housing related benefits/allowances as collected in the rent and housing benefit module and council tax and utilities module of the EHS interview questionnaire.
Type	Numeric
Measure	Scale
Value labels	100000.00 "£100,000 or more"
Missing data	None
File location	interviewfs19.sav and interview 18+19.sav
Validation	The SPSS code used for the modelling contains internal checks of input and interim variables. Checks are made throughout the income model to ensure missing data has been imputed correctly. Interim private and benefit income variables and the final variable are compared with results from previous years overall and by key household descriptor variables.
Key inputs	The raw EHS interview survey data collected from the module Subjective economic status ('QInter.QTecStat') on the EHS Interview Survey Questionnaire documentation to the module Savings and investments ('QSavings'), excluding the section on Income support and mortgage interest (QISMI).
Modelling	The calculation of this variable replicates the income procedure for the EHS Basic Income variable (see 'hhincx'), the difference being that the variable represents gross income as opposed to net income and therefore the Income Tax and National Insurance payable are not deducted as appropriate.
Imputation	There are two forms of imputation; 1. Imputation of missing income amounts for specified private income sources, benefit income sources and savings (for more details see the section on Household derived indicators, Chapter 5 of the English Housing Survey 2019-20 Technical Report Annex 5.4), and 2. Imputation of the overall household income. Households where the total HRP and partner income is missing are estimated using the median gross income for households as defined by working status, social economic group and whether a partner of the HRP is present in the household.
Key dependencies	'HYEARGRx' and ALLincx

Introduction date	2008
Date of key modelling changes	2008 – this variable is modelled in a very different way from the old SEH variable with the same name so results from the two surveys are not directly comparable. Since 2008, the definition of income has not changed for this variable but the income variable is updated periodically to incorporate new interview survey questionnaire changes, policy updates and modelling improvements. E.g. the EHS 2013-14 survey collected income information surrounding the newly introduced Universal Credit and Personal Independence Payment. These benefits have been incorporated into the calculation of state benefit income, an income component of JOINTINCx. Although key modelling changes would be noted in this section, details on relatively minor modelling changes are incorporated into the text of Chapter 5 of the English Housing Survey 2019-20 Technical Report Annex 5.4.
Other relevant information	This variable is the gross equivalent of the derived variable 'hhincx'. The source code used for modelling also calculates the variable 'hhincx'. For 2011/12 data, a new income variable 'ALLincx' was created that simply adds together the two derived variables annual gross income of the HRP and partner (JOINTINCx) to an annualised housing benefit/LHA amount (amthbenx*52). For anonymisation purposes, an annual income of £100,000 or more is capped at '£100,000 or more'. This variable is available in the datasets provided under the End User Licence and the Special Licence.
Previous variable	SEH – 'JOINTINC'

Lenown**Interviewfs19.sav, Interview 18+19.sav**

Variable name	Lenown
Variable label	Length of ownership (years)
Units	Years
Description	Number of years that the HRP and any partner have owned the property – continuous.
Type	Numeric
Measure	Scale
Value labels	-9 "does not apply" -8 "no answer" 54 "54 or more"
Missing data	None
File location	interviewfs19.sav and interview 18+19.sav
Validation	Frequency checks and consistency checks with tenure, who purchased from and 'lenres'.
Key inputs	'Fiyear', 'Fimonth', 'onbyyear', 'onbyseas', 'tenure8x'
Modelling	Calculated using responses to 'onbyyear' and 'onbyseas' and using 'tenure8x' to verify ownership.
Imputation	None. Not enough information was available for reliable imputation of the remaining few cases with missing data.
Key dependencies	'lenownb', 'mortwkx'
Introduction date	1996
Date of key modelling changes	2008
Other relevant information	From 2017, newly available variables 'fimonth' and 'fiyear' (month and year of interview) were used as inputs in place of 'fmonth' and 'fyear' (month and year of issue). For anonymisation purposes, a length of ownership of 54 years or more is capped at '54 or more'. This variable is only available in the dataset provided under the Special Licence.
Previous variable	Former EHCS variable 'Lenown' but this was modelled using input variables that are now obsolete

Lenownb**Interviewfs19.sav, Interview 18+19.sav**

Variable name	Lenownb
Variable label	Length of ownership to date of survey
Units	N/A
Description	Number of years that the HRP and any partner have owned the property – banded.
Type	Numeric
Measure	Ordinal
Value labels	-9 "does not apply" -8 "no answer" 1 "less than 1 year" 2 "one year" 3 "two years" 4 "3-4 years" 5 "5-9 years" 6 "10-19 years" 7 "20-29 years" 8 "30+ years"
Missing data	None
File location	interviewfs19.sav and interview 18+19.sav
Validation	Frequency checks and consistency checks.
Key inputs	'lenownb'
Modelling	None – recodes the above variable.
Imputation	None. Not enough information was available for reliable imputation of the remaining few cases with missing data.
Key dependencies	None
Introduction date	1996
Date of key modelling changes	2008
Other relevant information	This variable is available in the datasets provided under the End User Licence and the Special Licence.
Previous variable	Former EHCS variable 'Lenownb' but modelled using input variables that are now obsolete

Lenres**Interviewfs19.sav, Interview 18+19.sav**

Variable name	Lenres
Variable label	Length of residence (years)
Units	Years
Description	Number of years that the HRP and any partner have lived in the property – continuous.
Type	Numeric
Measure	Scale
Value labels	-9 "does not apply" -8 "no answer" 54 "54 or more"
Missing data	None
File location	interviewfs19.sav and interview 18+19.sav
Validation	Lots of crosschecks in modelling including ensuring that length residence does not exceed age of home etc
Key inputs	'restme2', 'HLong3', 'resmth2', 'yrbult3', 'fodlivey', 'fiyear', 'fimonth', 'tenure8x'
Modelling	Convert 'restme2' 'HLong3' 'Resmth2' from person level to household level variable. Based on hrp's answer to the variables first, then the partner's answer if hrp has refused to answer. If 'restme2' unknown use 'fodlivey'
Imputation	Where data is missing for 'restme3', data is taken from 'fodlivey' where available. For remaining cases, where length of residence is missing, if owner occupier, length of residence set to length of ownership (as long as not a right to buy owner); if tenant, length of residence set to average for tenure and age band of hrp.
Key dependencies	'lenresb'
Introduction date	1991 but not available in 2012/13 and 2013/14
Date of key modelling changes	2008
Other relevant information	Only banded residency data was collected in 2012/13 and 2013/14 and in these years, continuous residency data was modelled (and known as 'lenres2'). From 2014/15 collection of continuous residency data, and hence creation of 'lenres' resumed. From 2017, newly available variables 'fimonth' and 'fiyear' (month and year of interview) were used as inputs in place of 'fmonth' and 'fyear' (month and year of issue). For anonymisation purposes, a household that lived in their home for 54 or more years is capped at '54 or more'. This variable is only available in the dataset provided under the Special Licence.

Previous variable	Lenres2
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Lenresb**Interviewfs19.sav, Interview 18+19.sav**

Variable name	Lenresb
Variable label	Length of residence
Units	N/A
Description	Number of years that the HRP and any partner have lived in the property – banded.
Type	Numeric
Measure	Ordinal
Value labels	-9 "does not apply" -8 "no answer" 1 "less than 1 year" 2 "one year" 3 "two years" 4 "3-4 years" 5 "5-9 years" 6 "10-19 years" 7 "20-29 years" 8 "30+ years"
Missing data	None
File location	interviewfs19.sav and interview 18+19.sav
Validation	Frequency checks and consistency checks.
Key inputs	'lenres'
Modelling	None – recodes the above variable.
Imputation	Where data is missing for 'restme3', data is taken from 'fodlivey' when available. For remaining cases, where length of residence is missing, if owner occupier, length of residence set to length of ownership (as long as not a 'right to buy' owner); if tenant, length of residence set to average for tenure and age band of hrp.
Key dependencies	None
Introduction date	1991
Date of key modelling changes	2008
Other relevant information	This variable is available in the datasets provided under the End User Licence and the Special Licence.
Previous variable	Former EHCS variable – 'Lenresb' but modelled using input variables that are now obsolete

Lharqx**Interviewfs19.sav, Interview 18+19.sav**

Variable name	Lharqx
Variable label	No of bedrooms required by the household (2011 definition)
Units	Rooms
Description	Number of bedrooms that the household theoretically requires based on the guidelines below under modelling
Type	Numeric
Measure	Scale
Value labels	-9 "does not apply" -8 "no answer" 5 "5 or more"
Missing data	None
File location	interviewfs19.sav and interview 18+19.sav
Validation	Frequency checks.
Key inputs	'sex', 'age', and all relationship variables
Modelling	Uses an alternative definition of the bedroom standard to calculate the number of bedrooms households require: a) 1 bedroom for couple (no age base) b) 1 bedroom for single person aged 16 or more c) 1 bedroom for paired adolescents (10-15) same sex d) 1 bedroom for paired children under 10 e) 1 bedroom for unpaired adolescents and unpaired child of the same sex f) 1 bedroom for any remaining unpaired adolescent or child.
Imputation	Any missing cases for the relationship variables were imputed using other information in the data set.
Key dependencies	'Lhastdx'
Introduction date	2011, however previous data has been recalculated back to 2008 using this method
Date of key modelling changes	N/A
Other relevant information	Bedrqx variable used in the analysis and reports. Lharqx excludes those living in halls of residence unless they are the HRP or partner of the HRP. For anonymisation purposes, a household that requires 5 or more bedrooms is capped at '5 or more'. This variable is available in the datasets provided under the End User Licence and the Special Licence.
Previous variable	N/A

Lhastdx**Interviewfs19.sav, Interview 18+19.sav**

Variable name	Lhastdx
Variable label	Bedroom standard (2011 definition)
Units	N/A
Description	Whether household is overcrowded as measured by the bedroom standard
Type	Numeric
Measure	Ordinal
Value labels	-9 "does not apply" -8 "no answer" 1 "two or more below standard" 2 "one below standard" 3 "at standard" 4 "one above standard" 5 "two or more above standard"
Missing data	None
File location	interviewfs19.sav and interview 18+19.sav
Validation	Frequency checks and time series checks. .
Key inputs	Lharqx, nbedsx
Modelling	Deducts number of bedrooms each household has from number of bedrooms required based on the alternative Bedroom Standard definition (see 'Lharqx').
Imputation	None
Key dependencies	None
Introduction date	2011, however previous data has been recalculated back to 2008 using this method
Date of key modelling changes	None
Other relevant information	Bedstdx is the variable used in the analysis and reports. Lhastdx excludes those living in halls of residence unless they are the HRP or partner of the HRP. This variable is available in the datasets provided under the End User Licence and the Special Licence.
Previous variable	N/A

Loncoupx Interviewfs19.sav, Interview 18+19.sav

Variable name	Loncoupx
Variable label	Single householder or with partner
Units	N/A
Description	Whether HRP is single (with or without children) or lives with partner (with or without children).
Type	Numeric
Measure	Nominal
Value labels	-9 "does not apply" -8 "no answer" 0 "lone person" 1 "couple" 9 "unsure"
Missing data	None
File location	interviewfs19.sav and interview 18+19.sav
Validation	Frequency checks and consistency checks against time series plus crosschecks with other household composition variables.
Key inputs	All relationship variables, 'livwth', 'age' and employment status variables to establish if the person is a full-time student or not, (see 'emphrpx' for further details on modelling employment status for each person).
Modelling	None
Imputation	None
Key dependencies	'hhtype11', 'hhtype7', 'hhtype6', 'hhcomp'
Introduction date	1991
Date of key modelling changes	None
Other relevant information	Originates from EHCS. Same sex couples are treated as couples. This variable is only available in the dataset provided under the Special Licence
Previous variable	N/A

Market_rent Interviewfs19.sav, Interview 18+19.sav

Variable name	Market_rent
Variable label	Tenancy types of renters
Units	N/A
Description	Social and Private renting tenancy types
Type	Numeric
Measure	Nominal
Value labels	-9 "does not apply" 1 "local authority" 2 "housing association" 3 "private rented - market rents" 4 "private rented - non-market rents" 5 "private rented - rent type unknown"
Missing data	None
File location	interviewfs19.sav and interview 18+19.sav
Validation	Frequency check and consistency check with tenure4x.
Key inputs	'tenure4x', 'rentExS', 'tentype'
Modelling	None – based on above variables. Some cases coded as private rented in tenure4x where 'tentype' is incorrectly set as 'does not apply' are set to 'private rented – rent type unknown'.
Imputation	None
Key dependencies	None
Introduction date	2011 (MHCLG requested this variable was added to the file, although has always been produced)
Date of key modelling changes	N/A
Other relevant information	This variable has normally been produced in syntax when required, for example, for the web tables and household report. The definition for this variable changed in 2011 when MHCLG requested that the 'tenytype' variable was replaced with 'tentype' in the questionnaire- these variables have slightly different options. In 2013, there was a slight change in the wording of the question 'tentype' (applicable dates for tenure type, to be more explicit) which may have impacted on classification. This variable is available in the datasets provided under the End User Licence and the Special Licence.
Previous variable	N/A

Mortwkx**Interviewfs19.sav, Interview 18+19.sav**

Variable name	Mortwkx
Variable label	Weekly mortgage payments
Units	£'s
Description	Weekly mortgage payments for owner occupiers. (If information provided by household the weekly mortgage amount will reflect the mortgage type, for example it will contain amounts for a mixture of repayment and interest only cases.)
Type	Numeric
Measure	Scale
Value labels	-9.00 "not applicable – tenant" -8.00 "unknown" 0.00 "no payments - own outright"
Missing data	None
File location	interviewfs19.sav and interview 18+19.sav
Validation	There is no external source that could be used to help validate this data. All validation is therefore internal. It includes checks and cross checks of raw input and interim variables. They also include comparing the final variable with results from previous years overall and by key household descriptor variables.
Key inputs	'MorgPayU' and 'MorgPerL'
Modelling	It calculates the monthly mortgage payment using usual payment amount ('MorgPayU') and payment period ('MorgPerL'), deducting notional estimates for building and contents insurance and mortgage protection (if included). If the monthly amount is unknown, it is imputed using the value of the current main mortgage, length of current main mortgage and current interest rate. Finally, monthly amounts are converted into weekly amounts. For more details see the section on Household derived indicators, Chapter 5 of the English Housing Survey 2019-20 Technical Report Annex 4.
Imputation	If key inputs missing, use 'Onorgmrg', 'MorgLnth1' and current interest rate (from UK Finance table RL6). If 'MorgLnth1' unknown, use length of residence and age of HRP setting max length at 25 years and assume mortgage paid off by 60s. If 'onorgmrg' missing, use 'Onpurpc' minus deposit. If deposit missing, assume 10% of 'Onpurpc', if 'Onpurpc' missing, use 'prptval1'/'prptval2', 'OnbyYear' and CLG annual regional average house price data (ONS Table 33, and Table 24). If 'prptval1'/'prptval2' missing, use latest CLG average house prices by region and dwelling type (ONS Table 26) A repayment amount is modelled for Equity release mortgage where there is sufficient information.

Key dependencies	Although not dependent on 'mortwvx' itself, the equity variables ('Equityr' and 'Equityr5') are dependent on key variables used in the modelling of 'mortwvx'.
Introduction date	1996
Date of key modelling changes	As market value survey discontinued in 2010, household estimates of market value were used instead, and imputed using latest CLG average house prices by region and dwelling type (ONS Table 26) where unknown. From 2016 shared owners with no mortgage payment details were set to nominal amounts, for more details see the section on Household derived indicators, Chapter 5 of the English Housing Survey 2019-20 Technical Report Annex 4.
Other relevant information	Source code also calculates equity ('Equityr', 'Equityr5'). This variable is available in the datasets provided under the End User Licence and the Special Licence.
Previous variable	N/A

Nbatha**Interviewfs19.sav, Interview 18+19.sav**

Variable name	Nbatha
Variable label	If shares a bathroom, shower room or WC
Units	N/A
Description	Whether household shares a bathroom, shower room or WC with other households.
Type	Numeric
Measure	Nominal
Value labels	-9 "does not apply" -8 "no answer" 1 "yes" 2 "no"
Missing data	None
File location	interviewfs19.sav and interview 18+19.sav
Validation	Frequency checks. 'Not applicables' checked and re-coded to 'no'.
Key inputs	'nrms5', 'ShBthWc'
Modelling	None – based upon the above variables.
Imputation	None
Key dependencies	None
Introduction date	2011 (replaced Nbath)
Date of key modelling changes	None
Other relevant information	Nbatha is now based on nrms5 – up to 2011 it was based on nrmsEHS. Former SEH variable – 'Nbath'. ShRms5 is no longer asked which just looked at bathrooms and was used to create Nbath. ShRms5 was replaced by ShBthWC-which asks about bathrooms, shower rooms and WC. This variable is only available in the dataset provided under the Special Licence.
Previous variable	Nbath

NbedsX**Interviewfs19.sav, Interview 18+19.sav**

Variable name	NbedsX
Variable label	Total number of bedrooms household actually has
Units	Number
Description	Total number of bedrooms available to the household.
Type	Numeric
Measure	Scale
Value labels	-9 "does not apply" -8 "no answer" 5 "5 or more"
Missing data	None
File location	interviewfs19.sav and interview 18+19.sav
Validation	Frequency checks and consistency checks against time series.
Key inputs	'NrmsEHS'
Modelling	N/A
Imputation	Where NrmsEHS was missing, it was imputed based on grossed mean number of bedrooms for that dwelling type and dwelling age where data existed.
Key dependencies	'Bedstdx'
Introduction date	2008
Date of key modelling changes	None
Other relevant information	Former SEH variable – 'Nbeds' EHCS - 'NbedsX' For anonymisation purposes, a household with 5 or more bedrooms is capped at '5 or more'. This variable is available in the datasets provided under the End User Licence and the Special Licence.
Previous variable	N/A

Ncouple**Interviewfs19.sav, Interview 18+19.sav**

Variable name	Ncouple
Variable label	Number of couples in household
Units	Number
Description	Number of couples in household, i.e. including other family units in addition to HRP's family unit.
Type	Numeric
Measure	Scale
Value labels	-9 "does not apply" -8 "no answer"
Missing data	None
File location	interviewfs19.sav and interview 18+19.sav
Validation	Various crosschecks with other household composition variables.
Key inputs	'XMarSta', 'LivWth', 'FUT' (internal interim derived variable created using 'afam' and 'aacode', 'age' and 'hallres')
Modelling	If family consists of more than one person, and the oldest person in that family unit is identified as living with / married to someone, they are coded as a couple. Same sex couples are treated as couples.
Imputation	Any inconsistent 'afam' cases were corrected using other household information in the dataset.
Key dependencies	'hhcomp1'
Introduction date	2008
Date of key modelling changes	None
Other relevant information	Originates from SEH. Based on FUT (Family Unit Type) coding, which is carried out on the SEH DVs, in order to create variables including 'Nlpar', 'Nsing', 'Ncouple', and SEH based household composition variables. Information is taken from the oldest person (not necessarily the HRP) in each family unit. Excludes those living in halls of residence who are not HRP or partner. For more details see the section on Household derived indicators, Chapter 5 of the English Housing Survey 2019-20 Technical Report Annex 5.4. This variable is only available in the dataset provided under the Special Licence
Previous variable	N/A

Ndepchild**Interviewfs19.sav, Interview 18+19.sav**

Variable name	Ndepchild
Variable label	Number of dependent children in household
Units	Number of dependent children
Description	Number of children in household that are dependents. Dependent children are classed as those aged under 16 or those aged 16-18 who are still in full-time education.
Type	Numeric
Measure	Scale
Value labels – Special Licence	-9 "does not apply" -8 "no answer"
Value labels – End User Licence	-9 "does not apply" -8 "no answer" 5 "5 or more"
Missing data	None
File location	interviewfs19.sav and interview 18+19.sav
Validation	Various crosschecks with other household composition variables.
Key inputs	'hallres' 'persno', information on the relationship of child to the family, 'age', and employment status variables used to establish if the person is a full-time student or not, (see 'emphrpx' for further details on modelling employment status for each person), – all used to create internal derived 'depch'.
Modelling	Created by summing person level variable depch
Imputation	None
Key dependencies	'hhcomp1'
Introduction date	2008
Date of key modelling changes	None
Other relevant information	Originates from SEH. Note it is possible to be both a dependent child and a single family unit (nsing). For more details see the section on Household derived indicators, Chapter 5 of the English Housing Survey 2019-20 Technical Report Annex 4. For anonymisation purposes, a household with 5 or more dependent children is capped at '5 or more' for the End User Licence dataset. This variable is available in the datasets provided under the End User Licence and Special Licence.
Previous variable	'ndepch' in SEH

Nemp**Interviewfs19.sav, Interview 18+19.sav**

Variable name	Nemp
Variable label	Number of employed persons in household
Units	Persons
Description	Number of employed persons in household. This counts the number of people whose primary economic status is coded as one of the following: working full-time, working part-time or on a government training scheme
Type	Numeric
Measure	Scale
Value labels	-9 "does not apply" -8 "no answer"
Missing data	None
File location	interviewfs19.sav and interview 18+19.sav
Validation	Frequency checks and consistency checks.
Key inputs	Raw data on working status, EconAct (allows multiple responses: EconFull to EconOthr), 'age', 'sex'
Modelling	See 'emphrpx' for details on modelling employment status for each person.
Imputation	See 'emphrpx'
Key dependencies	None
Introduction date	2008
Date of key modelling changes	None
Other relevant information	See 'emphrpx' for details. This variable is only available in the dataset provided under the Special Licence.
Previous variable	Former SEH variable – 'Nemp1'

Ninac**Interviewfs19.sav, Interview 18+19.sav**

Variable name	Nlnac
Variable label	Number of economically inactive persons in household
Units	Persons
Description	Number of economically inactive persons in household. This counts the number of people whose primary economic status is coded as one of the following: not working/long term sick, not registered unemployed but seeking work, at home not seeking work or retired.
Type	Numeric
Measure	Scale
Value labels	-9 "does not apply" -8 "no answer"
Missing data	None
File location	interviewfs19.sav and interview 18+19.sav
Validation	Frequency checks and consistency checks.
Key inputs	Raw data on working status, EconAct (allows multiple responses: EconFull to EconOthr), 'age', 'sex'
Modelling	See 'emphrpx' for details on modelling employment status for each person.
Imputation	See 'emphrpx'
Key dependencies	None
Introduction date	2008
Date of key modelling changes	None
Other relevant information	See 'emphrpx' for details. This variable is only available in the dataset provided under the Special Licence.
Previous variable	SEH variable – 'Nlnac'

Nkita**Interviewfs19.sav, Interview 18+19.sav**

Variable name	Nkita
Variable label	If shares a kitchen
Units	N/A
Description	Whether household shares a kitchen with other households.
Type	Numeric
Measure	Nominal
Value labels	-9 "does not apply" -8 "no answer" 1 "yes" 2 "no"
Missing data	None
File location	interviewfs19.sav and interview 18+19.sav
Validation	Frequency checks. 'Not applicables' checked and re-coded to 'no'.
Key inputs	'shrms2', 'nrms2a'
Modelling	None
Imputation	None
Key dependencies	None
Introduction date	2011 (replaced NKit)
Date of key modelling changes	Same as SEH methodology
Other relevant information	Nkita is now based on nrms2a – up to 2011 it was based on nrmsEHS. From 2011 the Shrms2 definition changed-no longer collects data about size of kitchen (i.e. how many are over six and a half feet wide). It now just states how many kitchens are shared with other households. This variable is only available in the dataset provided under the Special Licence.
Previous variable	SEH variable – 'Nkit'

Nliving**Interviewfs19.sav, Interview 18+19.sav**

Variable name	Nliving
Variable label	If shares other room (living room)
Units	N/A
Description	Whether household shares a living room with other households.
Type	Numeric
Measure	Nominal
Value labels	-9 "does not apply" -8 "no answer" 1 "yes" 2 "no"
Missing data	None
File location	interviewfs19.sav and interview 18+19.sav
Validation	Frequency checks. Not applicables' checked and re-coded to 'no'.
Key inputs	'nrms4', 'shrms4',
Modelling	None
Imputation	None
Key dependencies	None
Introduction date	2011
Date of key modelling changes	N/A
Other relevant information	Nliving is now based on nrms4 – in 2011 it was based on nrmsEHS. This variable before 2011 was 'nother' - in 2011 this variable was changed to nliving as living room is the only room which feeds into this derived variable since Shrms6 (how many utility and other rooms does the household share) is no longer asked. This variable is only available in the dataset provided under the Special Licence.
Previous variable	'nother'

Nlpar**Interviewfs19.sav, Interview 18+19.sav**

Variable name	Nlpar
Variable label	Number of lone parent families in household
Units	Number
Description	Number of family units that contain more than one person and do not consist of a couple. These will normally consist of a single parent with one or more children.
Type	Numeric
Measure	Scale
Value labels	-9 "does not apply" -8 "no answer" 2 "2 or more"
Missing data	None
File location	interviewfs19.sav and interview 18+19.sav
Validation	Various crosschecks with other household composition variables.
Key inputs	'XMarSta', 'LivWth', 'FUT' (internal interim derived variable created using 'afam' and 'aacode', 'age' and 'hallres')
Modelling	None
Imputation	Manual recoding of some problem cases where 'nlpar' conflicts with 'depch' (an internally derived variable for the number of dependent children at person level). Any inconsistent 'afam' cases were corrected using other household information in the dataset.
Key dependencies	'hhcomp1'
Introduction date	2008
Date of key modelling changes	None
Other relevant information	Originates from SEH. Based on FUT (Family Unit Type) coding, which is carried out on the SEH DVs, in order to create variables including 'nlpar', 'nsing', 'ncouple', and the SEH household composition variables. Information is taken from the oldest person (not necessarily the HRP) in the main family unit. For more details see the section on Household derived indicators, Chapter 5 of the English Housing Survey 2019-20 Technical Report Annex 5.4. For anonymisation purposes, a household with 2 or more lone parent families is capped at '2 or more'. This variable is only available in the dataset provided under the Special Licence
Previous variable	N/A

NoUnits1**Interviewfs19.sav, Interview 18+19.sav**

Variable name	NoUnits1
Variable label	Banded number of family units in household
Units	Number
Description	Number of family units in the household – banded.
Type	Numeric
Measure	Ordinal
Value labels	-9 "does not apply" -8 "no answer" 1 "1" 2 "2 or more"
Missing data	None
File location	interviewfs19.sav and interview 18+19.sav
Validation	Frequency checks and consistency checks against time series.
Key inputs	'Famnumx'
Modelling	None
Imputation	None.
Key dependencies	None
Introduction date	1991
Date of key modelling changes	None
Other relevant information	Number of family units is computed excluding those in halls of residence who are not the HRP or partner (unlike old SEH definition). Also, same sex couples are counted as separate family units, despite being treated as a couple living together in other variables (loncoupx, bedstdx, household composition). Therefore, there may be some discontinuity when comparing these variables. See famnumx for more details. Originates from EHCS. This variable is available in the datasets provided under the End User Licence and Special Licence.
Previous variable	N/A

Nroomsa**Interviewfs19.sav, Interview 18+19.sav**

Variable name	Nroomsa
Variable label	Number of rooms available to household
Units	Rooms
Description	Number of rooms available to household (this includes rooms that are shared).
Type	Numeric
Measure	Scale
Value labels	-9 "does not apply" -8 "no answer" 14 "14 or more"
Missing data	None
File location	interviewfs19.sav and interview 18+19.sav
Validation	Frequency checks. Those with high values were crosschecked with physical survey and photos, where available and 3 input variables were also checked for misspunched data by the interviewer, and corrected where possible before 'nroomsa' recalculated.
Key inputs	'totkitsa', 'nrmsEHS', 'nrms4',
Modelling	Combines totals of the above variables
Imputation	Where NrmsEHS was missing, it was imputed based on grossed mean number of bedrooms for that dwelling type and dwelling age where data existed.
Key dependencies	'nrooms1a'
Introduction date	2011 (before this it was nrooms)
Date of key modelling changes	Same as SEH methodology
Other relevant information	Original SEH variable –'nrooms'. Nrooms changed to nroomsa in 2011. Nrms6 (the number of utility and other rooms) was dropped from 2011/12 interview survey. This has led to a difference in the information in the nrooms and nrooms1 variables. For anonymisation purposes, a household with 14 or more rooms is capped at '14 or more'. This variable is only available in the dataset provided under the Special Licence.
Previous variable	Nrooms

Nrooms1a Interviewfs19.sav, Interview 18+19.sav

Variable name	Nrooms1a
Variable label	Number of rooms available to household (grouped)
Units	Rooms
Description	Number of rooms available to household (this includes rooms that are shared).
Type	Numeric
Measure	Scale
Value labels	-9 "does not apply" -8 "no answer" 8 "8 or more"
Missing data	None
File location	interviewfs19.sav and interview 18+19.sav
Validation	Frequency checks
Key inputs	'nroomsa'
Modelling	'nroomsa' recoded so that 8 or more rooms are grouped together
Imputation	None
Key dependencies	None
Introduction date	2011 (before this nrooms1)
Date of key modelling changes	Same as SEH methodology
Other relevant information	Original SEH variable – 'nrooms1'. The definition for nrooms changed in 2011 and became nroomsa. This change to the key input necessitated a change to nrooms1. This variable is available in the datasets provided under the End User Licence and the Special Licence.
Previous variable	Nrooms1

Nshare**Interviewfs19.sav, Interview 18+19.sav**

Variable name	Nshare
Variable label	If shares any part of accommodation
Units	N/A
Description	Whether household shares any part of their accommodation with other households.
Type	Numeric
Measure	Nominal
Value labels	-9 "does not apply" -8 "no answer" 1 "shares" 2 "does not share"
Missing data	None
File location	interviewfs19.sav and interview 18+19.sav
Validation	Frequency checks
Key inputs	'share2'
Modelling	None
Imputation	None
Key dependencies	None
Introduction date	2008
Date of key modelling changes	Same as SEH methodology
Other relevant information	Original SEH variable – 'nshare' This variable is only available in the dataset provided under the Special Licence.
Previous variable	N/A

Nsing**Interviewfs19.sav, Interview 18+19.sav**

Variable name	Nsing
Variable label	Number of one-person family units in household
Units	Number
Description	Number of single person family units in household, i.e. including other family units in addition to HRP's family unit.
Type	Numeric
Measure	Scale
Value labels	-9 "does not apply" -8 "no answer" 5 "5 or more"
Missing data	None
File location	interviewfs19.sav and interview 18+19.sav
Validation	Various crosschecks with other household composition variables.
Key inputs	XMarSta, LivWth, FUT (internal interim derived variable created using 'afam' and 'aacode', 'age' and 'hallres').
Modelling	Based on SEH method
Imputation	Any inconsistent 'afam' cases were corrected using other household information in the dataset.
Key dependencies	hhcomp1
Introduction date	2008
Date of key modelling changes	None
Other relevant information	Based on FUT (Family Unit Type) coding, which is carried out on the SEH DVs, in order to create variables including Nlpar, Nsing, Ncouple, and SEH based household composition variables. Information is taken from the oldest person (not necessarily the HRP) in the family unit. Originates from SEH. Note it is possible to be a single family unit and an independent (NXdepch) or dependent child (Ndepchild). For more details see the section on Household derived indicators, Chapter 5 of the English Housing Survey 2019-20 Technical Report Annex 5.4. For anonymisation purposes, a household with 5 or more single person family units is capped at '5 or more'. This variable is only available in the dataset provided under the Special Licence
Previous variable	N/A

Nssech9**Interviewfs19.sav, Interview 18+19.sav**

Variable name	Nssech9
Variable label	NS-SEC Socio-economic Classification – HRP
Units	N/A
Description	NS-SEC Socio-economic Classification – HRP.
Type	Numeric
Measure	Nominal
Value labels	-9 "does not apply" -8 "no answer" 1 "higher managerial and professional occupations" 2 "lower managerial and professional occupations" 3 "intermediate occupations" 4 "small employers and own account workers" 5 "lower supervisory and technical occupations" 6 "semi-routine occupations" 7 "routine occupations" 8 "never worked and long term unemployed" 9 "other"
Missing data	Yes
File location	interviewfs19.sav and interview 18+19.sav
Validation	Frequency checks, consistency checks and time series checks
Key inputs	'nssec', 'persno', 'HRP'
Modelling	Combination of the above inputs
Imputation	If missing cases for 'age', and 'wrking', set to 'no answer for 'nssech9'.
Key dependencies	None
Introduction date	2014
Date of key modelling changes	In 2014, nssech9 replaced nssech as many categories were combined. Key modelling was unchanged. From 2012, certain cases assigned to 'does not apply' or 'no answer' whereas prior to this, such cases had been coded as 'not HRP'
Other relevant information	Same as EHCS methodology. This variable is available in the datasets provided under the End User Licence and the Special Licence.
Previous variable	Former EHS variable – 'nssech' Former SEH variable - 'nssec2' Prior to 2001, the EHCS and SEH variables were based on head of household (HOH) rather than household reference person (HRP). They were also based on a slightly different coding scheme for occupations.

Nssecp9**Interviewfs19.sav, Interview 18+19.sav**

Variable name	Nssecp9
Variable label	NS-SEC Socio-economic Classification – HRP's partner
Units	N/A
Description	NS-SEC Socio-economic Classification – HRP's partner.
Type	Numeric
Measure	Nominal
Value labels	-9 "does not apply" -8 "no answer" 0 "not partner" 1 "higher managerial and professional occupations" 2 "lower managerial and professional occupations" 3 "intermediate occupations" 4 "small employers and own account workers" 5 "lower supervisory and technical occupations" 6 "semi-routine occupations" 7 "routine occupations" 8 "never worked and long term unemployed" 9 "other"
Missing data	Yes
File location	interviewfs19.sav and interview 18+19.sav
Validation	Frequency checks, consistency checks and time series checks
Key inputs	'nssec', 'reltohrp2'
Modelling	Combination of the above inputs
Imputation	If missing cases for 'age', and 'wrking,' set to 'no answer for 'nssecp9'.
Key dependencies	None
Introduction date	2014
Date of key modelling changes	In 2014, nssecp9 replaced nssecp as many categories were combined. Key modelling was unchanged. From 2012, certain cases assigned to 'does not apply' or 'no answer' whereas prior to this, such cases had been coded as 'not partner'
Other relevant information	Same as EHCS methodology. This variable is available in the datasets provided under the End User Licence and the Special Licence.
Previous variable	EHS variable – 'nssecp'

Nstud Interviewfs19.sav, Interview 18+19.sav

Variable name	NStud
Variable label	Number of students in household
Units	Persons
Description	Number of full-time students in household. This counts the number of people whose primary economic status is coded as full-time student
Type	Numeric
Measure	Scale
Value labels	-9 "does not apply" -8 "no answer"
Missing data	None
File location	interviewfs19.sav and interview 18+19.sav
Validation	Frequency checks and consistency checks
Key inputs	Raw data on working status, EconAct (allows multiple responses: EconFull to EconOthr), 'age', 'sex'
Modelling	See 'emphrpx' for details on modelling employment status for each person.
Imputation	See 'emphrpx'
Key dependencies	None
Introduction date	2008
Date of key modelling changes	None
Other relevant information	See 'emphrpx' for details. This variable is only available in the dataset provided under the Special Licence.
Previous variable	SEH – 'NStud'

Nunemp**Interviewfs19.sav, Interview 18+19.sav**

Variable name	NUnemp
Variable label	Number of unemployed persons in household
Units	Persons
Description	Number of registered unemployed persons in household. This counts the number of people whose primary economic status is coded as registered unemployed
Type	Numeric
Measure	Scale
Value labels	-9 "does not apply" -8 "no answer"
Missing data	None
File location	interviewfs19.sav and interview 18+19.sav
Validation	Frequency checks and consistency checks
Key inputs	Raw data on working status, EconAct (allows multiple responses: EconFull to EconOthr), 'age', 'sex'
Modelling	See 'emphrpx' for details on modelling employment status for each person.
Imputation	See 'emphrpx'
Key dependencies	None
Introduction date	2008
Date of key modelling changes	None
Other relevant information	See 'emphrpx' for details. This variable is only available in the dataset provided under the Special Licence.
Previous variable	SEH variable – 'NUnemp'

Nxdepch**Interviewfs19.sav, Interview 18+19.sav**

Variable name	Nxdepch
Variable label	Number of non-dependent children in household
Units	Number of non-dependent children
Description	Number of children in household that are no longer classed as dependents. Independent children are classed as those aged over 18 or those aged 16-18 who are not in full-time education.
Type	Numeric
Measure	Scale
Value labels	-9 "does not apply" -8 "no answer"
Missing data	None
File location	interviewfs19.sav and interview 18+19.sav
Validation	Crosschecks with other household composition variables.
Key inputs	Information on the relationship of child to the family, 'age', employment status variables to establish if the person is a full-time student or not, (see 'emphrpx' for further details on modelling employment status for each person), 'hallres' "persno" – all used to create internal derived 'Xdepch'.
Modelling	Created by summing person level variable 'Xdepch'
Imputation	None
Key dependencies	'hhcomp1'
Introduction date	2008
Date of key modelling changes	In SEH independent children were all those aged 16 over whereas EHS and EHCS classed 16-18 year olds in full-time education as dependent children.
Other relevant information	This variable is only available in the dataset provided under the Special Licence.
Previous variable	'Nxdepch' in SEH

Olderx**Interviewfs19.sav, Interview 18+19.sav**

Variable name	Olderx
Variable label	Number of people aged 60 plus who are HRP or partner
Units	N/A
Description	Number of people in the household aged 60 or over who are HRP or partner.
Type	Numeric
Measure	Scale
Value labels	-9 "does not apply" -8 "no answer"
Missing data	None
File location	interviewfs19.sav and interview 18+19.sav
Validation	Frequency checks and consistency checks against time series.
Key inputs	'agehrpx', 'agepartx'
Modelling	None – recodes the above two variables.
Imputation	None
Key dependencies	None
Introduction date	2008
Date of key modelling changes	Same as EHCS methodology
Other relevant information	This variable is only available in the dataset provided under the Special Licence.
Previous variable	EHCS – 'olderx'

Otherfam**Interviewfs19.sav, Interview 18+19.sav**

Variable name	Otherfam
Variable label	Additional families present in household
Units	N/A
Description	Whether there is more than family unit in the household.
Type	Numeric
Measure	Nominal
Value labels	-9 "does not apply" -8 "no answer" 1 "yes" 2 "no"
Missing data	None
File location	interviewfs19.sav and interview 18+19.sav
Validation	Frequency checks and consistency checks against time series.
Key inputs	'famnumx'
Modelling	None
Imputation	None
Key dependencies	'othfam1p'
Introduction date	1991
Date of key modelling changes	None
Other relevant information	Excludes those in halls of residence who are not the HRP or partner. Same sex couples are treated as separate family units. See 'famnumx' for more details. Originates from the EHCS. This variable is only available in the dataset provided under the Special Licence.
Previous variable	N/A

Othfam1p**Interviewfs19.sav, Interview 18+19.sav**

Variable name	Othfam1p
Variable label	Type of additional families in household
Units	N/A
Description	Type of additional family units in the household.
Type	Numeric
Measure	Nominal
Value labels	-9 "does not apply" -8 "no answer" 1 "no additional families" 2 "one person additional families only" 3 "other additional families"
Missing data	None
File location	interviewfs19.sav and interview 18+19.sav
Validation	Frequency checks and consistency checks against time series.
Key inputs	'famnumx', 'otherfam'
Modelling	None
Imputation	None
Key dependencies	'hhtype11'
Introduction date	1991
Date of key modelling changes	None
Other relevant information	Excludes those in halls of residence who are not the HRP or partner. Same sex couples are treated as separate family units. See famnumx for more details. Originates from EHCS. This variable is only available in the dataset provided under the Special Licence.
Previous variable	N/A

Owntype **Interviewfs19.sav, Interview 18+19.sav**

Variable name	Owntype
Variable label	Type of ownership
Units	N/A
Description	Whether property is owned as freehold or leasehold.
Type	Numeric
Measure	Nominal
Value labels	-9 "does not apply" -8 "no answer" 1 "freeholder of house" 2 "leaseholder - no share of FH" 3 "leaseholder owning FH collectively" 4 "leaseholder owning FH of whole bldg" 5 "freeholder of flat - owning FH of flat only"
Missing data	Yes
File location	interviewfs19.sav and interview 18+19.sav
Validation	Frequency checks and consistency checks.
Key inputs	'Accom', 'Lease2', 'CHLease2', 'Leasothr', 'FreeHld1New', 'Tenure4', 'Tenure1'
Modelling	Combines the above variables to create the derived variable.
Imputation	Park homes and boats are recorded as no answer. Unknowns recoded to 'does not apply' if tenure1 says rented from employer
Key dependencies	None
Introduction date	2008
Date of key modelling changes	2008
Other relevant information	In 2017, the commonholder classification was dropped from the survey and recorded as freehold. This variable is available in the datasets provided under the End User Licence and the Special Licence.
Previous variable	EHCS – 'bgdfreeh' but modelled differently as questions asked were changed.

Prevtten**Interviewfs19.sav, Interview 18+19.sav**

Variable name	Prevtten
Variable label	Previous Tenure
Units	N/A
Description	Tenure of previous home of HRP. This is only recorded for those who have lived in their current home for less than 3 years.
Type	Numeric
Measure	Nominal
Value labels	-9 "does not apply" -8 "no answer" 1 "new household" 2 "owned outright" 3 "buying with a mortgage" 4 "owned, unknown if outright or mortgage" 5 "local authority" 6 "housing association" 7 "privately rented" 8 "rented, landlord unknown" 9 "household reference person resident 3 years or more"
Missing data	Yes
File location	interviewfs19.sav and interview 18+19.sav
Validation	Frequency checks and consistency checks.
Key inputs	'HRP', 'restme3', 'tempac', 'prevacn', 'prev00', 'prevac', 'prevr'
Modelling	Combination of above variables.
Imputation	None
Key dependencies	None
Introduction date	2008
Date of key modelling changes	None
Other relevant information	Based on HRP. This variable is available in the datasets provided under the End User Licence and the Special Licence.
Previous variable	N/A

Pyngbx**Interviewfs19.sav, Interview 18+19.sav**

Variable name	Pyngbx
Variable label	Age band of youngest person in household
Units	N/A
Description	Age of youngest person in household – banded.
Type	Numeric
Measure	Ordinal
Value labels	-9 "does not apply" -8 "no answer" 1 "0 - 4" 2 "5 - 10" 3 "11 - 15" 4 "16 - 24" 5 "25 - 59" 6 "60 - 74" 7 "75 - 84" 8 "85 or over"
Missing data	None
File location	interviewfs19.sav and interview 18+19.sav
Validation	Frequency checks and consistency checks against time series.
Key inputs	'pyngx'
Modelling	None – bands the variable pyngx.
Imputation	None
Key dependencies	None
Introduction date	2008
Date of key modelling changes	None
Other relevant information	Excludes those living in halls of residence unless they are the HRP or partner. This variable is available in the datasets provided under the End User Licence and the Special Licence.
Previous variable	SEH variable – 'minageg' EHCS variable 'pyngbx'

Pyngx**Interviewfs19.sav, Interview 18+19.sav**

Variable name	Pyngx
Variable label	Age of youngest person in household
Units	Years
Description	Age of youngest person in household.
Type	Numeric
Measure	Scale
Value labels	-9 "does not apply" -8 "no answer" 85 "85 or over"
Missing data	None
File location	interviewfs19.sav and interview 18+19.sav
Validation	Frequency checks and consistency checks against time series.
Key inputs	'age'
Modelling	None – identifies the minimum age of the household members.
Imputation	None
Key dependencies	'pyngbx'
Introduction date	2008
Date of key modelling changes	None
Other relevant information	Excludes those living in halls of residence unless they are the HRP or partner. For anonymisation purposes, where the age of the youngest person in the household is aged over 84, their age is capped at '85 or over'. This variable is only available in the dataset provided under the Special Licence.
Previous variable	SEH – 'Ageyt' EHCS – 'Pyngx'

RentExS**Interviewfs19.sav, Interview 18+19.sav**

Variable name	RentExS
Variable label	Total weekly rent excluding the cost of services
Units	£'s
Description	The total weekly rent payable for the property, excluding the cost of services e.g. heating, council tax etc. This variable only applies to households that rent their home, including shared owners.
Type	Numeric
Measure	Scale
Value labels	-9.00 "question not applicable - owner occupier and not shared ownership"
Missing data	None
File location	interviewfs19.sav and interview 18+19.sav
Validation	The SPSS code contains internal checks and cross checks of the raw input and interim variables. The final variable is compared with results from previous years overall and by key dwelling descriptor variables.
Key inputs	'rentwqx' (derived variable), 'SeriCTax', 'SeriHeat', 'SeriWter', 'SeriLght', 'SeriHWtr', 'SeriFuel', 'SeriMeal', 'SeriTvLi', 'SeriNone', 'RentHOLA' and 'RentHoWks'.
Modelling	During the EHS interview survey, renter householders are asked if the rent amount that they provided (used to produce 'rentwqx') included any services. If the householder's rent does not include any of the stated services then their total weekly rent payable for the property, excluding the cost of services ('rentExS') is set at the total weekly rent ('rentwqx'). If the householder's rent does include one or more of the stated services, then the net rent is modelled consistently for all cases by subtracting an estimated service charge from the total weekly rent payable for the property. See the 'Imputation' category for more detail.
Imputation	If services are included in the rental amount provided by the householder, then an estimated amount for the selected services are modelled and the variable 'rentExS' is calculated by subtracting the estimated service amount from the total weekly rent payable ('rentwqx'). For more detail see the section on Household derived indicators, Chapter 5 of the English Housing Survey 2019-20 Technical Report Annex 5.4 on the sources used for modelling service amounts.
Key dependencies	AHC equivalised weekly income ('AHCinceq')
Introduction date	2008
Date of key modelling changes	2018/19 – Prior to 2018 modelling, if the total rent amount provided by the householder did include one or more selected services, then the net rent (where possible) was based upon the

	householder's response to their rent amount excluding services as provided in the EHS interview survey. From 2018 modelling, the net rent is modelled consistently for all cases (see Modelling/Imputation above). The householder's response to their net rent amount is no longer used in modelling. As all net rents are now modelled consistently, the derived variable 'rentExSflg' that previously indicated the nature/extent of any net rent imputation is now longer required and has been removed from the derived interview dataset.
Other relevant information	The net rent questions were introduced into the EHS 2008 survey. The source code used for modelling also calculates the variables 'rentwkx', 'rentflg', 'amthbenx' and 'housbenx'. This variable is available in the datasets provided under the End User Licence and the Special Licence.
Previous variable	N/A

Rentflg

Interviewfs19.sav, Interview 18+19.sav

Variable name	Rentflg
Variable label	Rent/housing benefit changed/imputed
Units	N/A
Description	Whether the total rent and housing related benefit variables ('rentwkx' and 'amthbenx') were imputed as raw data was missing, inconsistent or implausible.
Type	Numeric
Measure	Nominal
Value labels	-9 "question not applicable - owner occupier and not shared ownership" 0 "raw rent/housing related benefits data used" 1 "rent/housing related benefits imputed as value missing" 2 "rent/housing related benefits overwritten"
Missing data	None
File location	interviewfs19.sav and interview 18+19.sav
Validation	Checks are made to ensure that missing/overwritten data has been coded to the correct value and that owner occupiers (excluding those on a shared ownership scheme) are set to -9. The overall flag is compared with results from previous years.
Key inputs	'rentwkx' and 'amthbenx'.
Modelling	If the case is owner occupier (excluding those on a shared ownership scheme) then "question not applicable - owner occupier" is assigned to the rent flag. For cases that rent their home (including shared ownership schemes); • If the rent/housing related benefit is based on amounts provided in the interview survey then the rent flag is set to "raw rent/housing related benefits data used", • Households that rent their home but do not provide sufficient rent and/or housing benefit information are assigned a theoretical guide rent amount and/or a theoretical housing benefit amount, for these cases the flag is set as "rent/housing related benefits imputed as value missing" and • Extreme rent outliers in the dataset are investigated case by case and if the values seem implausible then the rent values (and housing benefit value if applicable) are overwritten by the guide rent amount, for these cases the rent flag is set as "rent/housing related benefits overwritten". • For a selected number of Universal Credit cases, the weekly housing related benefit amount received by the household is calculated earlier in the modelling process when deriving household income, and then the modelled amount is brought into the rent and housing benefit modelling and used to calculate the final housing related benefit amount. For these

	selected cases, the rent flag is also set as "rent/housing related benefits overwritten".
Imputation	N/A
Key dependencies	None
Introduction date	1996
Date of key modelling changes	2019/20 – The value label 2 changed from 'rent/housing related benefits data overwritten as value implausible' to 'rent/housing related benefits overwritten'. From 2019/20 modelling, this category includes selected Universal Credit cases where the weekly housing related benefit amount received by the household is calculated earlier in the modelling process when deriving household income, and then the modelled amount is brought into the rent and housing benefit modelling and used to calculate the final housing related benefit amount.
Other relevant information	The source code used for modelling also calculates the variables 'rentwqx', 'rentExS', 'amthbenx' and 'housbenx'. This variable is only available in the dataset provided under the Special Licence.
Previous variable	N/A

Rentwvx**Interviews19.sav, Interview 18+19.sav**

Variable name	Rentwvx
Variable label	Total weekly rent payable (rent plus housing benefit)
Units	£'s
Description	Total weekly rent payable for the property, i.e. it includes the rent paid by the householder plus any housing benefit/Local Housing Allowance/Universal Credit received towards their rent as reported in the rent and housing benefit module of the EHS interview questionnaire. This variable only applies to households that rent their home, including households in a shared ownership scheme.
Type	Numeric
Measure	Scale
Value labels	-9.00 "question not applicable - owner occupier and not shared ownership"
Missing data	None
File location	interviews19.sav and interview 18+19.sav
Validation	The SPSS code contains internal checks of input and interim variables. Checks are made to ensure missing data has been imputed correctly and extreme rent outliers are identified and investigated on a case by case basis looking at property photos (where available) and dwelling characteristics such as tenure, tenancy, number of bedrooms, landlord, geographical location etc. The final variable is compared with results from previous years overall and by key dwelling descriptor variables.
Key inputs	Most of the Rent and Housing Benefit (QLandB) section in the EHS interview survey from the variable 'Ldjoint' to 'SerInc' (see the EHS questionnaire documentation for the full list) and also variables 'RentHOLA' and 'RentHOLWks'.
Modelling	'Rentwvx' and housing benefit/Local Housing Allowance/Universal Credit as selected in the rent and housing benefit module of the EHS questionnaire (derived variable 'amthbenx') are calculated in conjunction with one another in the modelling process. The variables are derived predominantly from the householder's response to the questions in the rent and housing benefit module. Households that rent their home but do not provide any rent/housing benefit information are assigned a theoretical total weekly rent payable, see the category 'imputation' below. Renting households that live rent-free are not asked the module on Rent and Housing benefit and for these cases their rent and housing benefit amount are set to zero. For more details see the section on Household derived indicators, Chapter 5 of the English Housing Survey 2019-20 Technical Report Annex 5.4.
Imputation	Households that rent their home but do not provide any rent/housing benefit information, or cases where the rent value provided by the householder is deemed as an

	outliers/implausible value, are assigned a total weekly rent payable made up of a theoretical guide rent and an estimated modelled amount for services e.g. heating and regular meals etc. that are included in the rent (where applicable as specified by the householder). Estimated total weekly guide rents are derived based upon tenure, number of bedrooms and Government Region (for private renters variables such as the type of landlord and furnishing are also included in the imputation) using the following data sources; average rents from the MHCLG Local Authority Housing Statistics and data drawn from the Statistical Data Return submitted by Private Registered Providers (for local authority and housing association properties respectively) and EHS data using the year in question and modelled accordingly for private rents.
Key dependencies	Total weekly rent excluding the cost of services 'rentExS'.
Introduction date	1996
Date of key modelling changes	2007 EHCS – The questions asked in the rent and housing benefit module was revised and the modelling was adjusted accordingly. 2008/09 data – The Survey of English Housing had previously been used to provide missing guide private rents. This survey discontinued in 2008, therefore from the 2008/09 missing private rents data in the EHS are imputed using the EHS private rent data from the year in question and modelled accordingly using multiple regression. 2011/12 data – Missing housing association guide rents are no longer imputed based upon data from the Regulatory and Statistical Returns Survey as this information is no longer available. For 2011/12 data was drawn from the Statistical Data Return submitted by Private Registered Providers. 2011/12 data – Missing values for the total weekly rent payable now also include an estimated modelled amount for services e.g. heating and regular meals etc. that are included in the rent (where applicable as specified by the householder). 2013/14 data – The average rents from the MHCLG Local Authority Housing Statistics (as used in the imputation of missing local authority rents) are no longer available for the year in question as the data is not published in time for EHS modelling purposes. Therefore, missing local authority rents in the EHS are imputed using average rents from the MHCLG Local Authority Housing Statistics for the previous financial year, uplifted to the current year using a factor generated from two successive years of weighted EHS local authority rent data. 2018/19 data and 2019/20 – The average rents supplied by the Regulator of Social Housing (as used in the imputation of missing housing association/shared owner rents) are no longer available for the year in question as the data is not published in time for EHS modelling purposes. However, the Welfare Reform and Work Act 2016 and amendment regulations require registered providers of social housing in England to reduce social housing rents by 1% a year for four years from a frozen

	2015 to 2016 baseline and to comply with maximum rent requirements for new tenancies. Therefore, for EHS 2018 and EHS 2019 rent modelling, the guide RSL rents were used for the previous modelling year with a 1% reduction applied to reflect the estimated 2018 and 2019 values respectively.
Other relevant information	The source code used for modelling also calculates the variables 'rentflg', 'rentExS', 'amthbenx' and 'housbenx'. From the EHS 2013-14, the interview survey was modified so that Universal Credit was included in the question on whether the household receives any help towards their rent in the rent and housing benefit module (Hbena2). This variable is available in the datasets provided under the End User Licence and the Special Licence.
Previous variable	N/A

Sexhrp**Interviewfs19.sav, Interview 18+19.sav**

Variable name	Sexhrp
Variable label	Sex of HRP
Units	N/A
Description	Sex of household reference person.
Type	Numeric
Measure	Nominal
Value labels	-9 "does not apply" -8 "no answer" 1 "male" 2 "female"
Missing data	None
File location	interviewfs19.sav and interview 18+19.sav
Validation	Frequency checks.
Key inputs	'reltohrp2', 'sex'
Modelling	None – based on the above variables.
Imputation	Missing cases for 'reltohrp2' imputed using other information in the dataset.
Key dependencies	None
Introduction date	2008
Date of key modelling changes	None
Other relevant information	This variable is available in the datasets provided under the End User Licence and the Special Licence.
Previous variable	Same variable name in former SEH Former EHCS variable name 'hrpsexx' In 1991 and 1996, the EHCS variable was based on head of household (HOH) rather than household reference person (HRP). Before 2001, the SEH variable was also based on HOH.

Sexprt**Interviewfs19.sav, Interview 18+19.sav**

Variable name	Sexprt
Variable label	Sex of HRP's partner
Units	N/A
Description	Sex of household reference person's partner.
Type	Numeric
Measure	Nominal
Value labels	-9 "does not apply" -8 "no answer" 1 "male" 2 "female"
Missing data	None
File location	interviewfs19.sav and interview 18+19.sav
Validation	Frequency checks.
Key inputs	'reltohrp2', 'sex'
Modelling	None – based on the above variables.
Imputation	Missing cases for 'reltohrp2' imputed using other information in the dataset.
Key dependencies	None
Introduction date	2008
Date of key modelling changes	None
Other relevant information	This variable is only available in the dataset provided under the Special Licence.
Previous variable	SEH – 'sexprt'

Sft**Interviewfs19.sav, Interview 18+19.sav**

Variable name	sft
Variable label	Number of full-time workers in household
Units	Persons
Description	Number of people in the household who work full-time. This counts the number of people whose primary economic status is coded as working full-time, where 'full-time' means working at least 30 hours per week.
Type	Numeric
Measure	Scale
Value labels – Special Licence	-9 "does not apply" -8 "no answer"
Value labels – End User Licence	-9 "does not apply" -8 "no answer" 5 "5 or more"
Missing data	None
File location	interviewfs19.sav and interview 18+19.sav
Validation	Frequency checks and consistency checks
Key inputs	Raw data on working status, EconAct (allows multiple responses: EconFull to EconOthr), 'age', 'sex'
Modelling	See 'emphrpx' for details on modelling employment status for each person.
Imputation	See 'emphrpx'
Key dependencies	None
Introduction date	2008
Date of key modelling changes	None
Other relevant information	Does not include people who are full-time students who also say they are working full-time. Also, does not include people over state retirement age who describe themselves as both 'retired' and 'working full-time'. See 'emphrpx' for further details. Originates from SEH. For anonymisation purposes, a household with 5 or more full-time workers is capped at '5 or more' for the End User Licence. This variable is available in the datasets provided under the End User Licence and Special Licence.
Previous variable	'SFT' from SEH

Sharer **Interviewfs19.sav, Interview 18+19.sav**

Variable name	Sharer
Variable label	If shares accommodation with other household
Units	N/A
Description	Whether the household shares any rooms with other households.
Type	Numeric
Measure	Nominal
Value labels	-9 "does not apply" -8 "no answer" 1 "shares accommodation" 2 "self-contained accommodation"
Missing data	None
File location	interviewfs19.sav and interview 18+19.sav
Validation	Frequency checks
Key inputs	'Share2'
Modelling	Renaming of 'share2' variable
Imputation	None
Key dependencies	'accomhh1'
Introduction date	2008
Date of key modelling changes	Same as SEH methodology
Other relevant information	This variable is available in the datasets provided under the End User Licence and the Special Licence.
Previous variable	SEH – 'sharer'

Srten2**Interviewfs19.sav, Interview 18+19.sav**

Variable name	Srten2
Variable label	Type of social sector tenancy
Units	N/A
Description	Type of social housing tenancy - 10 category
Type	Numeric
Measure	Nominal
Value labels	-9 "does not apply" -8 "no answer" 1 "local authority: secure tenancy" 2 "local authority: flexible tenancy (from April 2012)" 3 "local authority: introductory tenancy" 4 "local authority: other tenancy" 5 "housing association: assured tenancy" 6 "housing association: secure tenancy (offered up to and including 1988)" 7 "housing association: assured shorthold tenancy" 8 "housing association: fixed term tenancy (from April 2011)" 9 "housing association: starter tenancy" 10 "housing association: other tenancy"
Missing data	Yes
File location	interviewfs19.sav and interview 18+19.sav
Validation	Frequency checks and consistency checks.
Key inputs	'tentypsr2, srtenlen2, srtenlen2b, tenex'
Modelling	Information is taken from a combination of srtenlen2, srtenlen2b and tentypsr2 where tenure ('tenex') is classed as housing association or local authority
Imputation	None
Key dependencies	Hatentp3
Introduction date	2014
Date of key modelling changes	None
Other relevant information	This is created using different input variables to the previous variable 'srten2', and there is an extra category 'housing association: assured shorthold tenancy' so comparing these variables is not advisable. In 2017, a continuous length of social rented tenancy variable was introduced (srtenlen2) in addition to the existing banded variable (srtenlen2b).

	This variable is available in the datasets provided under the End User Licence and the Special Licence.
Previous variable	EHS variable 'srtentype'

Studhrp Interviewfs19.sav, Interview 18+19.sav

Variable name	Studhrp
Variable label	If HRP is a full-time student
Units	N/A
Description	Whether HRP is a full-time student. Based on primary economic status.
Type	Numeric
Measure	Nominal
Value labels	-9 "does not apply" -8 "no answer" 1 "full-time student" 2 "not full-time student"
Missing data	None
File location	interviewfs19.sav and interview 18+19.sav
Validation	Frequency checks and consistency checks
Key inputs	Raw data on working status, EconAct (allows multiple responses: EconFull to EconOthr), 'age', 'sex', 'reltohrp2',
Modelling	See 'emphrpx' for details on modelling employment status for each person.
Imputation	See 'emphrpx'
Key dependencies	None
Introduction date	2008
Date of key modelling changes	None
Other relevant information	See 'emphrpx' for details. Same as SEH method This variable is only available in the dataset provided under the Special Licence.
Previous variable	'studhrp' from SEH

Studprt Interviewfs19.sav, Interview 18+19.sav

Variable name	Studprt
Variable label	If HRP's partner is a full-time student
Units	N/A
Description	Whether partner of HRP is a full-time student. Based on 'primary' economic status.
Type	Numeric
Measure	Nominal
Value labels	-9 "does not apply" -8 "no answer" 1 "full-time student" 2 "not full-time student"
Missing data	None
File location	interviewfs19.sav and interview 18+19.sav
Validation	Frequency checks and consistency checks
Key inputs	Raw data on working status, EconAct (allows multiple responses: EconFull to EconOthr), 'age', 'sex', 'reltop2'
Modelling	See 'emphrpx' for details on modelling employment status for each person.
Imputation	See 'emphrpx'
Key dependencies	None
Introduction date	2008
Date of key modelling changes	None
Other relevant information	See 'emphrpx' for details. Same as SEH method. This variable is only available in the dataset provided under the Special Licence.
Previous variable	'studprt' from SEH

Tenex Interviewfs19.sav, Interview 18+19.sav

Variable name	Tenex
Variable label	Extended tenure of household
Units	N/A
Description	Tenure.
Type	Numeric
Measure	Nominal
Value labels	-9 "Does not apply" -8 "No answer" 1 "own with mortgage" 2 "own outright" 3 "privately rent" 4 "rent from local authority" 5 "rent from housing association"
Missing data	None
File location	interviewfs19.sav and interview 18+19.sav
Validation	Crosschecks with other tenure variables and consistency checks against time series.
Key inputs	'tenure2' ('ten1', 'llord', 'tenure1')
Modelling	None
Imputation	See tenure1
Key dependencies	'buypresh', 'mortwkx'
Introduction date	1991
Date of key modelling changes	2008
Other relevant information	Households who rent from the local authority where the HRP/partner also work for the local authority are coded as 'private rented'. Originates from EHCS. From 2008, created from SEH variable "tenure2". This variable is available in the datasets provided under the End User Licence and the Special Licence.
Previous variable	N/A

Tenure1**Interviewfs19.sav, Interview 18+19.sav**

Variable name	Tenure1
Variable label	Tenure group 1
Units	N/A
Description	Tenure.
Type	Numeric
Measure	Nominal
Value labels	-9 "does not apply" -8 "no answer" 1 "own outright" 2 "buying with mortgage" 3 "shared ownership" 4 "local authority tenant" 5 "housing association tenant" 7 "employer" 8 "organisation (inc property company)" 9 "relative/friend" 11 "individual" 12 "squatting"
Missing data	None
File location	interviewfs19.sav and interview 18+19.sav
Validation	Crosschecks with other tenure variables and consistency checks against time series.
Key inputs	'ten1', 'llord', 'tied'
Modelling	None
Imputation	In 2013 a new validation check was introduced into the interview which crosschecks the mortgage question (onoutmrg1) with Ten1. As this can't be corrected at the time of the interview, Ten1 is checked in the modelling and altered if necessary before deriving Tenure1. This feeds into all the other derived tenure variables. Please note Ten1 is not changed in the raw data set.
Key dependencies	'tenure2', 'tenure3', 'tenure4', 'tenex', 'tenure4x', 'hatentp3'
Introduction date	2008
Date of key modelling changes	None
Other relevant information	Households who rent from the local authority where the HRP/partner also work for the local authority are coded as renting from their employer and not from the local authority. Households who have equity release mortgages are recoded as 'buying with mortgage' if originally coded as 'own outright'. Originates from SEH.

	This variable is only available in the dataset provided under the Special Licence.
Previous variable	N/A

Tenure2

Interviewfs19.sav, Interview 18+19.sav

Variable name	Tenure2
Variable label	Tenure group 2
Units	N/A
Description	Tenure.
Type	Numeric
Measure	Nominal
Value labels	-9 "does not apply" -8 "no answer" 1 "own outright" 2 "buying with mortgage (including shared ownership)" 3 "local authority tenant" 4 "housing association tenant" 5 "private renter unfurnished (or unknown if furnished)" 6 "private renter furnished"
Missing data	None
File location	interviewfs19.sav and interview 18+19.sav
Validation	Crosschecks with other tenure variables and consistency checks against time series.
Key inputs	'tenure1' ('ten1', 'llord'), 'furn'
Modelling	None
Imputation	See tenure1
Key dependencies	'tenure4'
Introduction date	2008
Date of key modelling changes	None
Other relevant information	Households who rent from the local authority where the HRP/partner also work for the local authority are coded as 'private rented' Originates from SEH. This variable is available in the datasets provided under the End User Licence and the Special Licence.
Previous variable	N/A

Tenure3**Interviewfs19.sav, Interview 18+19.sav**

Variable name	Tenure3
Variable label	Tenure group 3
Units	N/A
Description	Tenure.
Type	Numeric
Measure	Nominal
Value labels	-9 "does not apply" -8 "no answer" 1 "own outright" 2 "buying with mortgage (including shared ownership)" 3 "local authority tenant" 4 "housing association tenant" 5 "rents from employer" 6 "other renter"
Missing data	None
File location	interviewfs19.sav and interview 18+19.sav
Validation	Crosschecks with other tenure variables and consistency checks against time series.
Key inputs	'tenure1' ('ten1', 'llord')
Modelling	None
Imputation	See tenure1
Key dependencies	None
Introduction date	2008
Date of key modelling changes	None
Other relevant information	Households who rent from the local authority where the HRP/partner also work for the local authority are coded as 'rents from employer'. Originates from SEH. This variable is only available in the dataset provided under the Special Licence.
Previous variable	N/A

Tenure4**Interviewfs19.sav, Interview 18+19.sav**

Variable name	Tenure4
Variable label	Tenure group 4
Units	N/A
Description	Tenure.
Type	Numeric
Measure	Nominal
Value labels	-9 "does not apply" -8 "no answer" 1 "owners" 2 "social sector" 3 "private renters"
Missing data	None
File location	interviewfs19.sav and interview 18+19.sav
Validation	Crosschecks with other tenure variables and consistency checks against time series.
Key inputs	'tenure2'
Modelling	None
Imputation	See tenure1
Key dependencies	'freeleas', 'owntype'
Introduction date	2008
Date of key modelling changes	None
Other relevant information	Households who rent from the local authority where the HRP/partner also work for the local authority are coded as 'private rented'. Originates from SEH. This variable is available in the datasets provided under the End User Licence and the Special Licence.
Previous variable	N/A

Totkitsa**Interviewfs19.sav, Interview 18+19.sav**

Variable name	Totkitsa
Variable label	Number of unshared kitchens
Units	Number
Description	Number of kitchens that are for the exclusive use of the household.
Type	Numeric
Measure	Scale
Value labels	-9 "does not apply" -8 "no answer" 2 "2 or more"
Missing data	None
File location	interviewfs19.sav and interview 18+19.sav
Validation	Frequency checks and consistency check against nkita. Frequency and plausibility check on shrms2.
Key inputs	'nrms2a', 'shrms2'
Modelling	Subtracts number of shared kitchens ('shrms2') from total number of kitchens 'nrms2a'
Imputation	None
Key dependencies	None
Introduction date	2011 (previously was totkits)
Date of key modelling changes	same as SEH methodology
Other relevant information	This variable changed due to changes in the questionnaire and the variable nrms2 being changed to nrms2a- since size of a kitchen was no longer relevant. Households with 0 kitchens include those in bedsit rooms with kitchen facilities but no separate kitchen. For anonymisation purposes, a household with 2 or more unshared kitchens is capped at '2 or more'. This variable is only available in the dataset provided under the Special Licence.
Previous variable	SEH – 'totkits'

Workless**Interviewfs19.sav, Interview 18+19.sav**

Variable name	Workless
Variable label	Household with no one of working age employed - ILO defn
Units	N/A
Description	Whether household has no one of working age in employment. The definition of 'employed' and unemployed' are those used by the ILO.
Type	Numeric
Measure	Nominal
Value labels	-9 "no one of working age in household" -8 "no answer" 0 "not workless" 1 "workless"
Missing data	None
File location	interviewfs19.sav and interview 18+19.sav
Validation	Frequency checks and consistency checks
Key inputs	Raw data on working status, 'EconAct' (allows multiple responses: 'EconFull' to 'EconOthr'), 'age', 'sex'
Modelling	Derived from the above variables. Workless households are defined as those that include at least one person of working age (aged 16 or over and below state retirement age) where no-one is in employment. Unlike other employment status variables, full-time students who are also working (full or part-time) are classed as in employment (i.e. not workless).
Imputation	See 'emphrpx'
Key dependencies	None
Introduction date	1996
Date of key modelling changes	None
Other relevant information	Excludes those living away from home in halls of residence if not HRP or HRP partner. This variable is available in the datasets provided under the End User Licence and the Special Licence.
Previous variable	EHCS variable 'workless' – derived in exactly the same way.

Chapter 3

Variables in Physical 18+19.sav

Alltypex

Physical 18+19.sav

Variable name	Alltypex
Variable label	Dwelling age and type
Units	N/A
Description	Dwelling age and type (10 categories).
Type	Numeric
Measure	Nominal
Value labels	-8 "unknown" 1 "purpose built flat, high rise" 2 "purpose built flat, low rise" 3 "converted flat" 4 "bungalow: all ages" 5 "detached house: pre 1919" 6 "detached house: post 1919" 7 "semi detached & terraced: pre 1919" 8 "semi detached & terraced: 1919-1944" 9 "semi detached & terraced: 1945-1964" 10 "semi detached & terraced: 1965 onwards"
Missing data	None
File location	physical 18+19.sav
Validation	Frequency checks and consistency checks
Key inputs	'dwtype8x', 'dwage4x'
Modelling	None – combines the above variables to create the dwelling age and type categories.
Imputation	None
Key dependencies	None
Introduction date	1991
Date of key modelling changes	None
Other relevant information	This variable is available in the datasets provided under the End User Licence and Special Licence.
Previous variable	N/A

Area3x**Physical 18+19.sav**

Variable name	Area3x
Variable label	Type of area
Units	N/A
Description	Type of area - loosely defined as the 'area around the survey dwelling of which the dwelling seems to be part'. Simplified version of variable 'arnatx'.
Type	Numeric
Measure	Nominal
Value labels	-8 "unknown" 1 "city and other urban centres" 2 "suburban residential areas" 3 "rural areas"
Missing data	None
File location	physical 18+19.sav
Validation	Frequency checks and checks with previous year.
Key inputs	'arnatx'
Modelling	None – recodes arnatx into the above categories.
Imputation	None
Key dependencies	None
Introduction date	1991
Date of key modelling changes	None
Other relevant information	This variable is only available in the dataset provided under the Special Licence. The variable was originally located in the general.sav file from 1991 to 2007 EHCS.
Previous variable	None

Arnatx**Physical 18+19.sav**

Variable name	Arnatx
Variable label	Nature of area
Units	N/A
Description	Nature of area - loosely defined as the 'area around the survey dwelling of which the dwelling seems to be part'.
Type	Numeric
Measure	Nominal
Value labels	-8 "unknown" 1 "city centre" 2 "other urban centre" 3 "suburban residential" 4 "rural residential" 5 "village centre" 6 "rural"
Missing data	None
File location	physical 18+19.sav
Validation	Frequency checks and checks with previous year.
Key inputs	'farnatur'
Modelling	None – based on the raw physical survey variable 'farnatur'.
Imputation	None
Key dependencies	area3x
Introduction date	1991
Date of key modelling changes	None
Other relevant information	This variable is only available in the dataset provided under the Special Licence. The variable was originally located in the general.sav file from 1991 to 2007 EHCS.
Previous variable	None

Attic

Physical 18+19.sav

Variable name	Attic
Variable label	Attic present in dwelling
Units	N/A
Description	Identifies whether the dwelling has a habitable attic space. To be classed as an attic, it must meet prescribed criteria e.g. the room in the loft must be no more than 1 metre below eaves level, have fixed serviceable stairs and natural light.
Type	Numeric
Measure	Nominal
Value labels	1 "yes" 2 "no"
Missing data	None
File location	physical 18+19.sav
Validation	Frequency checks and consistency checks against time series. Crosschecks against raw physical survey variables.
Key inputs	'Fshattic', 'Fdhmflrs', 'Fdhaflrs', 'Fdhmlev1 – 3'
Modelling	Created by the energy model using the above raw data crosschecked with the number and levels of floors in the main and additional structure to ensure that, e.g. the top habitable floor of a dwelling with a mansard roof has not been coded as an attic.
Imputation	None
Key dependencies	N/A
Introduction date	1991
Date of key modelling changes	None
Other relevant information	This variable relates to the dwelling and not the block or module. This variable is available in the datasets provided under the End User Licence and Special Licence.
Previous variable	N/A

Basement

Physical 18+19.sav

Variable name	Basement
Variable label	Basement present in dwelling
Units	N/A
Description	Identifies whether the dwelling has a habitable basement (rather than a cellar). To be classed as a basement, it must meet prescribed criteria e.g. be more than 1 metre or more below ground level
Type	Numeric
Measure	Nominal
Value labels	1 "yes" 2 "no"
Missing data	None
File location	physical 18+19.sav
Validation	Frequency checks and consistency checks against time series. Crosschecks against raw physical survey variables.
Key inputs	'Fshattic', 'Fshentr', 'Fdhmflrs', 'Fdhaflrs', 'Fdhmlev1 – 3'
Modelling	Created by the energy model using the raw data crosschecked with the number and level of floors in the main and additional structure and the entry floor (which could be the basement level).
Imputation	None
Key dependencies	N/A
Introduction date	1991
Date of key modelling changes	None
Other relevant information	This variable relates to the dwelling and not the block or module. This variable is only available in the dataset provided under the Special Licence.
Previous variable	N/A

Boiler

Physical 18+19.sav

Variable name	Boiler
Variable label	Type of boiler
Units	N/A
Description	The type of central heating boiler in dwellings that have central heating as the primary space heating system.
Type	Numeric
Measure	Nominal
Value labels	-9 "no boiler" 1 "standard boiler (floor or wall)" 2 "back boiler (to fire or stove)" 3 "combination boiler" 4 "condensing boiler" 5 "condensing-combination boiler"
Missing data	None
File location	physical 18+19.sav
Validation	Frequency checks and consistency checks against time series. Crosschecks against raw physical survey variables.
Key inputs	'Finchtyp', 'Finmhboi', 'Finchbcd'
Modelling	Created by the energy model primarily using the raw primary heating type variable, which is crosschecked against the heating group and the primary heating appliance code to check that the dwelling uses a boiler driven system for space heating. For more detail see the section on modelled indicators, Chapter 5 of the English Housing Survey 2019-20 Technical Report, Annex 6
Imputation	Where boiler information is missing, but the primary system clearly uses a boiler, a standard boiler code is used by default.
Key dependencies	N/A
Introduction date	2006
Date of key modelling changes	None
Other relevant information	This variable is available in the datasets provided under the End User Licence and Special Licence.
Previous variable	N/A

Constx**Physical 18+19.sav**

Variable name	Constx
Variable label	Construction type
Units	N/A
Description	Construction type from section 14 of physical survey form. The original construction type of the dwelling is recorded, (i.e. excludes extensions and improvements which may change the predominant construction type). The construction method refers to the main structural components supporting the roof and floors, and possibly walls. Any inconsistencies with wall structure (section 18 of the physical survey form) are examined and recoded if required.
Type	Numeric
Measure	Nominal
Value labels	1 "solid masonry" 2 "cavity masonry" 3 "crosswall masonry" 4 "timber frame" 5 "steel frame" 6 "concrete frame" 7 "concrete boxwall" 8 "other concrete"
Missing data	None
File location	physical 18+19.sav
Validation	Crosschecks made with wall structure (section 18 of the physical survey form) and photos examined where inconsistencies
Key inputs	'fmtconst', 'type', 'fexws1te', 'fexws2te', 'typewstr2', 'wallcavy', 'dwage6x'
Modelling	Based on 'fmtconst' and corrected if necessary where conflicts arise with the above variables, using photos as an extra aid.
Imputation	None
Key dependencies	None
Introduction date	2010
Date of key modelling changes	None
Other relevant information	A crosstab of predominant wall by wall type and by construction type may show inconsistencies. This is due to the predominance being calculated on tenths of area, whilst wall type makes use of modelled wall areas from dimensions data. This variable is only available in the dataset provided under the Special Licence.

Previous variable	N/A
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Cstactbx**Physical 18+19.sav**

Variable name	Cstactbx
Variable label	Basic repair costs (actual)
Units	£'s
Description	Estimated costs of carrying out all basic repairs that the surveyor indicated needed to be carried out to the dwelling in £. Costs are at current prices (i.e. costs for the 2019 data set are at 2019 prices) and take account of differences in building costs in the different tenures and regions. They do not include VAT or fees but do include additional costs such as scaffolding where this would be needed. 'Basic repairs' means all repair work that needs doing within five years and therefore includes all urgent repairs (see 'cstactux'). The costs do not include any necessary improvements or additional work required to address non-decency or to deal with Category 1 hazards.
Type	Numeric
Measure	Scale
Value labels – Special Licence	N/A – all amounts are coded in £. Where there are no repairs, this appears as £0.
Value labels – End User Licence	50,000.00 "£50,000 or more" All amounts are coded in £. Where there are no repairs, this appears as £0.
Missing data	None
File location	physical 18+19.sav
Validation	There is no other external data source that could be used to validate this variable. The DELPHI and SPSS code contain internal checks of input and interim variables. Consistency checks made with previous years overall and by key dwelling descriptor variables.
Key inputs	All variables relating to repairs needed in sections 5, 9, 18 and 21 of the physical survey form plus all outputs from dimensions model. Also data file with unit prices and inflation factors.
Modelling	Repair costs come from a suite of programs mainly written in DELPHI. For more detail see the section on modelled indicators, Chapter 5 of the English Housing Survey 2019-20 Technical Report, Annex 5.
Imputation	For more detail see the section on modelled indicators, Chapter 5 (Annex 5) of the English Housing Survey 2019-20 Technical Report.
Key dependencies	Some of the costs are used in modelling of costs to deal with HHSRS Category 1 hazards and costs to make decent.
Introduction date	1981
Date of key modelling changes	1986 – complete review of methodology 1996 – complete review of unit prices and overall structure of the model

	2005 – Changes to the dimensions model had some impact on all repair cost variables 2011 - alteration of read/write methods to incorporate the use of SQL. 2014 – public sector and private sector inflation factors were previously gathered from the BCIS website each year and represent an index of inflation for building works. The BCIS factors are based on a sample of construction industry jobs carried out throughout the year. These factors were not readily available from 2014, so projected factors based on the available BCIS indices from 2009 to 2013 were used. Regional inflation factors continued to be available from BCIS. Subsequent annual inflation factors for public, private sector and regions were updated using a projection from previous year.
Other relevant information	This variable takes into account variations in the price of building work by tenure, dwelling type and region and should not be used to compare the extent of disrepair in different groups of the stock. It does however indicate the likely realistic costs (excluding fees and VAT) of carrying out repair works and can be used, for example to see what proportion of the total repair bill relates to a particular sector or type of dwelling etc. For anonymisation purposes, costs of £50,000 or more are capped at £50,000 for the End User Licence. This variable is available in the datasets provided under the End User Licence and Special Licence.
Previous variable	N/A

Cstactcx**Physical 18+19.sav**

Variable name	Cstactcx
Variable label	Comprehensive repair costs (actual)
Units	£'s
Description	Estimated costs of all comprehensive repairs that the surveyor indicated needed to be carried out to the dwelling in £. Costs are at current prices (i.e. costs for the 2019 data set are at 2019 prices) and take account of differences in building costs in the different tenures and regions. They do not include VAT or fees but do include additional costs such as scaffolding where this would be needed. 'Comprehensive repairs' means all repair work that needs doing as specified by the variable 'cstactbx', plus replacing any exterior element whose remaining lifetime is less than or equal to 10 years. The costs do not include any necessary improvements or additional work required to address non-decency or to deal with Category 1 hazards.
Type	Numeric
Measure	Scale
Value labels – Special Licence	N/A – all amounts are coded in £. Where there are no repairs, this appears as £0.
Value labels – End User Licence	50,000.00 "£50,000 or more" All amounts are coded in £. Where there are no repairs, this appears as £0.
Missing data	None
File location	physical 18+19.sav
Validation	There is no other external data source that could be used to validate this variable. The DELPHI and SPSS code contains internal checks of input and interim variables. Consistency checks made with previous years overall and by key dwelling descriptor variables.
Key inputs	All variables relating to repairs needed in sections 5, 9, 18 and 21 of the physical survey form plus all outputs from dimensions model. Also data file with unit prices and inflation factors.
Modelling	Repair costs come from a suite of programs mainly written in DELPHI. For more detail see the section on modelled indicators, Chapter 5 of the English Housing Survey 2019-20 Technical Report, Annex 5.
Imputation	For more detail see the section on modelled indicators, Chapter 5 (Annex 5) of the English Housing Survey 2019-20 Technical Report.
Key dependencies	Some of the costs are used in modelling of costs to deal with HHSRS Category 1 hazards and costs to make decent.
Introduction date	1981

Date of key modelling changes	1986 – complete review of methodology 1996 – complete review of unit prices and overall structure of the model 2005 – Changes to the dimensions model had some impact on all repair cost variables 2011 - alteration of read/write methods so as to incorporate the use of SQL. 2014 – public sector and private sector inflation factors were previously gathered from the BCIS website each year and represent an index of inflation for building works. The BCIS factors are based on a sample of construction industry jobs carried out throughout the year. These factors were not readily available from 2014, so projected factors based on the available BCIS indices from 2009 to 2013 were used. Regional inflation factors continued to be available from BCIS. Subsequent annual inflation factors for public, private sector and regions were updated using a projection from previous year.
Other relevant information	This variable takes into account variations in the price of building work by tenure, dwelling type and region and should not be used to compare the extent of disrepair in different groups of the stock. It does however indicate the likely realistic costs (excluding fees and VAT) of carrying out repair works and can be used, for example to see what proportion of the total repair bill relates to a particular sector or type of dwelling etc. For anonymisation purposes, costs of £50,000 or more are capped at £50,000 for the End User Licence. This variable is available in the datasets provided under the End User Licence and Special Licence.
Previous variable	N/A

Cstactux**Physical 18+19.sav**

Variable name	Cstactux
Variable label	Urgent repair costs (actual)
Units	£'s
Description	Estimated costs of carrying out all urgent repairs that the surveyor indicated needed to be carried out to the dwelling in £. Costs are at current prices (i.e. costs for the 2019 data set are at 2019 prices) and take account of differences in building costs in the different tenures and regions. They do not include VAT or fees but do include additional costs such as scaffolding where this would be needed. 'Urgent repairs' means all repair work that needs doing in order to remove threats to health, safety, security and comfort to the occupants and to forestall further deterioration, plus all internal works and works to shared facilities and common areas. The costs do not include any necessary improvements or additional work required to address non-decency or to deal with Category 1 hazards.
Type	Numeric
Measure	Scale
Value labels – Special Licence	N/A – all amounts are coded in £. Where there are no repairs, this appears as £0.
Value labels – End User Licence	50,000.00 “£50,000 or more” All amounts are coded in £. Where there are no repairs, this appears as £0.
Missing data	None
File location	physical 18+19.sav
Validation	There is no other external data source that could be used to validate this variable. The DELPHI and SPSS code contains internal checks of input and interim variables. Consistency checks made with previous years overall and by key dwelling descriptor variables.
Key inputs	All variables relating to repairs needed in sections 5, 9, 18 and 21 of the physical survey form plus all outputs from dimensions model. Also data file with unit prices and inflation factors.
Modelling	Repair costs come from a suite of programs mainly written in DELPHI. For more detail see the section on modelled indicators, Chapter 5 of the English Housing Survey 2019-20 Technical Report, Annex 5
Imputation	For more detail see the section on modelled indicators, Chapter 5 (Annex 5) of the English Housing Survey 2019-20 Technical Report.
Key dependencies	Some of the costs are used in modelling of costs to deal with HHSRS Category 1 hazards and costs to make decent.
Introduction date	1981

Date of key modelling changes	1986 – complete review of methodology 1996 – complete review of unit prices and overall structure of the model 2005 – changes to the dimensions model had some impact on all repair cost variables 2011 - alteration of read/write methods so as to incorporate the use of SQL. 2014 – public sector and private sector inflation factors were previously gathered from the BCIS website each year and represent an index of inflation for building works. The BCIS factors are based on a sample of construction industry jobs carried out throughout the year. These factors were not readily available from 2014, so projected factors based on the available BCIS indices from 2009 to 2013 were used. Regional inflation factors continued to be available from BCIS. Subsequent annual inflation factors for public, private sector and regions were updated using a projection from previous year.
Other relevant information	This variable takes into account variations in the price of building work by tenure, dwelling type and region and should not be used to compare the extent of disrepair in different groups of the stock. It does however indicate the likely realistic costs (excluding fees and VAT) of carrying out repair works and can be used, for example to see what proportion of the total repair bill relates to a particular sector or type of dwelling etc. For anonymisation purposes, costs of £50,000 or more are capped at £50,000 for the End User Licence. This variable is available in the datasets provided under the End User Licence and Special Licence.
Previous variable	N/A

Cststdbx**Physical 18+19.sav**

Variable name	Cststdbx
Variable label	Basic repair costs (per square metre)
Units	£/m ²
Description	Estimated costs to carry out all basic repairs that the surveyor indicated needed to be carried out to the dwelling in £ per m ² of floor area (see 'floorx'). Costs are at current prices (i.e. costs for the 2019 data set are at 2019 prices) and DO NOT take account of differences in building costs in the different tenures and regions. They do not include VAT or fees but do include additional costs such as scaffolding where this would be needed. 'Basic repairs' means all repair work that needs doing within five years and therefore includes all urgent repairs (see 'cststdux'). The costs do not include any necessary improvements or additional work required to address non-decency or to deal with Category 1 hazards.
Type	Numeric
Measure	Scale
Value labels – Special Licence	N/A – all amounts are coded in £/m ² . Where there are no repairs, this appears as 0 £/m ² .
Value labels – End User Licence	500.00 "£500 or more" All amounts are coded in £/m ² . Where there are no repairs, this appears as 0 £/m ² .
Missing data	None
File location	physical 18+19.sav
Validation	There is no other external data source that could be used to validate this variable. The DELPHI and SPSS code contains internal checks of input and interim variables. Consistency checks made with previous years overall and by key dwelling descriptor variables.
Key inputs	All variables relating to repairs needed in sections 5, 9, 18 and 21 of the physical survey form plus all outputs from dimensions model. Also data file with unit prices and inflation factors.
Modelling	Repair costs come from a suite of programs mainly written in DELPHI. For more detail see the section on modelled indicators, Chapter 5 of the English Housing Survey 2019-20 Technical Report, Annex 5.
Imputation	For more detail see the section on modelled indicators, Chapter 5 (Annex 5) of the English Housing Survey 2019-20 Technical Report.
Key dependencies	Some of the costs are used in modelling of costs to deal with HHSRS Category 1 hazards and costs to make decent.
Introduction date	1981
Date of key modelling changes	1986 – complete review of methodology 1996 – complete review of unit prices and overall structure of

	the model 2005 – changes to the dimensions model had some impact on all repair cost variables 2011 - alteration of read/write methods so as to incorporate the use of SQL. 2014 – public sector and private sector inflation factors were previously gathered from the BCIS website each year and represent an index of inflation for building works. The BCIS factors are based on a sample of construction industry jobs carried out throughout the year. These factors were not readily available from 2014, so projected factors based on the available BCIS indices from 2009 to 2013 were used. Regional inflation factors continued to be available from BCIS. Subsequent annual inflation factors for public, private sector and regions were updated using a projection from previous year.
Other relevant information	This variable does not take into account variations in the price of building work by tenure or region and can be used to compare the extent of disrepair in different groups of the stock. Costs are not, however indicative of the likely realistic costs. For EHS Reporting, these costs for other survey years are rebased to 2016 prices using the Building Cost Information Service (BCIS) National Index, to allow for the assessment of disrepair over time. For anonymisation purposes, costs of 500 £/m² or more are capped at 500 £/m² for the End User Licence. This variable is available in the datasets provided under the End User Licence and Special Licence.
Previous variable	N/A

Cststdcx**Physical 18+19.sav**

Variable name	Cststdcx
Variable label	Comprehensive repair costs (per square metre)
Units	£/m ²
Description	Estimated costs of all comprehensive repairs that the surveyor indicated needed to be carried out to the dwelling in £ per m² of floor area (see 'floorx'). Costs are at current prices (i.e. costs for the 2019 data set are at 2019 prices) and DO NOT take account of differences in building costs in the different tenures and regions. They do not include VAT or fees but do include additional costs such as scaffolding where this would be needed. 'Comprehensive repairs' means all repair work that needs doing as specified by the variable cststdbx, plus replacing any exterior element whose remaining lifetime is less than or equal to 10 years. The costs do not include any necessary improvements or additional work required to address non-decency or to deal with Category 1 hazards.
Type	Numeric
Measure	Scale
Value labels – Special Licence	N/A – all amounts are coded in £/m ² . Where there are no repairs, this appears as 0 £/m ² .
Value labels – End User Licence	500.00 "£500 or more" All amounts are coded in £/m ² . Where there are no repairs, this appears as 0 £/m ² .
Missing data	None
File location	physical 18+19.sav
Validation	There is no other external data source that could be used to validate this variable. The DELPHI and SPSS code contains internal checks of input and interim variables. Consistency checks made with previous years overall and by key dwelling descriptor variables (by BRE).
Key inputs	All variables relating to repairs needed in sections 5, 9, 18 and 21 of the physical survey form plus all outputs from dimensions model. Also data file with unit prices and inflation factors.
Modelling	Repair costs come from a suite of programs mainly written in DELPHI. For more detail see the section on modelled indicators, Chapter 5 of the English Housing Survey 2019-20 Technical Report, Annex 5.
Imputation	For more detail see the section on modelled indicators, Chapter 5 (Annex 5) of the English Housing Survey 2019-20 Technical Report.
Key dependencies	Some of the costs are used in modelling of costs to deal with HHSRS Category 1 hazards and costs to make decent.
Introduction date	1981

Date of key modelling changes	1986 – complete review of methodology 1996 – complete review of unit prices and overall structure of the model 2005 – changes to the dimensions model had some impact on all repair cost variables 2011 - alteration of read/write methods so as to incorporate the use of SQL. 2014 – public sector and private sector inflation factors were previously gathered from the BCIS website each year and represent an index of inflation for building works. The BCIS factors are based on a sample of construction industry jobs carried out throughout the year. These factors were not readily available from 2014, so projected factors based on the available BCIS indices from 2009 to 2013 were used. Regional inflation factors continued to be available from BCIS. Subsequent annual inflation factors for public, private sector and regions were updated using a projection from previous year.
Other relevant information	This variable does not take into account variations in the price of building work by tenure or region and can be used to compare the extent of disrepair in different groups of the stock. Costs are not, however indicative of the likely realistic costs. For anonymisation purposes, costs of 500 £/m² or more are capped at 500 £/m². for the End User Licence. This variable is available in the datasets provided under the End User Licence and Special Licence.
Previous variable	N/A

Cststdux**Physical 18+19.sav**

Variable name	Cststdux
Variable label	Urgent repair costs (per square metre)
Units	£/m ²
Description	Estimated costs of carry out all urgent repairs that the surveyor indicated needed to be carried out to the dwelling in £ per m² of floor area (see 'floorx'). Costs are at current prices (i.e. costs for the 2019 data set are at 2019 prices) and DO NOT take account of differences in building costs in the different tenures and regions. They do not include VAT or fees but do include additional costs such as scaffolding where this would be needed. 'Urgent repairs' means all repair work that needs doing in order to remove threats to health, safety, security and comfort to the occupants and to forestall further deterioration, plus all internal works and works to shared facilities and common areas. The costs do not include any necessary improvements or additional work required to address non-decency or to deal with Category 1 hazards.
Type	Numeric
Measure	Scale
Value labels – Special Licence	N/A – all amounts are coded in £/m ² . Where there are no repairs, this appears as 0 £/m ² .
Value labels – End User Licence	500.00 "£500 or more" All amounts are coded in £/m ² . Where there are no repairs, this appears as 0 £/m ² .
Missing data	None
File location	physical 18+19.sav
Validation	There is no other external data source that could be used to validate this variable. The DELPHI and SPSS code contains internal checks of input and interim variables. Consistency checks made with previous years overall and by key dwelling descriptor variables.
Key inputs	All variables relating to repairs needed in sections 5, 9, 18 and 21 of the physical survey form plus all outputs from dimensions model. Also data file with unit prices and inflation factors.
Modelling	Repair costs come from a suite of programs mainly written in DELPHI. For more detail see the section on modelled indicators, Chapter 5 of the English Housing Survey 2019-20 Technical Report, Annex 5.
Imputation	For more detail see the section on modelled indicators, Chapter 5 (Annex 5) of the English Housing Survey 2019-20 Technical Report.
Key dependencies	Some of the costs are used in modelling of costs to deal with HHSRS Category 1 hazards and costs to make decent.

Introduction date	1981
Date of key modelling changes	1986 – complete review of methodology 1996 – complete review of unit prices and overall structure of the model 2005 – changes to the dimensions model had some impact on all repair cost variables 2011 - alteration of read/write methods so as to incorporate the use of SQL. 2014 – public sector and private sector inflation factors were previously gathered from the BCIS website each year and represent an index of inflation for building works. The BCIS factors are based on a sample of construction industry jobs carried out throughout the year. These factors were not readily available from 2014, so projected factors based on the available BCIS indices from 2009 to 2013 were used. Regional inflation factors continued to be available from BCIS. Subsequent annual inflation factors for public, private sector and regions were updated using a projection from previous year.
Other relevant information	This variable does not take into account variations in the price of building work by tenure or region and can be used to compare the extent of disrepair in different groups of the stock. Costs are not, however indicative of the likely realistic costs. For anonymisation purposes, costs of 500 £/m² or more are capped at 500 £/m². for the End User Licence. This variable is available in the datasets provided under the End User Licence and Special Licence.
Previous variable	N/A

Dampalf

Physical 18+19.sav

Variable name	Dampalf
Variable label	Dampness problems in one or more rooms
Units	N/A
Description	Whether there are any problems with rising damp, penetrating damp or serious condensation/mould present in any of the main rooms surveyed in the physical survey (the main living room, kitchen, main bedroom, and bathroom or circulation space).
Type	Numeric
Measure	Nominal
Value labels	0 "not present" 1 "problem present"
Missing data	None
File location	physical 18+19.sav
Validation	Frequency checks and time series checks
Key inputs	'findfxrd', 'findfxpd', 'findfxmo', 'type'
Modelling	Based on a combination of the above variables
Imputation	None
Key dependencies	None
Introduction date	2009
Date of key modelling changes	None
Other relevant information	This variable is available in the datasets provided under the End User Licence and Special Licence.
Previous variable	N/A

Dampcdf

Physical 18+19.sav

Variable name	Dampcdf
Variable label	Serious condensation in one or more rooms
Units	N/A
Description	Whether there are any problems with serious condensation/mould present in any of the main rooms surveyed in the physical survey (the main living room, kitchen, main bedroom, and bathroom or circulation space).
Type	Numeric
Measure	Nominal
Value labels	0 "not present" 1 "problem present"
Missing data	None
File location	physical 18+19.sav
Validation	Frequency checks and time series checks
Key inputs	'Findfxmo', 'type'
Modelling	Based on a combination of the above variables
Imputation	None
Key dependencies	None
Introduction date	2009
Date of key modelling changes	None
Other relevant information	This variable is available in the datasets provided under the End User Licence and Special Licence.
Previous variable	N/A

Damppnf**Physical 18+19.sav**

Variable name	Damppnf
Variable label	Penetrating damp in one or more rooms
Units	N/A
Description	Whether there are any problems with penetrating damp present in any of the main rooms surveyed in the physical survey (the main living room, kitchen, main bedroom, bathroom or circulation space)
Type	Numeric
Measure	Nominal
Value labels	0 "not present" 1 "problem present"
Missing data	None
File location	physical 18+19.sav
Validation	Frequency checks and time series checks
Key inputs	'Findfxpd', 'type'
Modelling	Based on a combination of the above variables
Imputation	None
Key dependencies	None
Introduction date	2009
Date of key modelling changes	None
Other relevant information	This variable is available in the datasets provided under the End User Licence and Special Licence.
Previous variable	N/A

Damprsf

Physical 18+19.sav

Variable name	Damprsf
Variable label	Rising damp in one or more rooms
Units	N/A
Description	Whether there are any problems with rising damp present in any of the main rooms surveyed in the physical survey (the main living room, kitchen, main bedroom, and bathroom or circulation space).
Type	Numeric
Measure	Nominal
Value labels	0 "not present" 1 "problem present"
Missing data	None
File location	physical 18+19.sav
Validation	Frequency checks and time series checks
Key inputs	'Findfxrd', 'type'
Modelling	Based on a combination of the above variables
Imputation	None
Key dependencies	None
Introduction date	2009
Date of key modelling changes	None
Other relevant information	This variable is available in the datasets provided under the End User Licence and Special Licence.
Previous variable	N/A

Dblglaz2**Physical 18+19.sav**

Variable name	Dblglaz2
Variable label	Extent of double glazing
Units	N/A
Description	Proportion of windows in the dwelling that are double glazed – 2 categories.
Type	Numeric
Measure	Ordinal
Value labels	-8 “unknown” 1 “less than 80% double glazed” 2 “80% or more double glazed”
Missing data	None
File location	physical 18+19.sav
Validation	Frequency checks and time series checks
Key inputs	‘fexwn1no’, ‘fexwn2no’, ‘type’
Modelling	Based on number of double glazed windows as a proportion of total number of windows (summed over front and back view).
Imputation	None
Key dependencies	None
Introduction date	1991
Date of key modelling changes	None
Other relevant information	This variable is available in the datasets provided under the End User Licence and Special Licence.
Previous variable	N/A

Dblglaz4**Physical 18+19.sav**

Variable name	Dblglaz4
Variable label	Extent of double glazing
Units	N/A
Description	Proportion of windows in the dwelling that are double glazed – 4 categories.
Type	Numeric
Measure	Ordinal
Value labels	-8 “unknown” 1 “no double glazing” 2 “half or less” 3 “more than half” 4 “entire house”
Missing data	None
File location	physical 18+19.sav
Validation	Frequency checks and time series checks
Key inputs	‘fexwn1no’, ‘fexwn2no’, ‘type’
Modelling	Based on number of double glazed windows as a proportion of total number of windows (summed over front and back view).
Imputation	None
Key dependencies	None
Introduction date	1991
Date of key modelling changes	None
Other relevant information	This variable is available in the datasets provided under the End User Licence and Special Licence.
Previous variable	N/A

Dhcosty

Physical 18+19.sav

Variable name	Dhcosty
Variable label	Cost to make decent (15 hazard HHSRS model)
Units	£'s
Description	Total cost of all works required to ensure that homes failing the decent homes standard meet the standard. Costs are at current prices (2018 prices for the 2018 data and 2019 prices for the 2019 data).
Type	Numeric
Measure	Scale
Value labels – Special Licence	-9 “not applicable – decent home”
Value labels – End User Licence	-9 “not applicable – decent home” 50,000.00 “£50,000 or more”
Missing data	None
File location	physical 18+19.sav
Validation	Consistency check with dhomesy. Average costs and distributions are compared with those from previous years overall and by key variables such as tenure.
Key inputs	All Decent Homes variables, several variables from repair cost model and interim component cost to make safe variables
Modelling	A suite of SPSS files that firstly creates flags for each item of work needed relating to the components of decent homes removing any double counting. It then costs these, using prices from the repair cost model for disrepair and HHSRS, and major repairs allowance prices for modernisation and thermal comfort. Access and uplift costs are also applied. The total costs are then adjusted for tenure, region and year of survey. For more detail see the section on modelled indicators, Chapter 5 of the English Housing Survey 2019-20 Technical Report, Annex 5.
Imputation	Checks are made for missing and implausible data and averages based on stereotypes are used where required.
Key dependencies	None
Introduction date	2005
Date of key modelling changes	None
Other relevant information	This is based on the 15 hazards HHSRS version of Decent Homes to be consistent with time series back to 2005. This variable is available in the datasets provided under the End User Licence and Special Licence. For anonymisation purposes, costs of £50,000 or more are capped at £50,000 for the End User Licence.

Previous variable	'dhcost' (which was when the Decent Homes standard included the fitness standard rather than HHSRS)
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Dhdisrx**Physical 18+19.sav**

Variable name	Dhdisrx
Variable label	Decent homes repair criterion
Units	N/A
Description	Whether the dwelling passes or fails the disrepair component of decent homes.
Type	Numeric
Measure	Nominal
Value labels	0 "pass" 1 "fail"
Missing data	None
File location	physical 18+19.sav
Validation	Results are compared with those from previous years overall and by key variables such as tenure.
Key inputs	Age of all external elements, bathroom, kitchen, heating distribution, boiler and storage heater plus raw data on the type and scale of repairs needed to these.
Modelling	Applies the criteria laid down in the Decent Homes Standard. Dwellings fail this component where at least 1 major element or 2 minor elements are classed as past their expected life AND in need of major repair or replacement. For more detail see the section on modelled indicators, Chapter 5 of the English Housing Survey 2019-20 Technical Report, Annex 5.
Imputation	If element ages are unknown, they are assumed to be original (same age as dwelling).
Key dependencies	'dhomesy', 'dhomesz'
Introduction date	1991
Date of key modelling changes	None
Other relevant information	This variable is available in the datasets provided under the End User Licence and Special Licence.
Previous variable	N/A

Dhhhsrsx**Physical 18+19.sav**

Variable name	Dhhhsrsx
Variable label	Decent homes HHSRS (15 hazard) criterion
Units	N/A
Description	Whether the dwelling meets the statutory minimum standard for housing by being free of Category 1 HHSRS hazards. This variable uses the 15 hazards version of HHSRS.
Type	Numeric
Measure	Nominal
Value labels	0 "pass" 1 "fail"
Missing data	None
File location	physical 18+19.sav
Validation	Results are compared with those from previous years overall and by key variables such as tenure, dwelling age and dwelling type.
Key inputs	'hsrall15' (in HHSRS 18+19_SL.sav and available under the Special Licence only)
Modelling	Some of the 15 hazards are fully measured and some are flagged when there is an extreme risk of harm. Other hazards are modelled using other data from the physical survey. Excess cold is based on the SAP rating of the dwelling. For more detail see the section on modelled indicators, Chapter 5 of the English Housing Survey 2019-20 Technical Report, Annex 5.
Imputation	None
Key dependencies	'dhomesy'
Introduction date	2006
Date of key modelling changes	2008 – hazards of damp and mould growth and noise were fully measured by surveyors rather than modelled from other data on the form. The hazards relating to personal hygiene, domestic hygiene, electrical safety and carbon monoxide were rated by surveyors rather than modelled from other data. From 2012, HHSRS noise no longer fully measured but assessed as an extreme risk. 2013 – 2017 HHSRS excess cold based on previous SAP12 methodology. SAP09 methodology used 2010-2012 for excess cold. From 2018, HHSRS excess cold based on new SAP12 methodology.
Other relevant information	This variable is available in the datasets provided under the End User Licence and Special Licence.

Previous variable	'dhunfitx' (when the Decent Homes standard used the fitness standard instead of HHSRS)
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Dhhhsrsy**Physical 18+19.sav**

Variable name	Dhhhsrsy
Variable label	Decent homes HHSRS (26 hazard) criterion
Units	N/A
Description	Whether the dwelling meets the statutory minimum standard for housing by being free of Category 1 HHSRS hazards. This variable uses the 26 hazards version of HHSRS.
Type	Numeric
Measure	Nominal
Value labels	0 "pass" 1 "fail"
Missing data	None
File location	physical 18+19.sav
Validation	Results are compared with those from previous years overall and by key variables such as tenure, dwelling age and dwelling type.
Key inputs	'hsrall26' (in HHSRS 18+19_SL.sav and available under the Special Licence only)
Modelling	Some of the 26 hazards are fully measured and some are flagged when there is an extreme risk of harm. Other hazards are modelled using other data from the physical survey. Excess cold is based on the SAP rating of the dwelling. For more detail see the section on modelled indicators, Chapter 5 of the English Housing Survey 2019-20 Technical Report, Annex 5.
Imputation	None
Key dependencies	'dhomesz'
Introduction date	2006
Date of key modelling changes	From 2012, 4 previously fully measured hazards (falls associated with baths, noise, collision and entrapment, and entry by intruders) assessed as to whether an extreme risk. 2013 – 2017 HHSRS excess cold based on previous SAP12 methodology. SAP09 methodology used 2010-2012 for excess cold. From 2018, HHSRS excess cold based on new SAP12 methodology.
Other relevant information	This variable is available in the datasets provided under the End User Licence and Special Licence.
Previous variable	'dhunfitx' (when the Decent Homes standard used the fitness standard instead of HHSRS)

Dhmodx**Physical 18+19.sav**

Variable name	Dhmodx
Variable label	Decent homes modern facilities criterion
Units	N/A
Description	Whether the dwelling passes or fails the modernisation component of decent homes.
Type	Numeric
Measure	Nominal
Value labels	0 "pass" 1 "fail"
Missing data	None
File location	physical 18+19.sav
Validation	Results are compared with those from previous years overall and by key variables such as tenure, dwelling age and dwelling type.
Key inputs	'finkitlr', 'finbatlr', 'dwage10x', 'finkdfsp', 'finkdfla', 'finbadlo', 'finbdflo', 'finlooin', tight space in common areas, noise problems in local area (section 24 of the physical survey form).
Modelling	Computes whether the dwelling fails each of the 6 components (modern kitchen, modern bathroom, whether defects with kitchen size or layout, bathroom location, common areas arrangement (where applicable) and problems with noise (external). Follows the MHCLG guidance and only classes those that fail on 3 or more of these aspects as failing on modernisation. For more detail see the section on modelled indicators, Chapter 5 of the English Housing Survey 2019-20 Technical Report, Annex 5.
Imputation	If element ages are unknown, they are assumed to be original (same age as dwelling).
Key dependencies	'dhomesy', 'dhomesz'
Introduction date	1991
Date of key modelling changes	2007 - the rating of space and layout of kitchens was changed from a 4 point scale to a simple question about whether these were defective or not. This change was incorporated into the modelling and resulted in a slight 'blip' in the overall time trend. 2011 - the questions on noise in neighbourhood were removed from interview survey in 2011, and hence cannot be used in the modelling. To compensate for the effect this has on numbers failing on noise, the criteria was tightened up for modelling noise data from the physical survey (section 24) 2012 - question on situation of block removed in 2012 (previously section 20 of the physical survey) and hence can no longer be used for modelling noise. Criteria for modelling noise was adjusted to compensate for this.

Other relevant information	This variable is available in the datasets provided under the End User Licence and Special Licence.
Previous variable	N/A

Dhnumy

Physical 18+19.sav

Variable name	Dhnumy
Variable label	Decent homes: number of criteria failed (15 hazard HHSRS model)
Units	N/A
Description	Number of components of decent homes that the dwelling fails on (using Decent Homes based on HHSRS 15 hazard model).
Type	Numeric
Measure	Ordinal
Value labels	0 "none failed- decent" 1 "one" 2 "two" 3 "three" 4 "four"
Missing data	None
File location	physical 18+19.sav
Validation	Results are compared with those from previous years overall and by key variables such as tenure.
Key inputs	'dhdisrx', 'dhmodx', 'dhthermy', 'dhhhsrsx'
Modelling	None – counts the number of above variables coded as 'fail'.
Imputation	None
Key dependencies	None
Introduction date	2005
Date of key modelling changes	2013 – 2017 thermal comfort and HHSRS components of decent homes based on SAP12 methodology. SAP09 methodology used 2010-2012. 2018 – thermal comfort based on new SAP12 methodology
Other relevant information	2008 – introduced 'dhnumz', using the alternative version decent homes 26 hazard HHSRS model 'Dhnumy' is the preferred version used by MHCLG in reporting This variable is available in the datasets provided under the End User Licence and Special Licence.
Previous variable	'dhnumx' (which was based on unfitness instead of HHSRS)

Dhnumz

Physical 18+19.sav

Variable name	Dhnumz
Variable label	Decent homes: number of criteria failed (26 hazard HHSRS model)
Units	N/A
Description	Number of components of decent homes that the dwelling fails on (using the 26 hazard HHSRS model)
Type	Numeric
Measure	Ordinal
Value labels	0 "none failed - decent" 1 "one" 2 "two" 3 "three" 4 "four"
Missing data	None
File location	physical 18+19.sav
Validation	Results are compared with those from previous years overall and by key variables such as tenure.
Key inputs	'dhdisrx', 'dhmodx', 'dhthermy', 'dhhhsrsy'
Modelling	None – counts the number of above variables coded as 'fail'
Imputation	None
Key dependencies	None
Introduction date	2008
Date of key modelling changes	2013 – 2017 thermal comfort and HHSRS components of decent homes based on previous SAP12 methodology. SAP09 methodology was used 2010-2012. 2018 thermal comfort and HHSRS based on new SAP12 methodology.
Other relevant information	This variable is <u>not</u> the one used in MHCLG EHS reporting. The preferred variable is 'dhnumy' This variable is available in the datasets provided under the End User Licence and Special Licence.
Previous variable	This is an alternative version to the decent homes variable 'dhnumy' based on the 15 hazard HHSRS model.

Dhomesy

Physical 18+19.sav

Variable name	Dhomesy
Variable label	Decent homes - overall standard (15 hazard HHSRS model)
Units	N/A
Description	Whether dwelling fails the decent homes standard based on the 15 hazard HHSRS model.
Type	Numeric
Measure	Nominal
Value labels	0 "decent" 1 "non-decent"
Missing data	None
File location	physical 18+19.sav
Validation	Results are compared with those from previous years overall and by key variables e.g. tenure, dwelling age, dwelling type.
Key inputs	'dhdisrx', 'dhmodx', 'dhthermy', 'dhhhsrsx'
Modelling	A suite of SPSS files model disrepair and modernisation. These elements are then looked at with thermal comfort (from the energy model) and HHSRS (from the HHSRS 15 hazard model) and if a dwelling fails on any of these items, it fails decent homes.
Imputation	None
Key dependencies	'Dhcosty'
Introduction date	2005
Date of key modelling changes	2005 – new dimensions model introduced and thermal comfort based on SAP05 methodology in place of SAP01 methodology 2005 – some adjustments were made to thermal comfort for flats not on top floor of block. 2010 – thermal comfort and HHSRS based on SAP09 methodology and not SAP05 methodology as in previous years 2011 – dhmodx modelled slightly differently due to the questions on noise in neighbourhood being removed from interview survey in 2011 (see dhmodx for further details) 2012 – dhmodx modelling amended slightly due to further question on noise being removed from physical survey in 2012 (see dhmodx for further details) 2013 – 2017 thermal comfort and HHSRS based on original SAP12 methodology and not SAP09 methodology 2018- thermal comfort and HHSRS based on revised SAP12 methodology
Other relevant information	This variable is used for reporting on trends in Decent homes from 2006 onwards rather than the alternative 'dhomesz' variable.

	This variable is available in the datasets provided under the End User Licence and Special Licence.
Previous variable	'dhomesx' (which was based on unfitnes instead of HHSRS)

Dhomesz**Physical 18+19.sav**

Variable name	Dhomesz
Variable label	Decent homes - overall standard (26 hazard HHSRS model)
Units	N/A
Description	Whether dwelling fails the decent homes standard based on the 26 hazard HHSRS model
Type	Numeric
Measure	Nominal
Value labels	0 "decent" 1 "non-decent"
Missing data	None
File location	physical 18+19.sav
Validation	Results are compared with those from previous years overall and by key variables such as tenure, dwelling age and dwelling type.
Key inputs	'dhdisrx', 'dhmodx', 'dhthermy', 'dhhhsrsy'
Modelling	A suite of SPSS files model disrepair and modernisation. These elements are then looked at with thermal comfort (from the energy model) and HHSRS (from the HHSRS 26 hazard model) and if a dwelling fails on any of these items, it fails decent homes.
Imputation	None
Key dependencies	None
Introduction date	2008
Date of key modelling changes	2010 – thermal comfort and HHSRS based on SAP09 methodology and not SAP05 methodology as in previous years 2011 – dhmodx modelled slightly differently due to the questions on noise in neighbourhood being removed from interview survey in 2011(see dhmodx for further details) 2012 – dhmodx modelling amended slightly due to further question on noise being removed from physical survey in 2012 (see dhmodx for further details) 2013 – 2017 thermal comfort and HHSRS based on original SAP12 methodology and not SAP09 methodology 2018- thermal comfort and HHSRS based on revised SAP12 methodology
Other relevant information	This variable is <u>not</u> used for EHS reporting on time trends for Decent Homes. The preferred variable is 'dhomesy'. This variable is available in the datasets provided under the End User Licence and Special Licence.
Previous variable	This is an alternative version to the decent homes variable 'dhomesy'.

Dhreasny

Physical 18+19.sav

Variable name	Dhreasny
Variable label	Decent homes criterion not met (15 hazard HHSRS model)
Units	N/A
Description	Reason for failing decent homes (using the 15 hazard HHSRS model).
Type	Numeric
Measure	Nominal
Value labels	-9 "decent" 1 "repair, hhsrs or modern facilities" 2 "thermal comfort only"
Missing data	None
File location	physical 18+19.sav
Validation	Results are compared with those from previous years overall and by key variables such as tenure, dwelling age and dwelling type.
Key inputs	'dhdisrx', 'dhmodx', 'dhthermy', 'dhhhsrsx', 'dhnumy'
Modelling	None – based on the above derived variables
Imputation	None
Key dependencies	None
Introduction date	2005
Date of key modelling changes	2008 – introduced 'dhreasnz', using the alternative version decent homes 26 hazard HHSRS model 2010 – thermal comfort and HHSRS components of decent homes based on SAP09 methodology and not SAP05 methodology as in previous years 2011 – dhmodx modelled slightly differently due to the questions on noise in neighbourhood being removed from interview survey in 2011(see dhmodx for further details) 2012 – dhmodx modelling amended slightly due to further question on noise being removed from physical survey in 2012 (see dhmodx for further details) 2013 – 2017 thermal comfort and HHSRS based on original SAP12 methodology and not SAP09 methodology 2018 - thermal comfort and HHSRS based on revised SAP12 methodology
Other relevant information	This variable is used in EHS reporting as it uses the 15 HHSRS hazard model which enables time comparisons. This variable is available in the datasets provided under the End User Licence and Special Licence.
Previous variable	'dhreasnx' (which was based on unfitness instead of HHSRS)

Dhreasnz**Physical 18+19.sav**

Variable name	Dhreasnz
Variable label	Decent homes criterion not met (26 hazard HHSRS model)
Units	N/A
Description	Reason for failing decent homes (using the 26 hazard HHSRS model).
Type	Numeric
Measure	Nominal
Value labels	-9 "decent" 1 "repair, hhsrs or modern facilities" 2 "thermal comfort only"
Missing data	None
File location	physical 18+19.sav
Validation	Results are compared with those from previous years overall and by key variables such as tenure, dwelling age and dwelling type.
Key inputs	'dhdisrx', 'dhmodx', 'dhthermy', 'dhhhsrsy', 'dhnumz'
Modelling	None – based on the above derived variables.
Imputation	None
Key dependencies	None
Introduction date	2008
Date of key modelling changes	2010 – thermal comfort and HHSRS based on SAP09 methodology and not SAP05 methodology as in previous years 2011 – dhmodx modelled slightly differently due to the questions on noise in neighbourhood being removed from interview survey in 2011(see dhmodx for further details) 2012 – dhmodx modelling amended slightly due to further question on noise being removed from physical survey in 2012 (see dhmodx for further details) 2013 – 2017 thermal comfort and HHSRS based on original SAP12 methodology (previously used SAP09 methodology 2010-2012) 2018 - thermal comfort and HHSRS based on revised SAP12 methodology
Other relevant information	This variable is <u>not</u> used for EHS reporting on trends in Decent Homes over time. The preferred variable is 'dhreasny'. This variable is available in the datasets provided under the End User Licence and Special Licence.
Previous variable	This is an alternative version to the decent homes variable 'dhreasny' based on the 15 hazard HHSRS model.

Dhtcacty**Physical 18+19.sav**

Variable name	Dhtcacty
Variable label	Requirement to pass decent homes thermal comfort criterion
Units	N/A
Description	What types of work are required to ensure that the dwelling passes the decent homes thermal comfort component.
Type	Numeric
Measure	Nominal
Value labels	-9 "meets thermal comfort criterion" 1 "gas central heating" 2 "storage heating" 3 "cavity wall insulation" 4 "50mm loft insulation" 5 "gas central heating & cavity wall insulation" 6 "gas central heating & 50mm loft insulation" 7 "200mm loft insulation" 8 "cavity ins & 200mm loft ins" 9 "cavity insulation and storage heating" 10 "200mm loft insulation & storage heating" 11 "cavity insulation & 200mm loft insulation & storage heating" 12 "cavity insulation & 50mm loft insulation & gas central htg"
Missing data	None
File location	physical 18+19.sav
Validation	Results are compared with those from previous years overall and by key variables such as tenure, dwelling age and dwelling type.
Key inputs	'dhthermy' and variables relating to heating type, whether wall insulation and thickness of loft insulation
Modelling	This variable originates from the energy model.
Imputation	Unknown values for heating, wall and loft insulation are imputed using a default stereotype database, providing a generic value derived from cases of a similar dwelling type, age and size.
Key dependencies	None
Introduction date	2005
Date of key modelling changes	2005 – new dimensions model introduced and thermal comfort based on SAP05 methodology in place of SAP01 methodology 2005 – some adjustments were made to thermal comfort for flats not on top floor of block 2013 – 2017 thermal comfort and HHSRS based on original SAP12 methodology (previously used SAP09 methodology 2010-2012)

	2018 - thermal comfort and HHSRS based on revised SAP12 methodology
Other relevant information	This variable is only available in the dataset provided under the Special Licence.
Previous variable	'dhtcactx' (which was based on unfitness instead of HHSRS). A pre 2005 version "dhtcact" was based on unfitness and used an older dimension model.

Dhtcreasy**Physical 18+19.sav**

Variable name	Dhtcreasy
Variable label	Reason for failing decent homes on thermal comfort?
Units	N/A
Description	Reason for failing the thermal comfort component of decent homes – summary.
Type	Numeric
Measure	Nominal
Value labels	-9 “meets thermal comfort criterion” 1 “failed on heating measures only” 2 “failed on heating and insulation measures” 3 “failed on insulation measures only”
Missing data	None
File location	physical 18+19.sav
Validation	Results are compared with those from previous years overall and by key variables such as tenure, dwelling age and dwelling type.
Key inputs	‘dhthermy’ and data relating to heating type, whether wall insulation and thickness of loft insulation
Modelling	This variable originates from the energy model.
Imputation	Unknown values for heating, wall and loft insulation are imputed using a default stereotype database, providing a generic value derived from cases of a similar dwelling type, age and size.
Key dependencies	None
Introduction date	2005
Date of key modelling changes	2005 – new dimensions model introduced and thermal comfort based on SAP05 methodology in place of SAP01 methodology Some adjustments were made to thermal comfort for flats not on top floor of block. 2010 – thermal comfort and HHSRS components of decent homes based on SAP09 methodology and not SAP05 methodology as in previous years 2013 – 2017 thermal comfort and HHSRS based on original SAP12 methodology (previously used SAP09 methodology 2010-2012) 2018 - thermal comfort and HHSRS based on revised SAP12 methodology
Other relevant information	This variable is available in the datasets provided under the End User Licence and Special Licence.
Previous variable	‘dhtcreasx’ (which was based on unfitness instead of HHSRS). A pre 2005 version dhtcreas was based on unfitness and used an older dimension model

Dhthermy

Physical 18+19.sav

Variable name	Dhthermy
Variable label	Decent homes thermal comfort criterion
Units	N/A
Description	Whether the dwelling passes or fails the thermal comfort component of decent homes.
Type	Numeric
Measure	Nominal
Value labels	0 "pass" 1 "fail"
Missing data	None
File location	physical 18+19.sav
Validation	Results are compared with those from previous years overall and by key variables such as tenure, dwelling age and dwelling type.
Key inputs	Raw data on heating type, whether wall insulation and thickness of loft insulation
Modelling	This variable originates from the energy model. It is modelled according to the MHCLG definition of the standard. Any flats that are not on the top floor of the block and have cavity walls are set as passing thermal comfort if they have gas or oil central heating. For more detail see the section on modelled indicators, Chapter 5 of the English Housing Survey 2019-20 Technical Report, Annex 5.
Imputation	Unknown values for heating, wall and loft insulation are imputed using a default stereotype database, providing a generic value derived from cases of a similar dwelling type, age and size.
Key dependencies	'dhomesy', 'dhomesz', 'dhnumy', 'dhnumz', 'dhreasny', 'dhreasnz', 'dhtcacty', 'dhtcreasy'
Introduction date	2006
Date of key modelling changes	2005 – thermal comfort based on SAP05 methodology in place of SAP01 methodology. Previous version ('dhthermx') did not set flats that were not on the top floor of the block with cavity walls to pass thermal comfort if they had gas or oil central heating 2010 – thermal comfort based on SAP09 methodology and not SAP05 methodology as in previous years 2013 – 2017 thermal comfort and HHSRS based on original SAP12 methodology (previously used SAP09 methodology 2010-2012) 2018 - thermal comfort and HHSRS based on revised SAP12 methodology

Other relevant information	From 2014, the physical survey records the % of Cavity Wall with insulation (CWI), estimated by the surveyor. This may increase the number of properties defined as having CWI. This variable is available in the datasets provided under the End User Licence and Special Licence.
Previous variable	'Dhthermx'

Dwage4x**Physical 18+19.sav**

Variable name	Dwage4x
Variable label	Dwelling age
Units	N/A
Description	Date of construction of the dwelling (4 bands).
Type	Numeric
Measure	Ordinal
Value labels	-8 "unknown" 1 "pre 1919" 2 "1919 to 1944" 3 "1945 to 1964" 4 "post 1964"
Missing data	None
File location	physical 18+19.sav
Validation	Checks with results from previous years overall and by key variables such as dwelling type and tenure.
Key inputs	'fodconst'
Modelling	None – recodes the above raw physical survey variable.
Imputation	None
Key dependencies	Most physical modelled variables – dimensions, repair costs, HHSRS, Decent Homes and SAP etc.
Introduction date	1991
Date of key modelling changes	None
Other relevant information	The date of construction relates to the oldest part of the dwelling. This variable is available in the datasets provided under the End User Licence and Special Licence.
Previous variable	N/A

Dwage5x**Physical 18+19.sav**

Variable name	Dwage5x
Variable label	Dwelling age
Units	N/A
Description	Date of construction of the dwelling (5 bands).
Type	Numeric
Measure	Ordinal
Value labels	-8 "unknown" 1 "pre 1919" 2 "1919 to 1944" 3 "1945 to 1964" 4 "1965 to 1980" 5 "post 1980"
Missing data	None
File location	physical 18+19.sav
Validation	Checks with results from previous years overall and by key variables such as dwelling type and tenure.
Key inputs	'fodconst'
Modelling	None – recodes the above raw physical survey variable.
Imputation	None
Key dependencies	Most physical modelled variables – dimensions, repair costs, HHSRS, Decent Homes and SAP etc.
Introduction date	1991
Date of key modelling changes	None
Other relevant information	The date of construction relates to the oldest part of the dwelling. This variable is available in the datasets provided under the End User Licence and Special Licence.
Previous variable	N/A

Dwage6x**Physical 18+19.sav**

Variable name	Dwage6x
Variable label	Dwelling age
Units	N/A
Description	Date of construction of the dwelling (6 bands).
Type	Numeric
Measure	Ordinal
Value labels	-8 "unknown" 1 "pre 1919" 2 "1919 to 1944" 3 "1945 to 1964" 4 "1965 to 1980" 5 "1981 to 1990" 6 "post 1990"
Missing data	None
File location	physical 18+19.sav
Validation	Checks with results from previous years overall and by key variables such as dwelling type and tenure.
Key inputs	'dwage10x'
Modelling	None – recodes the above derived variable.
Imputation	None
Key dependencies	Most physical modelled variables – dimensions, repair costs, HHSRS, Decent Homes and SAP, 'constx' etc.
Introduction date	2001
Date of key modelling changes	None
Other relevant information	The date of construction relates to the oldest part of the dwelling. This variable is available in the datasets provided under the End User Licence and Special Licence.
Previous variable	N/A

Dwage7x**Physical 18+19.sav**

Variable name	Dwage7x
Variable label	Dwelling age
Units	N/A
Description	Date of construction of the dwelling (7 bands).
Type	Numeric
Measure	Ordinal
Value labels	-8 "unknown" 1 "pre 1919" 2 "1919 to 1944" 3 "1945 to 1964" 4 "1965 to 1980" 5 "1981 to 1990" 6 "1991 to 2002" 7 "post 2002"
Missing data	None
File location	physical 18+19.sav
Validation	Checks with results from previous years overall and by key variables such as dwelling type and tenure.
Key inputs	'dwage10x'
Modelling	None – recodes the above derived variable.
Imputation	None
Key dependencies	Most physical modelled variables – dimensions, repair costs, HHSRS, Decent Homes and SAP, 'constx' etc.
Introduction date	2015
Date of key modelling changes	None
Other relevant information	The date of construction relates to the oldest part of the dwelling. This variable is available in the datasets provided under the End User Licence and Special Licence.
Previous variable	N/A

Dwage10x**Physical 18+19.sav**

Variable name	Dwage10x
Variable label	Dwelling age
Units	N/A
Description	Date of construction of the dwelling (9 bands).
Type	Numeric
Measure	Ordinal
Value labels	-8 "unknown" 1 "pre 1850" 2 "1850 to 1899" 3 "1900 to 1918" 4 "1919 to 1944" 5 "1945 to 1964" 6 "1965 to 1974" 7 "1975 to 1980" 8 "1981 to 1990" 9 "1991 to 2002" 10 "post 2002"
Missing data	None
File location	physical 18+19.sav
Validation	Checks with results from previous years overall and by key variables such as dwelling type and tenure.
Key inputs	'fodconst'
Modelling	None – recodes the above raw physical survey variable.
Imputation	None
Key dependencies	Most physical modelled variables – dimensions, repair costs, HHSRS, Decent Homes and SAP and dwage6x etc.
Introduction date	2015
Date of key modelling changes	None
Other relevant info	The date of construction relates to the oldest part of the dwelling. This variable is only available in the dataset provided under the Special Licence.
Previous variable	Dwage9x

Dwtype3x**Physical 18+19.sav**

Variable name	Dwtype3x
Variable label	Dwelling type
Units	N/A
Description	Dwelling type (3 categories).
Type	Numeric
Measure	Nominal
Value labels	-8 "unknown" 1 "house or bungalow" 2 "low rise flat" 3 "high rise flat"
Missing data	None
File location	physical 18+19.sav
Validation	Plausibility checks based on dwelling type and comparison with results from previous years.
Key inputs	'dwtype8x'
Modelling	None – recode of above derived variable.
Imputation	None
Key dependencies	All key variables relating to dimensions, repair costs, SAP and energy related variables, HHSRS, Decent Homes and 'housex'.
Introduction date	1991
Date of key modelling changes	2015 - The derivation was amended to place a small number of converted flats with 6 or more storeys in the 'high-rise flat' category. Previously, these were 'low-rise' by default.
Other relevant information	This variable is available in the datasets provided under the End User Licence and Special Licence.
Previous variable	N/A

Dwtype7x**Physical 18+19.sav**

Variable name	Dwtype7x
Variable label	Dwelling type
Units	N/A
Description	Dwelling type (7 categories).
Type	Numeric
Measure	Nominal
Value labels	-8 "unknown" 1 "end terrace" 2 "mid terrace" 3 "semi detached" 4 "detached" 5 "purpose built" 6 "converted" 7 "non residential"
Missing data	None
File location	physical 18+19.sav
Validation	Plausibility checks based on dwelling type and comparison with results from previous years.
Key inputs	'foddtype'
Modelling	None – based on the above raw physical survey variable.
Imputation	None
Key dependencies	All key variables relating to dimensions, repair costs, SAP and energy related variables, HHSRS, Decent Homes and 'dwtype8x', 'dwtypenx'.
Introduction date	1991
Date of key modelling changes	None
Other relevant information	This variable is only available in the dataset provided under the Special Licence.
Previous variable	N/A

Dwtype8x**Physical 18+19.sav**

Variable name	Dwtype8x
Variable label	Dwelling type
Units	N/A
Description	Dwelling type (8 categories).
Type	Numeric
Measure	Nominal
Value labels	-8 "unknown" 1 "small terraced house" 2 "medium/large terraced house" 3 "semi-detached house" 4 "detached house" 5 "bungalow" 6 "converted flat" 7 "purpose built flat, low rise" 8 "purpose built flat, high rise"
Missing data	None
File location	physical 18+19.sav
Validation	Plausibility checks based on dwelling type and comparison with results from previous years.
Key inputs	'dwtype7x', 'storeyx', 'floorx'
Modelling	None – based on the above derived variables.
Imputation	None
Key dependencies	All key variables relating to dimensions, repair costs, SAP and energy related variables, HHSRS, Decent Homes and 'dwtype3x', 'dwtypenx'.
Introduction date	1991
Date of key modelling changes	None
Other relevant information	This variable is available in the datasets provided under the End User Licence and Special Licence.
Previous variable	N/A

Dwtypenx**Physical 18+19.sav**

Variable name	Dwtypenx
Variable label	Dwelling type
Units	N/A
Description	Dwelling type (8 categories).
Type	Numeric
Measure	Nominal
Value labels	-8 "unknown" 1 "end terrace" 2 "mid terrace" 3 "semi detached" 4 "detached" 5 "bungalow" 6 "converted flat" 7 "purpose built flat, low rise" 8 "purpose built flat, high rise"
Missing data	None
File location	physical 18+19.sav
Validation	Plausibility checks based on dwelling type and comparison with results from previous years.
Key inputs	'foddtype', 'storeyx'
Modelling	None – based on the above derived variables.
Imputation	None
Key dependencies	All key variables relating to dimensions, repair costs, SAP and energy related variables, HHSRS and Decent Homes.
Introduction date	1991
Date of key modelling changes	None
Other relevant information	This variable is available in the datasets provided under the End User Licence and Special Licence.
Previous variable	N/A

EPceeb12e**Physical 18+19.sav**

Variable name	EPceeb12e
Variable label	Energy efficiency rating band (SAP 2012)
Units	N/A
Description	The energy efficiency rating of the dwelling - groups from A to G that are used in the Energy Performance Certificate. 'A' is the most efficient and 'G' is the least efficient. Based on the most recent SAP 2012 methodology.
Type	Numeric
Measure	Ordinal
Value labels – Special Licence	1 "A" 2 "B" 3 "C" 4 "D" 5 "E" 6 "F" 7 "G"
Value labels - End User Licence	2 "A/B" 3 "C" 4 "D" 5 "E" 6 "F" 7 "G"
Missing data	None
File location	physical 18+19.sav
Validation	Comparisons with results from previous years overall and by key dwelling characteristics.
Key inputs	'sap12'
Modelling	For more detail see the section on modelled indicators, Chapter 5 of the English Housing Survey 2019-20 Technical Report Annex 6.
Imputation	N/A
Key dependencies	N/A
Introduction date	2013, however all earlier data as far back as 1996 has been remodelled using the current method (previous versions were 'EPceeb09e' and 'EPceeb05e'))
Date of key modelling changes	2010 – SAP 2009 methodology adopted. Previous data recalculated using this method. 2013 – SAP 2012 methodology adopted. Previous data recalculated using this method. 2019 – Updated RdSAP methodology to v9.93. See SAP12.

Other relevant information	For anonymisation purposes, A and B rated properties have been grouped for the End User Licence. As A rated properties become more prevalent in the stock, the separation of A and B rated dwellings will be reviewed. This variable is available in the datasets provided under the End User Licence and Special Licence.
Previous variable	'EPceeb09e' – based on the SAP09 methodology 'EPceeb05e' – based on the SAP05 methodology

EPceib12e**Physical 18+19.sav**

Variable name	EPceib12e
Variable label	Environmental impact rating band (SAP 2012)
Units	N/A
Description	The environmental impact rating of the dwelling - groups from A to G used in the Energy Performance Certificate. 'A' is the best with the lowest CO ₂ emissions and 'G' is the worst with the highest CO ₂ emissions. Based on the most recent SAP 2012 methodology.
Type	Numeric
Measure	Ordinal
Value labels – Special Licence	1 "A" 2 "B" 3 "C" 4 "D" 5 "E" 6 "F" 7 "G"
Value labels - End User Licence	2 "A/B" 3 "C" 4 "D" 5 "E" 6 "F" 7 "G"
Missing data	None
File location	physical 18+19.sav
Validation	Comparisons with results from previous years overall and by key dwelling characteristics.
Key inputs	'EPceir12e'
Modelling	For more detail see the section on modelled indicators, Chapter 5 of the English Housing Survey 2019-20 Technical Report Annex 6.
Imputation	None
Key dependencies	None
Introduction date	2013 (originally included from 2006 as EPceib05e and from 2010 as EPceib09e – see dates of modelling changes)
Date of key modelling changes	2010 – SAP 2009 methodology adopted. 2013 – SAP 2012 methodology adopted. 2019 – Updated RdSAP methodology to v9.93. See SAP12.
Other relevant information	For anonymisation purposes, A and B rated properties have been grouped for the End User Licence. As A rated properties become more prevalent in the stock, the separation of A and B rated dwellings will be reviewed.

	This variable is available in the datasets provided under the End User Licence and Special Licence.
Previous variable	'EPceib09e' – based on SAP09 methodology 'EPceib05e' – based on SAP05 methodology

EPceir12e**Physical 18+19.sav**

Variable name	EPceir12e
Variable label	Environmental impact rating (SAP 2012)
Units	N/A
Description	The environmental impact rating (based on the modelled CO ₂ emissions) of the dwelling on a normalised scale from 1 to 100, where 1 is the worst/has the highest CO ₂ emissions. Based on the most recent SAP 2012 methodology.
Type	Numeric
Measure	Scale
Value labels - Special Licence	N/A
Value labels - End User Licence	91.00 "91 or more"
Missing data	None
File location	physical 18+19.sav
Validation	Extensive validation throughout the modelling process at key stages of the calculation. Comparisons with previous data across key dwelling characteristics and checks against manual calculations using the SAP methodology.
Key inputs	Raw physical data relating to heating and services, insulation information, built form and age, construction materials and derived dimensions variables.
Modelling	Created by the energy model. It follows the SAP 2012 methodology for calculating domestic CO ₂ emissions per m ² as closely as the EHS data allows, with imputed values or assumptions based on the type of dwelling used where data is missing or not collected in the EHS. A logarithmic scale converts the CO ₂ emissions into a normalised rating. For more detail see the section on modelled indicators, Chapter 5 of the English Housing Survey 2019-20 Technical Report Annex 6.
Imputation	Missing dimensions data or data not collected in the EHS are imputed using a default stereotype database, providing a generic value derived from cases of a similar dwelling type, age and size. Other missing inputs, e.g. boiler efficiency are given a conservative estimate for that dwelling type.
Key dependencies	'EPceib12e'
Introduction date	2013 (originally included from 2006 as EPceib05e and from 2010 as EPceib09e – see dates of modelling changes)
Date of key modelling changes	2010 – SAP 2009 methodology adopted. 2013 – SAP 2012 methodology adopted. 2019 – Updated RdSAP methodology to v9.93. See SAP12.
Other relevant information	For anonymisation purposes, ratings of 91 or above are capped at 91 or more for the End User Licence. This variable is available in the datasets provided under the End User Licence and Special Licence.

Previous variable	'EPceir09e' – based on SAP09 methodology. 'EPceir05e' – based on SAP05 methodology.
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Floor5x**Physical 18+19.sav**

Variable name	Floor5x
Variable label	Usable floor area - original EHS definition
Units	m ²
Description	Total usable floor area of the dwelling using the original EHS definition of floor area (banded). See 'floorx' for more detail.
Type	Numeric
Measure	Ordinal
Value labels	-8 "unknown" 1 "less than 50 sqm" 2 "50 to 69 sqm" 3 "70 to 89 sqm" 4 "90 to 109 sqm" 5 "110 sqm or more"
Missing data	None
File location	physical 18+19.sav
Validation	There is no other external data source that could be used to validate this variable. The DELPHI and SPSS code contains internal checks of input variables. Also outlier cases are examined individually, and findings checked for consistency with previous years overall and by key dwelling descriptor variables.
Key inputs	'floorx'
Modelling	None – recodes the derived variable 'floorx'.
Imputation	None
Key dependencies	None
Introduction date	1981
Date of key modelling changes	1986 – complete review of methodology 1996 – conversion of model from Fortran to Pascal 2005 – re-write of model to improve efficiency and allow more detailed outputs 2011 alteration of read/write methods so as to incorporate the use of SQL.
Other relevant information	See EHS Technical Report, 2019-20, Chapter 5, Annex 7 for more details on 'floorx' and dimensions modelling. This variable is available in the datasets provided under the End User Licence and the Special Licence.
Previous variable	N/A

Floor5y**Physical 18+19.sav**

Variable name	Floor5y
Variable label	Usable floor area - building regulations definition
Units	m ²
Description	Total usable floor area of the dwelling using the Building Regulations definition of floor area (banded). See 'floory' for more detail.
Type	Numeric
Measure	Ordinal
Value labels	-8 "unknown" 1 "less than 50 sqm" 2 "50 to 69 sqm" 3 "70 to 89 sqm" 4 "90 to 109 sqm" 5 "110 sqm or more"
Missing data	None
File location	physical 18+19.sav
Validation	There is no other external data source that could be used to validate this variable. The DELPHI and SPSS code contains internal checks of input variables. Also outlier cases are examined individually, and findings checked for consistency with previous years overall and by key dwelling descriptor variables.
Key inputs	'floory'
Modelling	None – recodes the derived variable 'floory'.
Imputation	None
Key dependencies	None
Introduction date	2013
Date of key modelling changes	N/A.
Other relevant information	Introduced as an alternative measure to 'floor5x' to reflect the review of Building Regulations See EHS Technical Report, 2019-20, Chapter 5, Annex 7 for more details on 'floory' and dimensions modelling. This variable is available in the datasets provided under the End User Licence and the Special Licence.
Previous variable	N/A.

Floorx

Physical 18+19.sav

Variable name	Floorx
Variable label	Usable floor area (sqm) - original EHS definition
Units	m ²
Description	The usable floor area supplied in the EHS data represents such floor space as could be reasonably used for habitation. It therefore represents all area within the footprint of the dwelling, minus the three following areas • The area under the external walls • The area under internal walls • The area occupied by staircases. The area left should thus represent the total of all room areas, hallways and circulation space (other than stairs) including cupboards, integral balconies and integral garages. Loft space is not included unless the loft is habitable, with a fixed stair in place to access it.
Type	Numeric
Measure	Scale
Value labels – Special Licence	N/A – all amounts are coded in m ² .
Values labels –End User Licence	500.00 “500 or more” All amounts are coded in m ² .
Missing data	None
File location	physical 18+19.sav
Validation	There is no other external data source that could be used to validate this variable. The DELPHI and SPSS code contains internal checks of input variables. Also we examine outlier cases by hand, and check consistency with previous years overall and by key dwelling descriptor variables.
Key inputs	All raw data relating to dimensions in sections 8 and 13 of the physical survey form plus measured room area, number of rooms and their floor levels presence of attic or basement and type of roof.
Modelling	The dimensions are converted from external to internal by subtracting the appropriate wall thickness from the width and depth measurements of the main and additional part. Exposed walls are considered to have a thickness equal to the AvWallThick variable (in dimensions 18+19_SL.sav) and party walls are considered to have a thickness of 0.125m. The floor area excluding external walls is calculated by multiplying the derived internal width and depth dimensions for each storey. Where a storey has an additional part, the area of that additional part is also calculated and added to the floor area of the main part. In the event that a dwelling has more than three storeys the surveyor will only be able to record the widths and depths of three of the storeys. The areas

	of the remaining storeys are assumed to be the same as those of the nearest measured storey. Floor areas for each storey are then summed. In order to calculate the area lost to internal walls, it is necessary to work out the perimeter of each room. The EHS does not measure the dimensions of all rooms, so the following approach is adopted. The number of rooms on each level is used to estimate the amount of circulation space needed. The total area of the level is reduced by the area needed for circulation, plus the area of any rooms that have been measured. The remaining space is then divided among the remaining rooms in order to work out the floor area of each room. Once the total area of all rooms has been calculated, the perimeter of each room is determined using the assumption that each room will be a rectangle with a standardised width to length ratio. The total perimeter length of all rooms are then added together, the perimeter of the external walls is subtracted, and divided by two to get the length of internal wall, and this is multiplied by an assumed internal wall thickness of 0.1 metres to estimate the footprint area under the internal walls. This amount is then subtracted from floor area excluding walls The area off the stairs, also to be subtracted, is assumed to be 3m² per staircase. This means that the area beneath the lowest staircases is considered habitable, as it is likely to be usable for storage. For more details see the section on Household derived indicators, Chapter 5 of the English Housing Survey 2019-20 Technical Report Annex 7.
Imputation	None
Key dependencies	'dwtype8x', 'floor5x' and all derived variables relating to dimensions, repair costs and energy.
Introduction date	1981
Date of key modelling changes	1986 - complete review of methodology 1996 - conversion of model from Fortran to Pascal 2005 - re-write of model to improve efficiency and allow more detailed outputs. Also, following a review of dimensions, a number of changes took place in the derivation of floor area from the EHS 2005/06 data onwards, including; - 0.2m was used as the thickness of the wall in the calculation of external to internal dimensions (previously 0.3m was used), - integral garages and integral balconies were included in the calculation of usable floor area (previously these were excluded) and - exclusion of the area underneath internal partition walls (previously this area was included in the calculation of usable floor area). 2011 - alteration of read/write methods so as to incorporate the use of SQL. 2013 - newly collected information on wall thickness is used to calculate average wall thickness for the determination of internal dimensions.

	2018 - newly collected bathroom measurements were used to calculate bathroom dimensions as well as internal wall areas (previously bathrooms were assumed to be 4m ²).
Other relevant information	For anonymisation purposes, floor area is capped at 500m ² or more for the End User Licence. This variable is available in the datasets provided under the End User Licence and the Special Licence.
Previous variable	N/A

Floory

Physical 18+19.sav

Variable name	Floory
Variable label	Usable floor area (sqm) - building regulations definition
Units	m ²
Description	This definition of dwelling floor area is aligned with Building Regulations. It therefore represents the area within the footprint of the dwelling, minus the three following areas • The area under the external walls • The area of integral balconies • The area of integral garages The area left should thus represent the total of all room areas, hallways and circulation space including cupboards and stairs. The area under partition walls is also included. Loft space is not included unless the loft is habitable, with a fixed stair in place to access it.
Type	Numeric
Measure	Scale
Value labels – Special Licence	N/A – all amounts are coded in m ² .
Values labels – End User Licence	500.00 “500 or more” All amounts are coded in m ² .
Missing data	None
File location	physical 18+19.sav
Validation	There is no other external data source that could be used to validate this variable. The DELPHI and SPSS code contains internal checks of input variables. Also we examine outlier cases by hand, and check consistency with previous years overall and by key dwelling descriptor variables.
Key inputs	All raw data relating to dimensions in sections 8 and 13 of the physical survey form plus measured room area, number of rooms and their floor levels presence of attic or basement and type of roof.
Modelling	The dimensions are converted from external to internal by subtracting the appropriate wall thickness from the width and depth measurements of the main and additional part. Exposed walls are considered to have a thickness equal to the AvWallThick variable (in dimensions 18+19_SL.sav) and party walls are considered to have a thickness of 0.125m. The floor area excluding external walls is calculated by multiplying the derived internal width and depth dimensions for each storey. Where a storey has an additional part, the area of that additional part is also calculated and added to the floor area of the main part. In the event that a dwelling has more than three storeys the surveyor will only be able to record the widths and depths of three of the storeys. The areas of the remaining storeys are assumed to be the same as those

	of the nearest measured storey. Floor areas for each storey are then summed. If there is a balcony present, a notional 3m² is subtracted. If there is an integral garage, a notional 10m² is subtracted. For more details see the section on Household derived indicators, Chapter 5 of the English Housing Survey 2019-20 Technical Report Annex 7.
Imputation	None
Key dependencies	'dwtype8x', 'floor5y' and all derived variables relating to dimensions, repair costs and energy.
Introduction date	2013
Date of key modelling changes	N/A
Other relevant information	Introduced as an alternative measure to 'floorx' to reflect the review of Building Regulations For anonymisation purposes, floor area is capped at 500m² or more for the End User Licence. This variable is available in the datasets provided under the End User Licence and the Special Licence.
Previous variable	N/A

Fuelx

Physical 18+19.sav

Variable name	Fuelx
Variable label	Main fuel type
Units	N/A
Description	Type of fuel used for the main or primary space heating system (specified in 'heat7x', 'heat4x').
Type	Numeric
Measure	Nominal
Value labels	-8 "not identified - communal systems" 1 "gas fired system" 2 "oil fired system" 3 "solid fuel fired system" 4 "electrical system"
Missing data	None
File location	physical 18+19.sav
Validation	Frequency checks and consistency checks against time series.
Key inputs	'mainfuel'
Modelling	Created by banding the above derived variable. See 'mainfuel' for further details.
Imputation	See 'mainfuel'
Key dependencies	N/A
Introduction date	1991
Date of key modelling changes	None
Other relevant information	Communal systems are coded as missing values as the specific fuel cannot be identified in these cases. 'Gas fired systems' includes LPG and bottled gas. This variable is available in the datasets provided under the End User Licence and Special Licence.
Previous variable	N/A

Heat4x**Physical 18+19.sav**

Variable name	Heat4x
Variable label	Main heating system
Units	N/A
Description	Type of main or primary space heating system – 4 summary types.
Type	Numeric
Measure	Nominal
Value labels	1 “central heating” 2 “storage heater” 3 “fixed room heating” 4 “portable heating only”
Missing data	None
File location	physical 18+19.sav
Validation	Frequency checks and consistency checks against time series. Crosschecks against raw physical survey variables as part of modelling validation.
Key inputs	‘heat7x’
Modelling	Created by combining categories from ‘heat7x’. ‘Central heating’ contains boiler, warm air, communal, heat pumps and other systems.
Imputation	See ‘heat7x’
Key dependencies	None
Introduction date	1991
Date of key modelling changes	2002 – Separate variables were created for type of system, type of fuel and boiler type. Until then, there was one combined primary heating variable and a separate variable for programmable heating. Categories for other heating expanded.
Other relevant information	For more detail see the section on modelled indicators, Chapter 5 of the English Housing Survey 2019-20 Technical Report Annex 6. This variable is available in the datasets provided under the End User Licence and Special Licence.
Previous variable	N/A

Heat7x**Physical 18+19.sav**

Variable name	Heat7x
Variable label	Main heating system
Units	N/A
Description	Type of main or primary space heating system – 7 types.
Type	Numeric
Measure	Nominal
Value labels	1 “boiler system with radiators” 2 “storage radiators” 3 “warm air system” 4 “room heater” 5 “other systems” 6 “communal” 7 “portable heaters only”
Missing data	None
File location	physical 18+19.sav
Validation	Frequency checks and consistency checks against time series. Crosschecks against raw physical survey variables as part of modelling validation.
Key inputs	‘Finchtyp’, ‘finchbcd’, ‘finohtyp’
Modelling	Created by the energy model mainly using the primary heating group variable. This is checked against the heating appliance code. If no primary system has been identified the other heating category becomes the primary system, typically coded in the room heater category.
Imputation	Where no primary heating exists and the other heating is unknown, electric room heaters are assumed.
Key dependencies	‘heat4x’
Introduction date	1991
Date of key modelling changes	2002 – Separate variables were created for type of system, type of fuel and boiler type. Until then, there was one combined primary heating variable and a separate variable for programmable heating. Categories for other heating expanded.
Other relevant information	‘Other systems’ are typically electric ceiling or underfloor systems. For more detail see the section on modelled indicators, Chapter 5 of the English Housing Survey 2019-20 Technical Report Annex 6. This variable is available in the datasets provided under the End User Licence and Special Licence.
Previous variable	N/A

Heatsec

Physical 18+19.sav

Variable name	Heatsec
Variable label	Secondary heating type
Units	N/A
Description	Type of secondary space heating system – 8 types.
Type	Numeric
Measure	Nominal
Value labels	-9 “no secondary heating” 1 “gas fire - open flue” 2 “mains gas fire - higher efficiency” 3 “mains gas fire - live effect” 4 “mains gas fire - decorative” 5 “fixed electric heater (including single storage heater)” 6 “solid fuel open fire” 7 “solid fuel stove/space heater” 8 “portable heater”
Missing data	None
File location	physical 18+19.sav
Validation	Frequency checks and consistency checks. Crosschecks against raw physical survey variables as part of modelling validation.
Key inputs	‘Finchtyp’, ‘finchbcd’, ‘finohtyp’
Modelling	Created from raw services data, taking into account changes introduced during detailed energy modelling. If no primary system has been identified the other heating category becomes the primary system, typically coded in the room heater category.
Imputation	Where no primary heating exists, and the other heating is unknown, electric room heaters are assumed.
Key dependencies	‘heat4x’
Introduction date	2013
Date of key modelling changes	2002 – Separate variables were created for type of system, type of fuel and boiler type. Until then, there was one combined primary heating variable and a separate variable for programmable heating. Categories for other heating expanded.
Other relevant information	This variable is only available in the dataset provided under the Special Licence.
Previous variable	N/A

Housex

Physical 18+19.sav

Variable name	Housex
Variable label	Dwelling type
Units	N/A
Description	Whether the dwelling is a house or flat.
Type	Numeric
Measure	Nominal
Value labels	-8 "unknown" 1 "house or bungalow" 2 "flat"
Missing data	None
File location	physical 18+19.sav
Validation	Plausibility checks based on dwelling type.
Key inputs	'dwtype3x'
Modelling	None – recode of dwtype3x
Imputation	None
Key dependencies	Most physical modelled variables – dimensions, repair costs, HHSRS, Decent Homes and SAP etc.
Introduction date	1991
Date of key modelling changes	N/A
Other relevant information	This variable is available in the datasets provided under the End User Licence and Special Licence.
Previous variable	N/A

Loftins4

Physical 18+19.sav

Variable name	Loftins4
Variable label	Loft insulation thickness
Units	N/A
Description	Thickness of loft insulation – 4 bands.
Type	Numeric
Measure	Ordinal
Value labels	-9 “not applicable - no roof directly above” 1 “none” 2 “less than 100mm” 3 “100 up to 150mm” 4 “150mm or more”
Missing data	None
File location	physical 18+19.sav
Validation	Frequency checks and consistency checks against time series.
Key inputs	‘loftinsx’
Modelling	See ‘loftinsx’
Imputation	Values are imputed where the amount of insulation is unknown or the dwelling has a flat roof directly above.
Key dependencies	None
Introduction date	1991
Date of key modelling changes	see ‘loftinsx’
Other relevant information	Information on the amount of loft insulation is only collected for houses and top floor flats. Other dwelling types are coded as ‘not applicable – no roof directly above’. For more detail see the section on modelled indicators, Chapter 5 of the English Housing Survey 2019-20 Technical Report Annex 6. This variable is available in the datasets provided under the End User Licence and Special Licence.
Previous variable	N/A

Loftins6**Physical 18+19.sav**

Variable name	Loftins6
Variable label	Loft insulation thickness
Units	N/A
Description	Thickness of loft insulation – 6 bands.
Type	Numeric
Measure	Ordinal
Value labels	-9 “not applicable - no roof directly above” 1 “none” 2 “less than 50mm” 3 “50 up to 99mm” 4 “100 up to 149mm” 5 “150 up to 199mm” 6 “200mm or more”
Missing data	None
File location	physical 18+19.sav
Validation	Frequency checks and consistency checks against time series.
Key inputs	‘loftinsx’
Modelling	see ‘loftinsx’
Imputation	Values are imputed where the amount of insulation is unknown or the dwelling has a flat roof directly above.
Key dependencies	None
Introduction date	1991
Date of key modelling changes	see ‘loftinsx’
Other relevant information	Information on the amount of loft insulation is only collected for houses and top floor flats. Other dwelling types are coded as ‘not applicable – no roof directly above’. For more detail see the section on modelled indicators, Chapter 5 of the English Housing Survey 2019-20 Technical Report Annex 6. This variable is available in the datasets provided under the End User Licence and Special Licence.
Previous variable	N/A

Loftinsu**Physical 18+19.sav**

Variable name	Loftinsu
Variable label	Loft insulation thickness with unknowns
Units	Mm
Description	Thickness of loft insulation with cases with flat roofs directly above and unknown amounts of insulation coded as 'unknown'. In the related variable 'loftinsx' (and 'loftins4' and 'loftins6' that are derived from this) these cases all have a value imputed for thickness of loft insulation.
Type	Numeric
Measure	Scale
Value labels	-9 "not applicable – no roof directly above" -8 "flat roof or unknown" 0 "no insulation"
Missing data	Yes
File location	physical 18+19.sav
Validation	Crosschecks against raw physical survey variables as part of modelling validation.
Key inputs	'Finlopos', 'flitypes', 'fliinsul', 'flithick'
Modelling	Created by the energy model using the raw variable 'Flithick' but removing values that had been imputed for missing data in the creation of 'loftinsx'. Cross checked with other loft survey variables to ensure whether there is a roof directly above has been properly assigned.
Imputation	None
Key dependencies	'loftu4'
Introduction date	2010
Date of key modelling Changes	There have been a number of changes to the data collected and how it is processed. See 'loftinsx' for more details.
Other relevant information	For more detail see the section on modelled indicators, Chapter 5 of the English Housing Survey 2019-20 Technical Report Annex 6. This variable is available in the datasets provided under the End User Licence and Special Licence.
Previous variable	N/A

Loftinsx

Physical 18+19.sav

Variable name	Loftinsx
Variable label	Loft insulation thickness
Units	Mm
Description	Thickness of loft insulation – continuous. Values are imputed where the amount of insulation is unknown or the dwelling has a flat roof directly above.
Type	Numeric
Measure	Scale
Value labels	-9 “not applicable – no roof directly above” 0 “no insulation”
Missing data	None
File location	physical 18+19.sav
Validation	Frequency checks and consistency checks against time series. Crosschecks against raw physical survey variables as part of modelling validation.
Key inputs	‘Finlopos’, ‘flitypes’, ‘fliinsul’, ‘flithick’
Modelling	Created by the energy model using the raw variable ‘Flithick’. Cross checking with other loft survey variables to ensure the identification whether the dwelling has a roof directly above is valid.
Imputation	Unknown values in dwellings with a roof above are imputed using a default stereotype database, providing a generic value derived from cases of a similar dwelling type, age and size. However, if the dwelling has a flat roof or a room in the roof, unknown values are imputed in accordance to appendix S of the SAP 2009 information booklet.
Key dependencies	‘loftins4’, ‘loftins6’
Introduction date	1991
Date of key modelling changes	2002 – Physical survey question was asked for all houses and top floor flats and not just pre 1980 houses as in 2001 and earlier. Also additional bands were added to the ‘flithick’ question on the physical survey covering those with 25mm, 125mm and those over 150mm 2003 – A question on dwelling type was introduced and the question on type of loft was slightly changed to make it easier to identify dwellings without a roof directly above. 2010 – The default stereotype data base used for imputing missing data was updated. Assumptions associated with missing loft insulation values for flat roofs, or dwellings with a room in the roof were aligned with appendix S of the SAP 2012 information booklet. 2015 - Changes to the energy modelling assumptions used to create this variable. These include aligning imputed loft insulation thickness values with those used in RdSAP methodology. As imputed unknowns for 2014/15 loftinsx are

	not aligned with RdSAP, the 2014/15 and 2015/16 combined variable contains a combination of these modelling methods. From combined survey years 2015/16 and 2016/17 onwards, the 'loftinsx' variable is based solely on this new method.
Other relevant information	Information on the amount of loft insulation is only collected for houses and top floor flats. Other dwelling types are coded as 'not applicable – no roof directly above'. For more detail see the section on modelled indicators, Chapter 5 of the English Housing Survey 2019-20 Technical Report Annex 6. This variable is available in the datasets provided under the End User Licence and Special Licence.
Previous variable	N/A

Loftu4**Physical 18+19.sav**

Variable name	Loftu4
Variable label	Loft insulation thickness with unknowns
Units	N/A
Description	Thickness of loft insulation – banded. Cases with flat roofs directly above and unknown amounts of insulation are coded as 'unknown'. In the related variable 'loftins4' these cases all have a value imputed for thickness of loft insulation.
Type	Numeric
Measure	Ordinal
Value labels	-9 "not applicable – no roof directly above" -8 "flat roof or unknown" 1 "none" 2 "less than 100mm" 3 "100 up to 150mm" 4 "150mm or more"
Missing data	Yes
File location	physical 18+19.sav
Validation	Frequency checks and consistency checks against time series.
Key inputs	'loftinsu'
Modelling	Created by banding the continuous variable 'loftinsu'.
Imputation	None
Key dependencies	None
Introduction date	2010
Date of key modelling changes	There have been a number of changes to the data collected and how it is processed. See 'loftinsx' for more details.
Other relevant information	Information on the amount of loft insulation is only collected for houses and top floor flats. Other dwelling types are coded as 'not applicable – no roof directly above'. This variable is available in the datasets provided under the End User Licence and Special Licence.

Previous variable	N/A
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Lv1upkpx**Physical 18+19.sav**

Variable name	Lv1upkpx
Variable label	Poor quality environment - upkeep problems
Units	N/A
Description	Whether there are any significant problems of upkeep/misuse in the local environment as assessed by the surveyor.
Type	Numeric
Measure	Nominal
Value labels	0 "no" 1 "yes"
Missing data	None
File location	physical 18+19.sav
Validation	Frequency checks and comparison with results from previous years.
Key inputs	'farlittr', 'fargraff', 'farvanda', 'fargrdns', 'farbldgs', 'farparks', 'farexcre', 'farcond', 'farroads'.
Modelling	All of the above are coded on a 5-point scale where 1=no problems and 5=major problems. If any of the above variables are coded as 4 or 5, then 'lv1upkpx' = yes.
Imputation	None
Key dependencies	None
Introduction date	1991
Date of key modelling changes	None
Other relevant information	The variable was originally located in the general.sav file from 1991 to 2007. This variable is available in the datasets provided under the End User Licence and Special Licence.
Previous variable	None

Lv2trafx**Physical 18+19.sav**

Variable name	Lv2trafx
Variable label	Poor quality environment - traffic problems
Units	N/A
Description	Whether there are any significant problems associated with traffic (and other transport noise) in the local environment as assessed by the surveyor.
Type	Numeric
Measure	Nominal
Value labels	0 "no" 1 "yes"
Missing data	None
File location	physical 18+19.sav
Validation	Frequency checks and comparison with results from previous years.
Key inputs	'farrails', 'farmotor', 'farairqu', 'fartraff'.
Modelling	All of the above are coded on a 5-point scale where 1=no problems and 5=major problems. If any of the above variables are coded as 4 or 5, then 'lv2trafx' = yes.
Imputation	None
Key dependencies	None
Introduction date	1991
Date of key modelling changes	None
Other relevant information	The variable was originally located in the general.sav file from 1991 to 2007. This variable is available in the datasets provided under the End User Licence and Special Licence.
Previous variable	None

Lv3utilx**Physical 18+19.sav**

Variable name	Lv3utilx
Variable label	Poor quality environment - utilisation problems
Units	N/A
Description	Whether there are any significant problems associated with non-residential uses and/or abandoned dwellings in the local environment as assessed by the surveyor.
Type	Numeric
Measure	Nominal
Value labels	0 "no" 1 "yes"
Missing data	None
File location	physical 18+19.sav
Validation	Frequency checks and comparison with results from previous years.
Key inputs	'farsites', 'farindus', 'farnocon', 'farvacnt'.
Modelling	All of the above are coded on a 5-point scale where 1=no problems and 5=major problems. If any of the above variables are coded as 4 or 5, then 'lv3utilx' = yes.
Imputation	None
Key dependencies	None
Introduction date	1991
Date of key modelling changes	None
Other relevant information	The variable was originally located in the general.sav file from 1991 to 2007. This variable is available in the datasets provided under the End User Licence and Special Licence.
Previous variable	None

Lvanyx**Physical 18+19.sav**

Variable name	Lvanyx
Variable label	Poor quality environment
Units	N/A
Description	Whether there are any significant problems in the local environment from the 3 categories: upkeep, traffic and utilisation.
Type	Numeric
Measure	Nominal
Value labels	0 "no" 1 "yes"
Missing data	None
File location	physical 18+19.sav
Validation	Frequency checks and comparison with results from previous years.
Key inputs	'lv1upkpx', 'lv2trafx', 'lv3utilx'
Modelling	If any of the above variables are coded as 'yes', then 'lvanyx'=yes.
Imputation	None
Key dependencies	None
Introduction date	1991
Date of key modelling changes	None
Other relevant information	The variable was originally located in the general.sav file from 1991 to 2007. This variable is available in the datasets provided under the End User Licence and Special Licence.
Previous variable	None

Lvnumx**Physical 18+19.sav**

Variable name	Lvnumx
Variable label	Number of liveability problems present
Units	N/A
Description	Number of problems in the local environment from the 3 categories: upkeep, traffic and utilisation.
Type	Numeric
Measure	Scale
Value labels	N/A
Missing data	None
File location	physical 18+19.sav
Validation	Frequency checks and comparison with results from previous years.
Key inputs	'lv1upkpx', 'lv2trafx', 'lv3utilx'
Modelling	None – derived from the counting the incidence of the above variables.
Imputation	None
Key dependencies	None
Introduction date	1991
Date of key modelling changes	None
Other relevant information	The variable was originally located in the general.sav file from 1991 to 2007. This variable is available in the datasets provided under the End User Licence and Special Licence.
Previous variable	None

Mainfuel

Physical 18+19.sav

Variable name	Mainfuel
Variable label	Main heating fuel
Units	N/A
Description	Type of fuel used for main or primary space heating system (specified in 'heat7x', 'heat4x') - 17 types
Type	Numeric
Measure	Nominal
Value labels	1 "gas (mains)" 2 "bulk LPG" 3 "bottled gas - propane" 4 "heating oil" 5 "house coal" 6 "smokeless fuel" 7 "anthracite nuts" 8 "anthracite grains" 9 "biomass" 10 "electricity (7 hr. on peak)" 11 "electricity (7 hr. off peak)" 12 "electricity (standard tariff)" 13 "electricity (10 hr. on peak)" 14 "electricity (10 hr. off peak)" 15 "electricity (24 hr. heating tariff)" 16 "community heating from boilers" 17 "community heating from CHP/waste heat"
Missing data	None
File location	physical 18+19.sav
Validation	Frequency checks and consistency checks against time series. Crosschecks against raw physical survey variables as part of modelling validation.
Key inputs	'Finmhfue', 'finchbcd', 'finohtyp', 'finmhbit'
Modelling	Created by the energy model primarily using the primary heating fuel variable. This is checked against the heating appliance code. If no primary system has been identified the other heating category becomes the primary system and the corresponding fuel is assigned as the primary fuel.
Imputation	Where no primary heating exists and the other heating is unknown, electric room heaters are assumed.
Key dependencies	'fuelx'
Introduction date	1991
Date of key modelling changes	2002 – Separate variables were created for type of system, type of fuel and boiler type. Until then, there was one combined primary heating variable and a separate variable for

	programmable heating. Categories for other heating were expanded.
Other relevant information	From 2015, value '9' was relabelled from wood to biomass. Biomass includes wood (chips, logs or pellets) but also includes any biomass gas or biomass oil. There are currently very few cases with this 'new' fuel type. This variable is only available in the dataset provided under the Special Licence.
Previous variable	N/A

Mainsgas

Physical 18+19.sav

Variable name	Mainsgas
Variable label	Mains gas supply present
Units	N/A
Description	Whether the dwelling has mains gas present
Type	Numeric
Measure	Nominal
Value labels	0 "no" 1 "yes"
Missing data	None
File location	physical 18+19.sav
Validation	Frequency checks and consistency checks against time series. Crosschecks against raw physical survey variables as part of modelling validation.
Key inputs	'Fingaspr', 'Fingasms', 'Finmhfue', 'Finohtyp', 'Finwhoty', 'Finwhxty', 'Finwsity', 'Finwdity', 'Finwspty', 'Finwempty'
Modelling	Derived in the energy model, based on whether mains gas is present as a heating fuel, or as part of the dwelling's primary services.
Imputation	None
Key dependencies	None
Introduction date	2019
Date of key modelling changes	None
Other relevant information	This variable is available in the datasets provided under the End User Licence and the Special Licence.
Previous variable	N/A

Neivisx**Physical 18+19.sav**

Variable name	Neivisx
Variable label	Appearance of area
Units	N/A
Description	Overall visual quality of local area as assessed by the surveyor using a scale.
Type	Numeric
Measure	Nominal
Value labels	-8 "unknown" 1 "satisfactory" 2 "some problems" 3 "poor"
Missing data	None
File location	physical 18+19.sav
Validation	Frequency check and comparisons with results from previous years.
Key inputs	'farquali'
Modelling	None – recodes the above variable.
Imputation	None
Key dependencies	None
Introduction date	1991. The variable was originally located in the general.sav file from 1991 to 2007.
Date of key modelling changes	None
Other relevant information	This variable is available in the datasets provided under the End User Licence and Special Licence.
Previous variable	None

Parking

Physical 18+19.sav

Variable name	Parking
Variable label	Parking provision of survey dwelling
Units	N/A
Description	Parking provision of the survey dwelling (based on the physical survey).
Type	Numeric
Measure	Nominal
Value labels	-8 "unknown" 1 "inadequate street parking" 2 "adequate street parking" 3 "garage" 4 "other off street parking" 5 "no parking provision"
Missing data	None
File location	physical 18+19.sav
Validation	Frequency checks and comparisons with previous years overall and by key dwelling descriptor variables. There is no other external data source that could be used to validate this variable.
Key inputs	'fcustr', 'fcuintpr', 'fcuattpr', 'fcudetpr', 'fcuporpr', 'fcuspapr'
Modelling	The physical survey records all types of parking available to the dwelling from observation and from asking the occupants about access to a garage or private parking space. The parking does not have to be on the plot of the dwelling – for example, the garage could be in a block of garages further down the road. Using the above raw variables, where there is more than one type of parking available, the 'best option' is chosen using the order as follows: garage, other off street parking, adequate street parking, inadequate street parking, and none.
Imputation	None needed.
Key dependencies	None
Introduction date	1991
Date of key modelling changes	None
Other relevant information	This variable is available in the datasets provided under the End User Licence and Special Licence.
Previous variable:	N/A

Pcavwallx**Physical 18+19.sav**

Variable name	Pcavwallx
Variable label	Percentage of external wall area that is cavity masonry
Units	N/A
Description	Banded percentage of total external wall area that is masonry cavity
Type	Numeric
Measure	Ordinal
Value labels	1 "all - 100%" 2 "75-99.9%" 3 "50-74.9%" 4 "25-49.9%" 5 "1-24.9%" 6 "none"
Missing data	None
File location	physical 18+19.sav
Validation	Frequency checks and consistency checks against derived variable 'wallcavy'. Comparisons with previous years.
Key inputs	Variable from energy model relating to proportion of wall that is cavity.
Modelling	The energy model combines derived dimensions data with tenths of external wall type and calculates the area of external wall that is of cavity construction. This is then banded to create 'pcavwallx'.
Imputation	None
Key dependencies	'constx'
Introduction date	2010
Date of key modelling changes	None
Other relevant information	For more detail see the section on modelled indicators, Chapter 5 of the English Housing Survey 2019-20 Technical Report Annex 6. This variable is available in the datasets provided under the End User Licence and Special Licence.
Previous variable:	N/A

Rdsap09**Physical 18+19.sav**

Variable name	rdsap09
Variable label	Reduced data energy efficiency rating (SAP 2009 based)
Units	SAP rating
Description	The energy efficiency rating of the dwelling on a normalised scale from 1 to 100, where 1 is the least efficient. The calculation uses the 2009 SAP methodology and reduced data assumptions.
Type	Numeric
Measure	Scale
Value labels – Special Licence	N/A
Value labels - End User Licence	91.00 “91 or more”
Missing data	None
File location	physical 18+19.sav
Validation	Extensive validation throughout the modelling process at key stages of the calculation. Checks against manual calculations using the RdSAP methodology.
Key inputs	See modelling
Modelling	Created by the energy model using raw data on heating and services, insulation, built form, age, construction and materials together with derived dimensions data. The model follows the SAP 2009 methodology for calculating domestic energy cost per m ² and uses the reduced data assumptions given in Appendix S of the SAP 2009 document. A logarithmic scale converts the cost per m ² into a normalised rating.
Imputation	Missing dimensions data or data not collected in the EHS are imputed using a default stereotype database, providing a generic value derived from cases of a similar dwelling type, age and size. Other missing inputs, e.g. boiler efficiency, are given a conservative estimate for that dwelling type.
Key dependencies	None
Introduction date	2013
Date of key modelling changes	N/A
Other relevant information	For anonymisation purposes, ratings are capped at 91 or more for the End User Licence. This variable is available in the datasets provided under the End User Licence and Special Licence.
Previous variable	N/A

Sap12

Physical 18+19.sav

Variable name	sap12
Variable label	Energy efficiency (SAP 2012) rating
Units	SAP rating
Description	The energy efficiency rating of the dwelling on a normalised scale from 1 to 100, where 1 is the least efficient. The calculation uses the methods and assumptions in the most recent SAP 2012 methodology.
Type	Numeric
Measure	Scale
Value labels – Special Licence	N/A
Value labels - End User Licence	91.00 “91 or more”
Missing data	None
File location	physical 18+19.sav
Validation	Extensive validation throughout the modelling process at key stages of the calculation. Comparisons with previous data across key dwelling characteristics and checks against manual calculations using the SAP methodology.
Key inputs	See modelling
Modelling	Created by the energy model using raw data on heating and services, insulation, built form, age, construction and materials together with derived dimensions data. The model follows the SAP 2012 methodology for calculating domestic energy cost per m ² as closely as the EHS data allows, with imputed values or assumptions based on the type of dwelling used where data is missing or not collected in the EHS. A logarithmic scale converts the cost per m ² into a normalised rating. For more detail see the section on modelled indicators, Chapter 5 of the English Housing Survey 2019-20 Technical Report. Annex 6.
Imputation	Missing dimensions data or data not collected in the EHS are imputed using a default stereotype database, providing a generic value derived from cases of a similar dwelling type, age and size. Other missing inputs, e.g. boiler efficiency, are given a conservative estimate for that dwelling type.
Key dependencies	‘sap412’, ‘EPceeb12e’
Introduction date	2013, however all earlier data as far back as 1996 has been remodelled using the current method (previous versions were ‘SAP09’, ‘SAP05’ and SAP01).
Date of key modelling changes	2002 – SAP 2001 methodology used and retrospectively calculated for 1991 to 2001 data. 2006 – SAP 2005 methodology adopted. Previous data recalculated using this method.

	2010 – SAP 2009 methodology adopted. Previous data recalculated using this method. 2011 – Incorporated EHS 2011 physical survey form changes to refine the SAP09 calculations, such as building orientation and window fenestration area of flats. 2013 - SAP 2012 methodology adopted. Previous data recalculated using this method. 2014 – Incorporated EHS 2014 physical survey form changes to refine the SAP2012 calculations. This includes improvements to PV generation, biomass heating, wall construction and wall insulation. 2019 – Updated RdSAP methodology to version 9.93, which revises the U-values applied to pre-1976 cavity solid and stone walls. This update is applied to dwellings from the 2018-19 survey year onwards.
Other relevant information	For anonymisation purposes, ratings are capped at 91 or more for the End User Licence. This variable is available in the datasets provided under the End User Licence and Special Licence.
Previous variable	'sap01', 'sap05' 'sap09'

Sap412**Physical 18+19.sav**

Variable name	sap412
Variable label	Energy efficiency (SAP 2012) rating
Units	N/A
Description	Banded SAP rating based on the most recent SAP 2012 methodology.
Type	Numeric
Measure	Ordinal
Value labels	1 "less than 30" 2 "30 to 50" 3 "51 to 70" 4 "more than 70"
Missing data	None
File location	physical 18+19.sav
Validation	Comparisons with previous data across key dwelling characteristics.
Key inputs	'sap12'
Modelling	See 'sap12'.
Imputation	N/A
Key dependencies	N/A
Introduction date	2013, however all earlier data as far back as 1996 has been remodelled using the current method (previous versions included 'sap401', 'sap405' and 'sap409' – see dates of modelling changes)
Date of key modelling changes	2002 – SAP 2001 methodology used and retrospectively calculated for 1991 to 2001 data. 2006 – SAP 2005 methodology adopted. Previous data recalculated using this method. 2010 – SAP 2009 methodology adopted. Previous data recalculated using this method. 2013 – SAP 2012 methodology adopted. Previous data recalculated using this method. 2019 – Updated RdSAP methodology to v9.93. See SAP12.
Other relevant information	This variable is available in the datasets provided under the End User Licence and Special Licence.
Previous variable	'sap401', 'sap405' 'sap409'

Secure**Physical 18+19.sav**

Variable name	Secure
Variable label	Secure windows and doors
Units	N/A
Description	Whether the main entrance door and all accessible windows are secure. This is based on the assessment by the physical surveyor.
Type	Numeric
Measure	Nominal
Value labels	-8 "unknown" 0 "not fully secure" 1 "secure"
Missing data	None
File location	physical 18+19.sav
Validation	Frequency checks and comparisons with previous years. There is no other external data source that could be used to validate this variable.
Key inputs	'finsecme' and 'finsecwn'
Modelling	Coded as fully secure only where the security of the main entrance door and accessible windows (where present) are classed as high or fairly high.
Imputation	None
Key dependencies	None
Introduction date	1991
Date of key modelling changes	2001 – security question on physical form changed slightly
Other relevant information	This variable is available in the datasets provided under the End User Licence and Special Licence.
Previous variable	N/A

Storeyx

Physical 18+19.sav

Variable name	Storeyx
Variable label	Number of floors above ground
Units	Floors
Description	Number of floors in the house or block (if a flat) excluding basement.
Type	Numeric
Measure	Scale
Value labels – Special Licence	-8 “unknown”
Value labels - End User Licence	-8 “unknown” 6 “6 or more”
Missing data	None
File location	physical 18+19.sav
Validation	Frequency checks and plausibility checks with dwelling type.
Key inputs	‘fdhmfllrs’, ‘fdhafllrs’
Modelling	None – based on the maximum of the above 2 variables.
Imputation	None
Key dependencies	‘dwtype8x’, ‘dwtypenx’, ‘dwtype3x’ and all derived variables relating to dimensions, repair costs, decent homes and energy.
Introduction date	1991
Date of key modelling changes	In 2017, the coding was improved to correct storeyx for a handful of cases where both a main and additional part was recorded in the dimensions but a basement was recorded in only the main or additional part and not both parts.
Other relevant information	For anonymisation purposes, a dwelling with greater than 5 storeys is capped at 6 or more for the End User Licence. This variable is available in the datasets provided under the End User Licence and the Special Licence.
Previous variable	N/A

Sysage

Physical 18+19.sav

Variable name	Sysage
Variable label	Age of heating system
Units	N/A
Description	Age of primary space heating system. For central heating systems, this is the age of the boiler.
Type	Numeric
Measure	Ordinal
Value labels	-8 "missing" 1 "less than 3 years" 2 "3-12 years" 3 "more than 12 years"
Missing data	None
File location	physical 18+19.sav
Validation	Frequency checks and consistency checks against time series.
Key inputs	'finchbag', 'finohage'
Modelling	Created by the energy model using raw data on the age of the primary heating appliance. Where no primary system is coded, the other heating becomes the primary system and the other heating age variable is used.
Imputation	Where the heating age is unknown, the oldest category is assumed.
Key dependencies	N/A
Introduction date	1991
Date of key modelling changes	None
Other relevant information	This variable is available in the datasets provided under the End User Licence and the Special Licence.
Previous variable	N/A

Typercov

Physical 18+19.sav

Variable name	Typercov
Variable label	Predominant type of roof covering
Units	N/A
Description	Type of roof covering. Where there is more than one type, the majority type is taken.
Type	Numeric
Measure	Nominal
Value labels	0 "mixed types" 1 "natural slate/stone/shingle/thatch" 2 "man made slate" 3 "clay tile" 4 "concrete tile" 5 "asphalt" 6 "felt" 7 "glass/metal/laminate"
Missing data	None
File location	physical 18+19.sav
Validation	Checks with results from previous years.
Key inputs	'fexrc1te', 'fexrc2te', 'type'
Modelling	The number of tenths of each material is summed across the front and back views and the majority type is taken. Where there is no majority type e.g. 50% concrete tile and 50% asphalt, it is classed as 'mixed types'.
Imputation	None
Key dependencies	None
Introduction date	1991
Date of key modelling changes	N/A
Other relevant information	'Thatch roof' category is combined with 'natural slate/stone/shingle'. This variable is only available in the dataset provided under the Special Licence.
Previous variable	N/A

Typersr**Physical 18+19.sav**

Variable name	Typersr
Variable label	Predominant type of roof structure
Units	N/A
Description	Type of roof structure. Where there is more than one type, the majority type is taken.
Type	Numeric
Measure	Nominal
Value labels	0 "mixed types" 1 "pitched" 2 "mansard" 3 "flat" 4 "chalet"
Missing data	None
File location	physical 18+19.sav
Validation	Checks with results from previous years.
Key inputs	'fexrs1te', 'fexrs2te', 'type'
Modelling	The number of tenths of each material is summed across the front and back views and the majority type is taken. Where there is no majority type e.g. 50% pitched and 50% flat, it is classed as 'mixed types'.
Imputation	None
Key dependencies	None
Introduction date	1991
Date of key modelling changes	N/A
Other relevant information	This variable is only available in the dataset provided under the Special Licence.
Previous variable	N/A

Typewfin

Physical 18+19.sav

Variable name	Typewfin
Variable label	Predominant type of wall finish
Units	N/A
Description	Type of wall finish. Where there is more than one type, the majority type is taken.
Type	Numeric
Measure	Nominal
Value labels	0 "mixed types" 1 "masonry pointing" 2 "non-masonry natural" 3 "rendered" 4 "shiplap timber" 5 "tile hung" 6 "slip/tile faced" 7 "wood/metal/plastic panels"
Missing data	None
File location	physical 18+19.sav
Validation	Checks with results from previous years.
Key inputs	'fexwf1te', 'fexwf2te', 'type'
Modelling	The number of tenths of each material is summed across the front and back views and the majority type is taken. Where there is no majority type e.g. 50% masonry pointing and 50% rendered, it is classed as 'mixed types'.
Imputation	None
Key dependencies	None
Introduction date	1991
Date of key modelling changes	None
Other relevant information	This variable is only available in the dataset provided under the Special Licence.
Previous variable	N/A

Typewin

Physical 18+19.sav

Variable name	Typewin
Variable label	Predominant type of window
Units	N/A
Description	Type of windows. Where there is more than one type, the majority type is taken.
Type	Numeric
Measure	Nominal
Value labels	0 "mixed types" 1 "single-glazed- wood casement" 2 "single-glazed- wood sash" 3 "single-glazed- UPVC" 4 "single-glazed- metal" 5 "double-glazed- wood" 6 "double-glazed- UPVC" 7 "double-glazed- metal"
Missing data	None
File location	physical 18+19.sav
Validation	Checks with results from previous years.
Key inputs	'fexwn1no', 'fexwn2no', 'type'
Modelling	The number of windows of each type is summed across the front and back views and the majority type is taken. Where there is no majority type e.g. there are 4 single-glazed wood casement and 4 double-glazed UPVC, it is classed as 'mixed'.
Imputation	None
Key dependencies	None
Introduction date	1991
Date of key modelling changes	None
Other relevant information	This variable is only available in the dataset provided under the Special Licence.
Previous variable	N/A

Typewstr2

Physical 18+19.sav

Variable name	Typewstr2
Variable label	Predominant type of wall structure
Units	N/A
Description	Type of wall structure. Where there is more than one type, the majority type is taken.
Type	Numeric
Measure	Nominal
Value labels	0 "mixed types" 1 "masonry cavity" 2 "masonry single leaf" 3 "9 inch solid" 4 "greater than 9 inch solid" 5 "in situ concrete" 6 "concrete panels" 7 "timber/metal/plastic panels"
Missing data	None
File location	physical 18+19.sav
Validation	Checks with results from previous years.
Key inputs	'fexws1te', 'fexws2te', 'type'
Modelling	The number of tenths of each material is summed across the front and back views and the majority type is taken. Where there is no majority type e.g. 50% masonry cavity and 50% 9" solid, it is classed as 'mixed'.
Imputation	None
Key dependencies	'constx'
Introduction date	2011
Date of key modelling changes	2011
Other relevant information	Due to changes in wall structure categories in 2011 (section 18 of physical form), timber panels and metal sheet walls are now a combined category. A crosstab of predominant wall by wall type and by original construction type may show inconsistencies. This is due to the predominance being calculated on tenths of area, whilst wall type makes use of modelled wall areas from dimensions data. This variable is only available in the dataset provided under the Special Licence.
Previous variable	Typewstr

Wallcavy

Physical 18+19.sav

Variable name	Wallcavy
Variable label	Type of wall
Units	N/A
Description	Predominant type of external walling, distinguishing between cavity and solid walls
Type	Numeric
Measure	Nominal
Value labels	1 "cavity wall" 2 "solid wall" 3 "other"
Missing data	None
File location	physical 18+19.sav
Validation	Frequency checks and consistency checks against time series.
Key inputs	'Fexws1te', 'Fexws2te' and derived wall areas from dimensions model
Modelling	The energy model combines derived dimensions data with tenths of external wall type and calculates the area of external wall that is of cavity construction. If this area is at least 50% of the total external wall area the dwelling is coded as "cavity wall". If the area of solid and system build wall is more than 50% of the total external wall area the dwelling is coded as "solid wall".
Imputation	None
Key dependencies	'pcavwallx', 'constx'
Introduction date	2013 (however for all earlier data as far back as 1991, there is a similar variable, "wallcavx", combining "solid wall" and "other" wall type categories).
Date of key modelling changes	None
Other relevant information	All dwellings where the external walling is at least 50% cavity brickwork are coded as 'cavity wall'. This will include a number of dwellings with timber, steel or concrete frames. For more detail see the section on modelled indicators, Chapter 5 of the English Housing Survey 2019-20 Technical Report Annex 6. A crosstab of predominant wall by wall type and by original construction type may show inconsistencies. This is due to the predominance being calculated on tenths of area, whilst wall type makes use of modelled wall areas from dimensions data. This variable is available in the datasets provided under the End User Licence and the Special Licence.
Previous variable	'wallcavx'

Wallinsz**Physical 18+19.sav**

Variable name	Wallinsz
Variable label	Type of wall and insulation
Units	N/A
Description	Summary variable indicating whether the external walling is predominantly cavity or solid and whether it is insulated.
Type	Numeric
Measure	Nominal
Value labels	1 "cavity with insulation" 2 "cavity uninsulated" 3 "solid with insulation" 4 "solid uninsulated" 5 "other"
Missing data	None
File location	physical 18+19.sav
Validation	Frequency checks and consistency checks against time series.
Key inputs	'Fexws1te', 'Fexws2te', 'Felcavlf', 'Felcavff', 'Felcavrf', 'Felcavbf', 'fcwipres' plus derived wall areas from dimensions model
Modelling	The energy model combines derived dimensions data with tenths of external wall type in wallstru.sav and calculates the area of external wall that is of cavity construction. If this area is at least 50% of the total external wall area, the dwelling is coded as cavity wall. If the area of solid and system build wall is more than 50% of the total external wall area, the dwelling is coded as solid wall. The model uses the surveyor's assessment of whether the dwelling has wall insulation, including evidence from the external wall faces, to decide whether at least 50% of the wall area is insulated. If so, the dwelling is coded as '... with insulation', other walls are coded as '... uninsulated'. In addition to the surveyor's assessment of the existence of insulation, an assumption is made that all dwellings constructed after 1995 were built with insulation. This assumption is consistent with RdSAP (www.bre.co.uk/sap2012).
Imputation	None
Key dependencies	'wins95x'
Introduction date	2014. Between 2012 and 2014 a similar variable "wallinsy" existed. However, for all earlier data as far back as 1991, there was a variable called "wallinsx" combining "cavity with insulation", "cavity uninsulated" and "other" wall type categories).
Date of key modelling changes	2008 - a series of new check questions were introduced to encourage surveyors to systematically look for evidence of

	CWI throughout the dwelling (loft, air-brick etc.). The briefing to surveyors also changed so that they only recorded CWI on the elevation features if they saw evidence of retro-fit (drill holes). A new summary section was also introduced along to enable the surveyor to draw together the evidence and make an overall assessment about the presence and extent of CWI in the house or module. 2010 - Homes built after 2002 with cavity walls are assumed to have full cavity wall insulation in line with Building Regulations. 2014 – Assumption about existence of CWI at time of construction extended to include dwellings built after 1995 in line with RdSAP. Under “wallinsy” homes built after 2002 with cavity walls were assumed to have full cavity wall insulation in line with Building Regulations.
Other relevant information	All dwellings where the external walling is at least 50% cavity brickwork are coded in one of the ‘cavity wall’ categories. This will include a number of dwellings with timber, steel or concrete frames. For more detail see the section on modelled indicators, Chapter 5 of the English Housing Survey 2019-20 Technical Report Annex 6. This variable is available in the datasets provided under the End User Licence and the Special Licence.
Previous variable	‘wallinsy’

Watercyl

Physical 18+19.sav

Variable name	Watercyl
Variable label	Hot water cylinder present
Units	N/A
Description	Presence of a water cylinder used to provide hot water to the dwelling.
Type	Numeric
Measure	Nominal
Value labels	0 "no" 1 "yes"
Missing data	None
File location	physical 18+19.sav
Validation	Frequency checks and consistency checks against time series. Crosschecks against raw physical survey variables as part of energy modelling validation.
Key inputs	'Finwheat', 'Finwhcpr', 'Finwhopr', 'Finwhxpr', 'Finwsipr', 'Finwdipr', 'Finwhlpr', 'Finwhcyl', 'Finchtyp', 'Finmhboi'
Modelling	Created by the energy model using the water heating section. The presence of a hot water cylinder is modelled to be consistent with other aspects of heating within the dwelling (i.e. if hot water is provided by a combination boiler then a cylinder will not be modelled as being present).
Imputation	A water cylinder is assumed to be present for standard boilers, back boilers, heat pumps and communal heating systems, where hot water is provided to the dwelling through the main heating system.
Key dependencies	'Watersys', 'Boiler'
Introduction date	2019
Date of key modelling changes	None
Other relevant information	This variable is available in the datasets provided under the End User Licence and the Special Licence.
Previous variable	N/A

Watersys

Physical 18+19.sav

Variable name	Watersys
Variable label	Water heating system
Units	N/A
Description	Type of water heating system. Where there is more than one type, the system used for analysis and modelling is selected in the order of the categories below (first is central heating, second dedicated etc.)
Type	Numeric
Measure	Nominal
Value labels	1 "with central heating" 2 "dedicated boiler" 3 "electric immersion heater" 4 "instantaneous" 5 "other"
Missing data	None
File location	physical 18+19.sav
Validation	Frequency checks and consistency checks against time series. Crosschecks against raw physical survey variables as part of energy modelling validation.
Key inputs	'Finwheat', 'Finwhcpr', 'Finwhopr', 'Finwhxpr', 'Finwsipr', 'Finwdipr', 'Finwsppr', 'Finwmppr', 'Finwhlpr', 'Finwotpr'
Modelling	Created by the energy model using the water heating section. Surveyors code all water heating systems present and the model uses an algorithm to decide on the primary type where there are multiple systems. For more detail see the section on modelled indicators, Chapter 5 of the English Housing Survey 2019-20 Technical Report Annex 6.
Imputation	Where the water heating is unknown, an electric immersion is assumed.
Key dependencies	None
Introduction date	2006
Date of key modelling changes	None
Other relevant information	This variable is available in the datasets provided under the End User Licence and the Special Licence.
Previous variable	N/A

Wins95x**Physical 18+19.sav**

Variable name	Wins95x
Variable label	Type of wall - post 1995 assumption
Units	N/A
Description	Overall wall type and whether cavity walls are insulated. This was created following changes to the energy modelling; with CWI assumed as built at time of construction being extended to include dwellings built after 1995 in line with RdSAP. Solid wall dwellings are included in the “other” category. A separate category is included for this variable that identifies cavity walled dwellings built after 1995 where no evidence of cavity wall insulation was seen by the surveyors and where no assumptions have been made based on the construction date. Data users can therefore make their own assumptions about what proportion of these homes would have had cavity wall insulation fitted at the time of construction.
Type	Numeric
Measure	Nominal
Value labels	1 “cavity walls with evidence of insulation” 2 “cavity walls with insulation assumed as built (post-1995 cavity walls with no evidence of insulation)” 3 “cavity uninsulated” 4 “other”
Missing data	None
File location	physical 18+19.sav
Validation	Frequency checks and consistency checks against time series.
Key inputs	‘wallinsz’, ‘fodconst’
Modelling	See ‘wallinsz’
Imputation	None
Key dependencies	None
Introduction date	2014
Date of key modelling changes	See ‘wallinsz’
Other relevant information	All dwellings where the external walling is at least 50% cavity brickwork are coded in one of the ‘cavity wall’ categories. This

	will include a number of dwellings with timber, steel or concrete frames. This variable is available in the datasets provided under the End User Licence and the Special Licence.
Previous variable	Wins90x (homes built after 2002 with cavity walls were assumed to have full cavity wall insulation in line with Building Regulations so not directly comparable)